

CMPE 413

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Lab Final Project Part 2: Cache VHDL Code and Layout

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Section 1 Mon/Wed 1:00 PM–2:15 PM

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1 Introduction

1.1 Design Strategies

The cache cell and cache block designs were pretty straightforward as the cache cell components were given and the cache block was just 4 cache cells with enables in them. The cache array is where a lot of the logic of the cache comes from. The cache array has the built in decoder and muxes so when an address is inputted the decoder would enable the designated block. This strategy was used throughout the project. Where enables were used to determine what should happen. We first did the cache FSM as behavioral so we could get a better understanding of how the logic was supposed to work. The FSM behaviorally was a moore machine where when a state was reached an output/action would occur. After we got the cache FSM to fully work, we then turned it to structural code. Turning it structurally used a lot of flip flops and muxes. The best way to store data at clock edges is through flip flops and the best way to get conditional statements is through enables and muxes.

1.2 Troubleshooting

There were several changes with this project. Most include just coding what should happen. There were also timing issues where we would just add or remove states till the timing requirements were met. The structural code for the cache array and cache FSM were challenging because we didn't have much experience with coding complex structural modules. Another challenge was getting the project to run on cadence. We used the EDA playground to test everything first and then we transferred the files to cadence. Because EDA playground is much different from cadence, some things that worked in the EDA would not work in cadence. Troubleshooting these issues was a big challenge.

2 Cache Diagrams

2.1 Block Diagram of Hierarchical Design

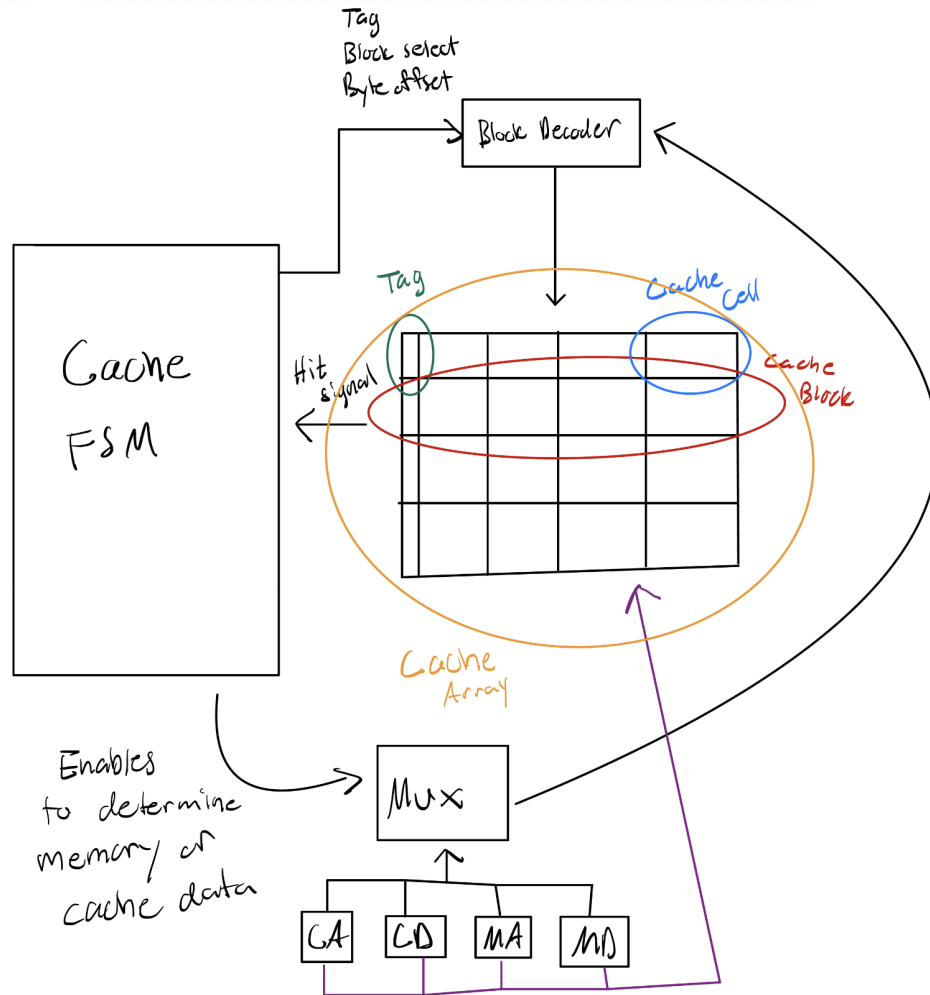


Figure 1: Cache Design

Figure 1 shows the full design of the cache. The cache array (inside the orange circle) is composed of cache blocks which are composed of cache tags and cache cells. The cache array is where the data is stored. The cache array has a decoder and mux going into them because the decoder decides which block should be looked at when an address is passed in. The mux decides what addresses and data is going to be checked/inputted memory or cache. The cache FSM handles the timing and actions taken to perform these operations.

2.2 Cache Cell Diagram

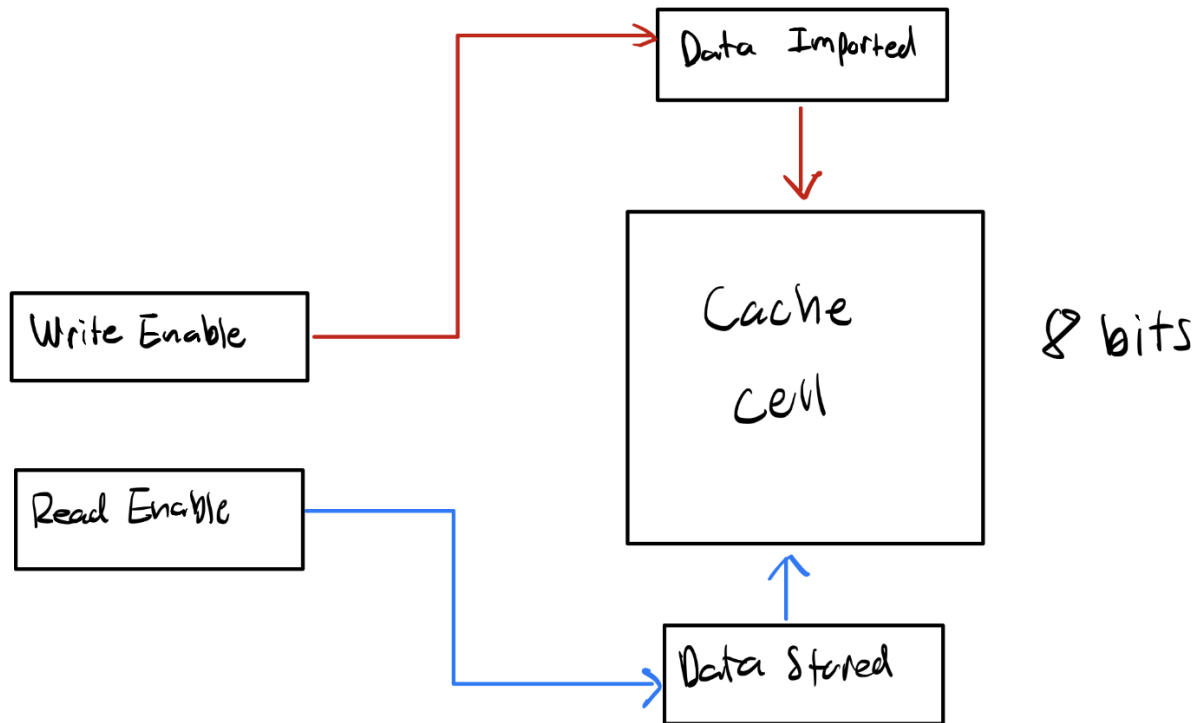


Figure 2: Cache Cell Diagram

Figure 2 shows the design idea of the cache cell. Write and read enablers are used to determine whether data should be written into the cells or not. Write enable enables the data passed in to be stored in the cells. Read enable enables the data in the cells to be read and output to the cache.

2.3 Cache Block Diagram

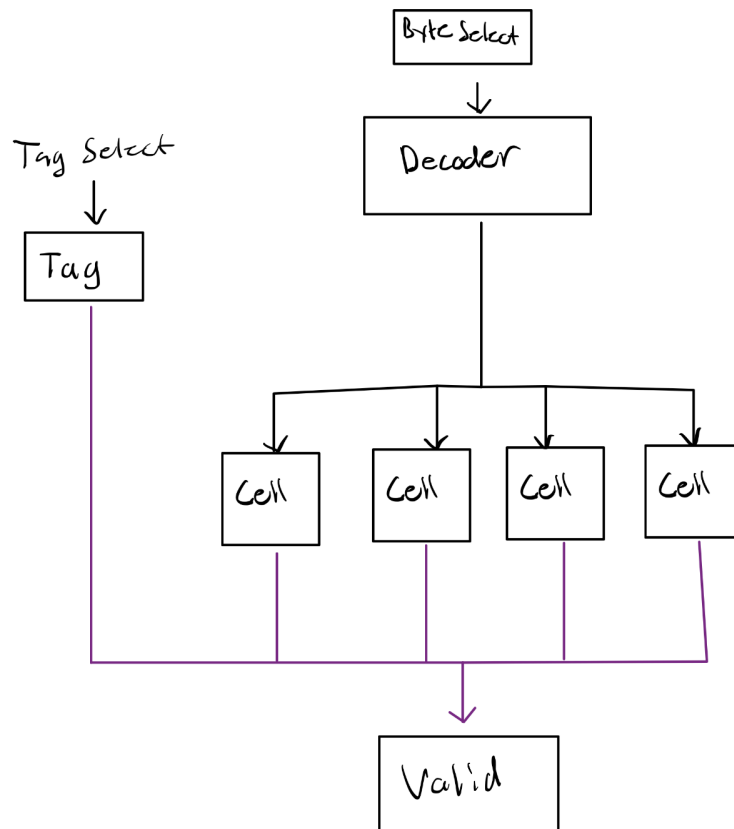


Figure 3: Cache Block Diagram

Figure 3 shows the cache block design. The cache block has 4 cache cells and there is a decoder that takes in the byte select to determine which cell to operate on. There is a tag select that takes in the inputted tag and stores it. The tag select and byte select will be used to check whether a block is valid or not. The valid state will be used by the cache array.

2.4 Cache Array Diagram

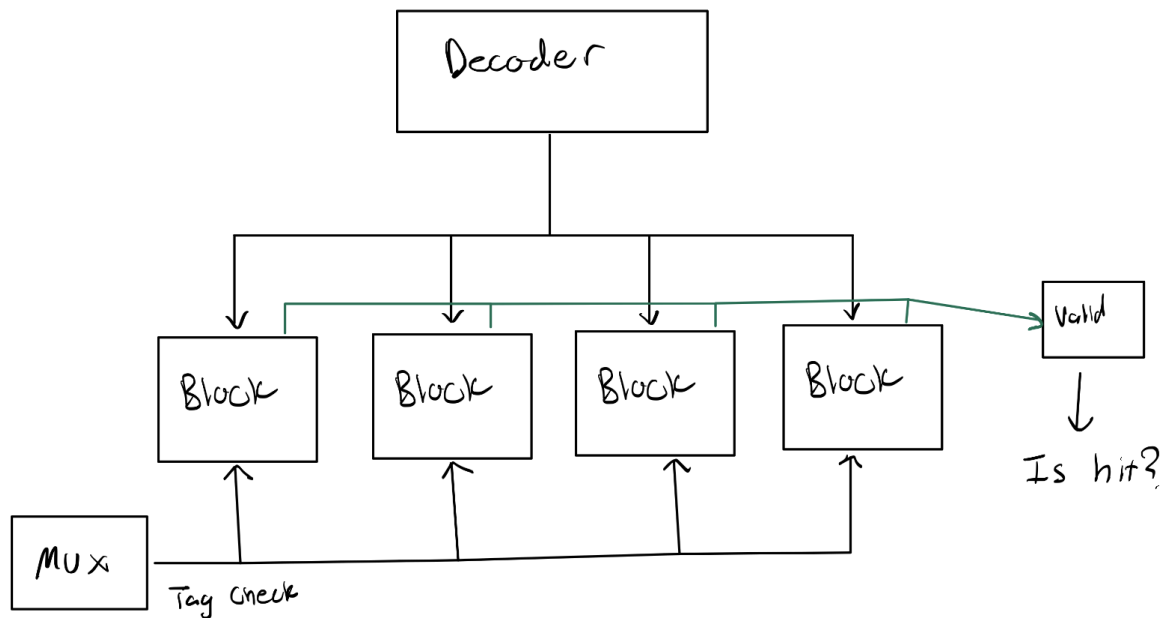


Figure 4: Cache Array Diagram

Figure 4 shows the cache array design idea. The cache array contains 4 cache blocks. There is a decoder that decides which block should be accessed. The decoder would take in the block select and whichever block is selected will have a byte selected imported as well. There is mux that determines which tag should be chosen. The valid from the chosen block and if tag matches will be used to determine whether the address is a hit.

2.5 Cache FSM Diagram

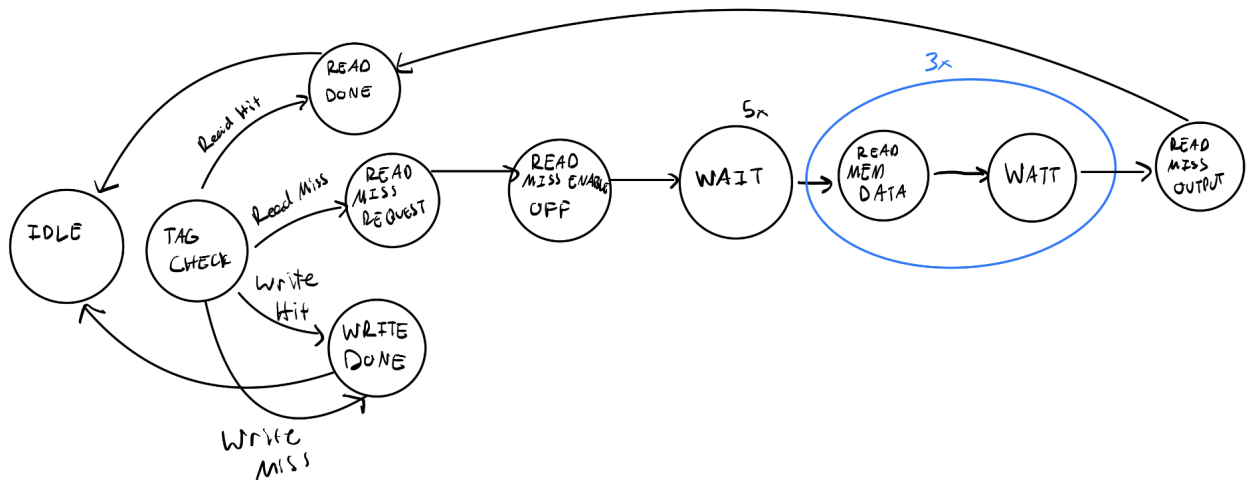


Figure 5: Cache FSM State Diagram

Figure 5 shows the state machine used for the cache. Tag check determines where a read miss, read hit, write miss, or write hit has occurred. Read miss first goes to read miss state which goes to a read miss request state that turns enable off and the next state will turn off the enable. After that state then a wait occurs to match the correct timing and then the memory data is written to the cache and then read after. Read hit goes directly to read done where data from an address is read. Write miss goes to a write miss state where nothing happens and then to write done. Write hit goes to write done where data is written to the correct address. All operations go to idle when complete.

3 Cache Functionality

3.1 Read Miss

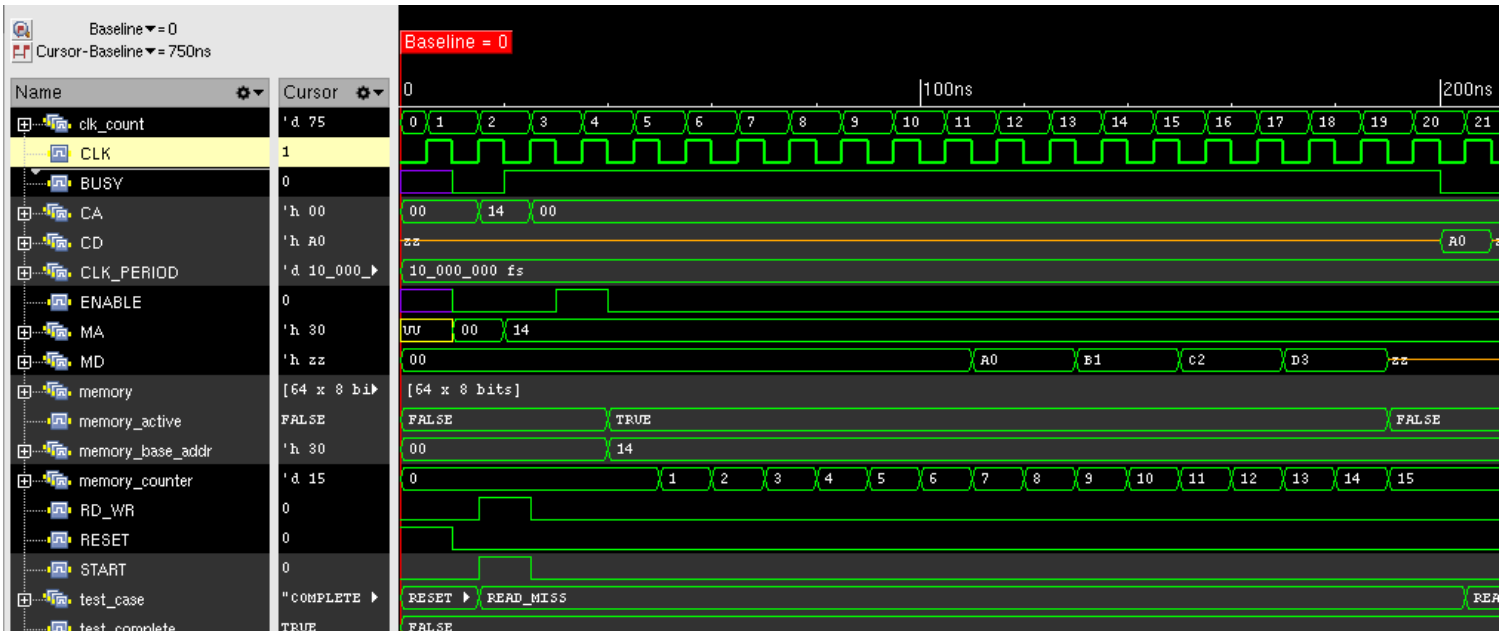


Figure 6: Read Miss Waveform

Figure 6 shows what the waveform looks like when a read miss occurs. The first data from memory is first imported at the 11'th clock cycle (the clk_count is off by an edge but the number of cycles matches the example waveform). Then after the 11'th clock the next 3 data from memory are imported every 2 cycles. The BUSY signal rises high at the negative edge in the middle of when START goes high. BUSY stays high until the read miss is finished. As soon as the busy goes low the first byte from the memory imported is present in the CD. The clk_count shows the read miss finish at 20 cycles (clk_count is off by an edge but clk is not). The ENABLE signal is the memory enable, where the memory will begin to be imputed after 8 cycles.

3.2 Read Hit

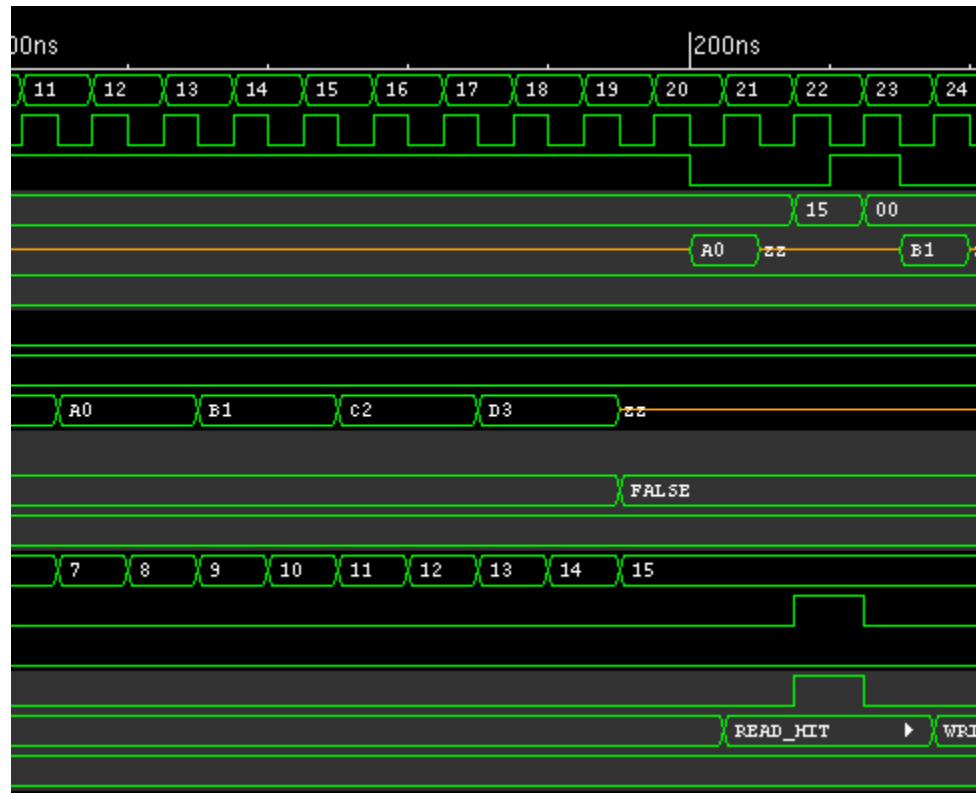


Figure 7 : Read Hit Waveform

Figure 7 shows what the waveform looks like for a read hit. For read hit the busy goes high for only one clock cycle. This can be seen on the graph above the “15” and “00”. Then right after the busy goes low “B1” appears on the cache data showing that the data was read. Address 15 is the second byte of the block containing address 14. This means address 15 should contain B1 because that was the second input from the read miss memory input. The second memory input corresponds to byte offset “01” which is just one more than 14. This shows that the data was read correctly.

3.3 Write Hit

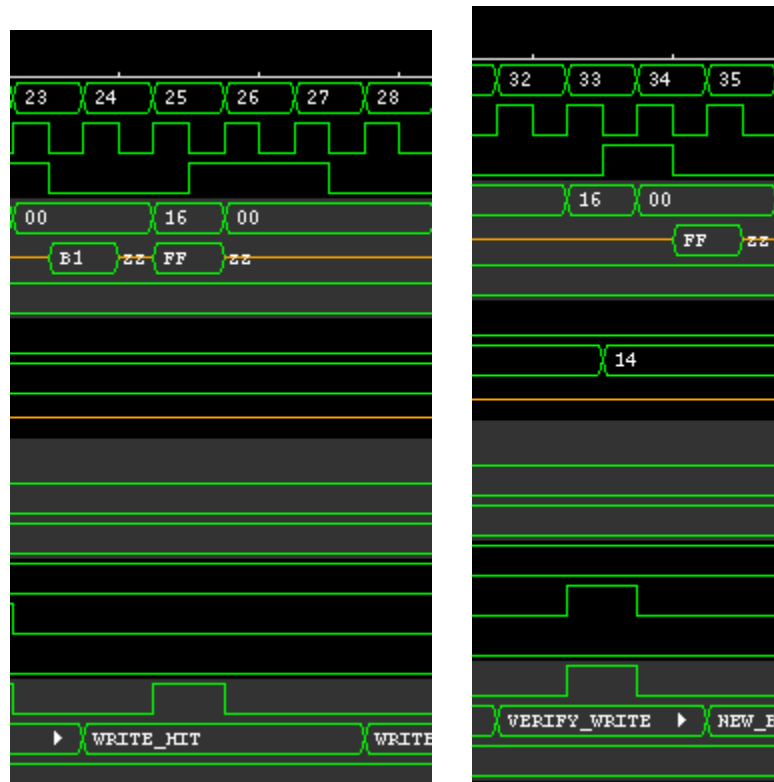


Figure 8: Write Hit Waveform

Figure 8 shows the waveform for a write hit. The first part of Figure 8 shows when the write hit occurs. The cache data “FF” was written to the cache address “16”. The BUSY signal goes high for 2 clock cycles in the middle of the START signal (signal above the WRITE_HIT and VERIFY_WRITE). No data is seen in the cache data because it's just righting the data. A read would need to occur to confirm the data. The second part of Figure 8 shows what was at the address where the write hit occurred. The second part shows a read hit on the address where the write hit wrote the data. After the read hit “FF” was shown on the cache data, this shows that the write hit was able to correctly write the data at the address.

3.4 Write Miss

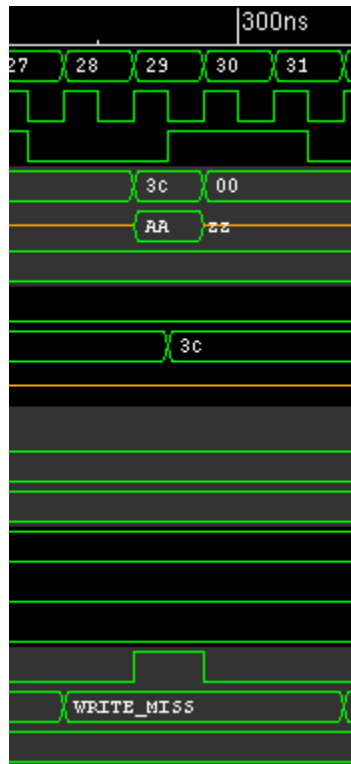


Figure 9 : Write Miss Waveform

Figure 9 shows the waveform for a write miss. The BUSY signal can be seen to only go high for two clock cycles in the middle of the START signal. The BUSY signal stays high for two clock cycles. This is a read miss because the address 3c was not populated by the read miss earlier in the waveform. Write miss doesn't do anything so nothing is changed when it is called. Only 3 clock cycles pass when write miss occurs.

4 Cache Layouts

4.1 Basic Logic Gates

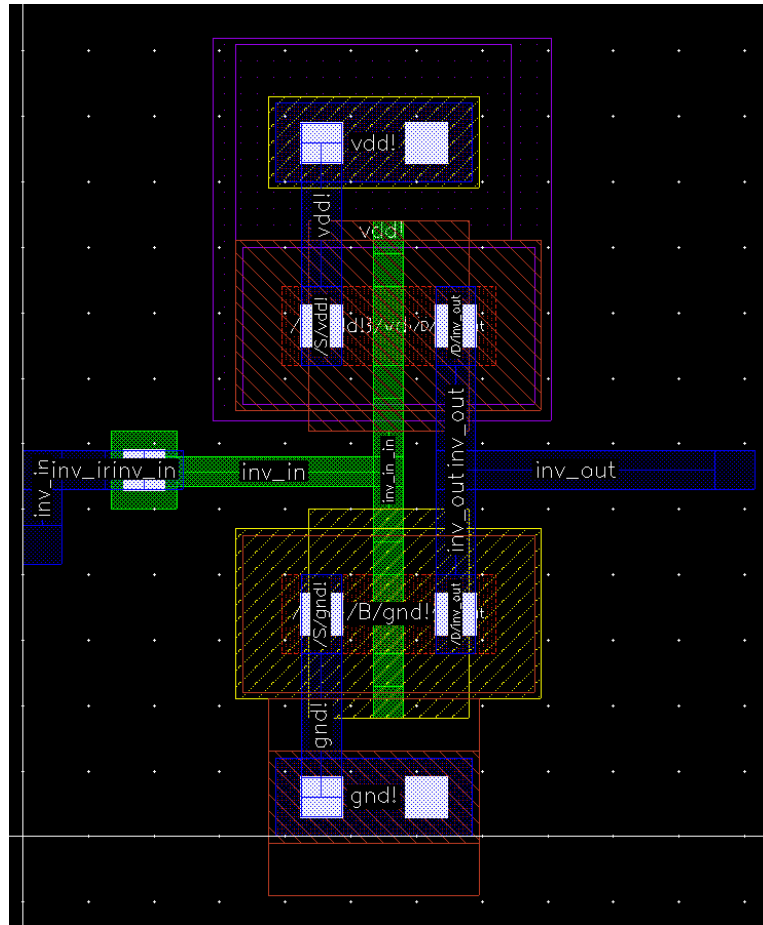


Figure 10: Inverter Layout

Figure 10 shows the layout for the inverter.

```
INFO: Rule ESD_8_2 completed with no violations, 562/562
INFO: Status: Runset 100% complete as of 2025-11-23 22:21:55 Elapsed: 00:00:02
Worker: Active CPUs Total Mem(MB) Active Mem(MB) CPU time(s)
0: 1 / 1 22 12 1
INFO: Status: Finishing as of 2025-11-23 22:21:55 Elapsed: 00:00:02 CPU time[s]: 2
INFO: Resource usage by worker at exit:
CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 1 2 1 47.63 22 12
Worker resource usage summary: Total CPU[s]: 1 Run duration[s]: 2 Max peak memory[MB]: 22
Worker CPU utilization summary: 1 of 2 available CPU seconds (50.00%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/inverter.sum
INFO: All selected rules completed
Total CPU Time : 1(s)
Total Real Time : 3(s)
Total Original Geometry : 58(58)
Total DRC RuleChecks : 562
Total DRC Results : 0 (0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/inverter.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:21:55
```

Figure 11: Inverter DRC Test

Figure 11 shows the DRC Test for the inverter layout and it was able to run with 0 errors.

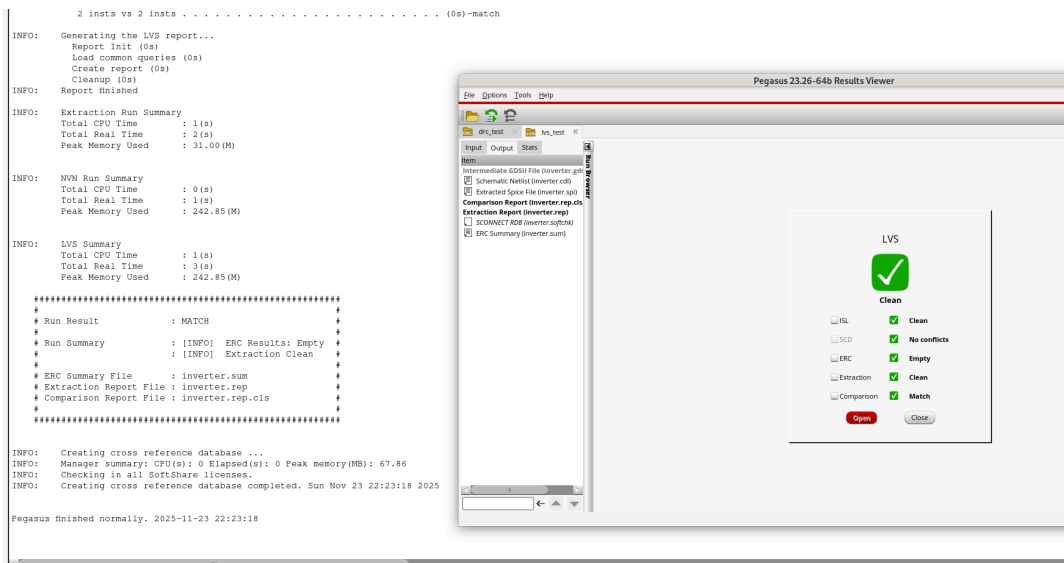


Figure 12: Inverter LVS Test

Figure 12 shows the LVS test for the inverter layout and it was able to pass.

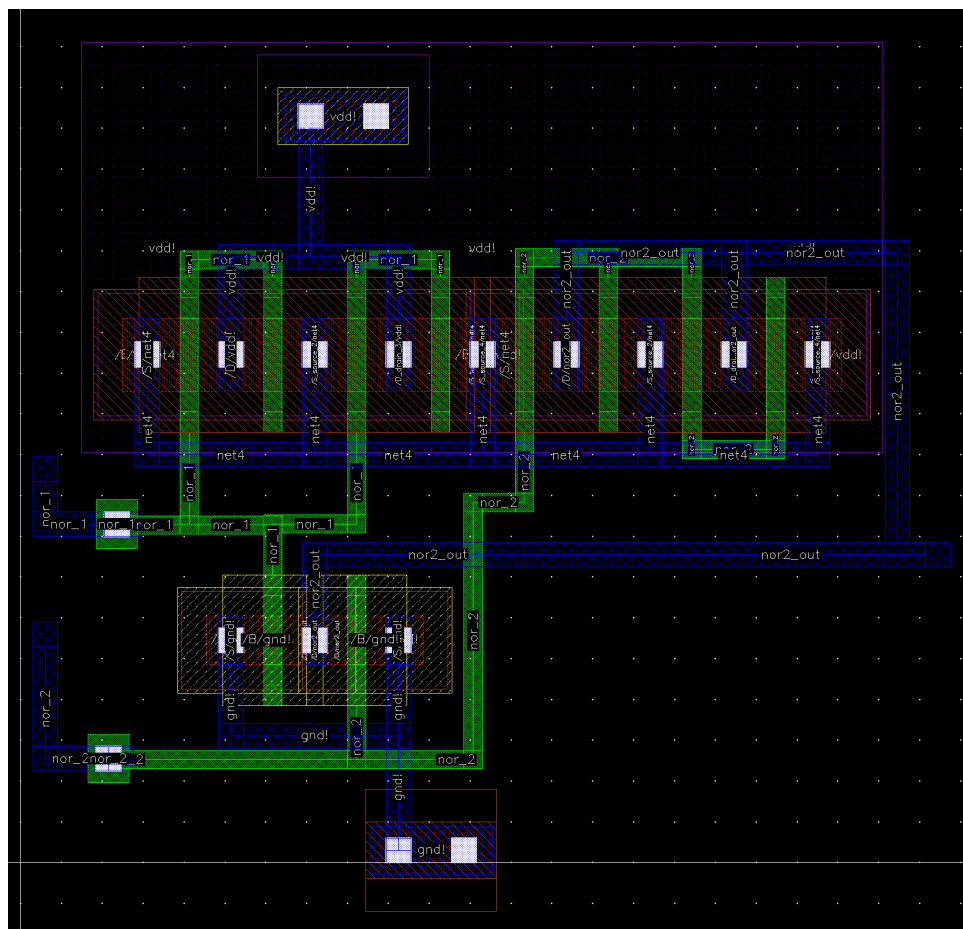


Figure 13: NOR 2-input Layout

Figure 13 shows the layout for the nor gate with 2 inputs.

```

INFO: Rule ESD.8_2 completed with no violations, 562/562
INFO: Status: Runset 100% complete as of 2025-11-23 22:24:45 Elapsed: 00:00:03
Worker: Active CPUs Total Mem(MB) Active Mem(MB) CPU time(s)
      0: 1 / 1 22 12 2
INFO: Status: Finishing as of 2025-11-23 22:24:45 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
      0: 2 3 1 62.58 22 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 22
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/nor2.sum
INFO: All selected rules completed
      Total CPU Time : 2(s)
      Total Real Time : 4(s)
      Total Original Geometry : 182(185)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/nor2.drc_errors.asci
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:24:45

```

Figure 14: NOR 2-input DRC Test

Figure 14 shows the DRC Test for the nor 2 input layout and it was able to run with 0 errors.

```

Report Init (0s)
Load common queries (0s)
Create report (0s)
Cleanup (0s)
INFO: Report finished
INFO: Extraction Run Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 31.00(M)
INFO: NVN Run Summary
      Total CPU Time : 0(s)
      Total Real Time : 1(s)
      Peak Memory Used : 243.91(M)
INFO: LVS Summary
      Total CPU Time : 1(s)
      Total Real Time : 3(s)
      Peak Memory Used : 243.91(M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
# ERC Summary File : nor2.sum #
# Extraction Report File : nor2.rep #
# Comparison Report File : nor2.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 71.00
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:25:39 2025

Pegasus finished normally. 2025-11-23 22:25:39

```

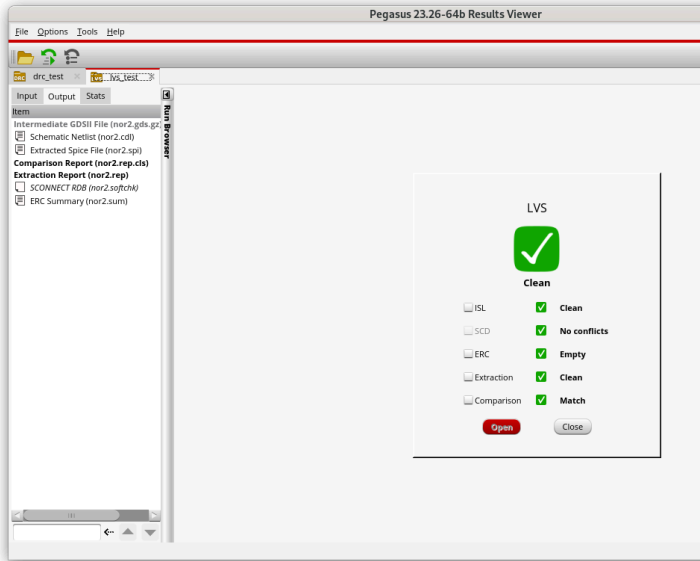


Figure 15: NOR 2-input LVS Test

Figure 15 shows the LVS test for the nor 2 input layout and it was able to pass.

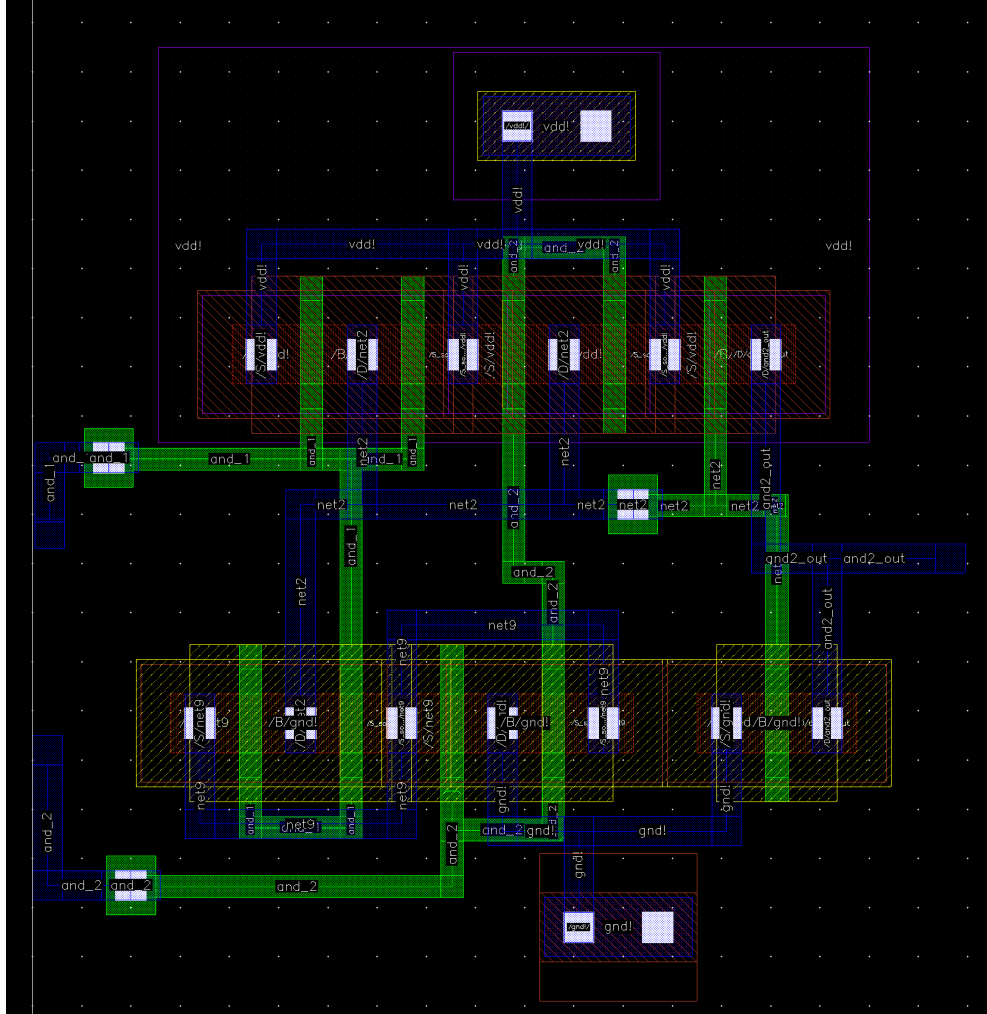


Figure 16: AND 2-input Layout

Figure 16 shows the layout for the 2 input and gate.

```
INFO: Rule DSD_6_2 completed with no violations, 302/302
INFO: Status: Runset 100% complete as of 2025-11-23 22:27:16 Elapsed: 00:00:02
Worker: Active CPUs Total Mem(MB) Active Mem(MB) CPU time(s)
0: 1 / 1 22 12 1
INFO: Status: Finishing as of 2025-11-23 22:27:16 Elapsed: 00:00:02 CPU time[s]: 2
INFO: Resource usage by worker at exit:
CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 1 2 1 48.98 22 12
Worker resource usage summary: Total CPU[s]: 1 Run duration[s]: 2 Max peak memory[MB]: 22
Worker CPU utilization summary: 1 of 2 available CPU seconds (50.00%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/and2.sum
INFO: All selected rules completed
Total CPU Time : 1(s)
Total Real Time : 3(s)
Total Original Geometry : 193(199)
Total DRC RuleChecks : 562
Total DRC Results : 0 (0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/and2.drc_errors.ascii
INFO: Checking in all SoftShare licenses.
Pegasus finished normally. 2025-11-23 22:27:16
```

Figure 17: AND 2-input DRC Test

Figure 17 shows the DRC Test for the 2 input AND layout and it was able to run with 0 errors.

```

INFO: Generating the LVS report...
      Report Init (0s)
      Load common queries (0s)
      Create report (0s)
      Cleanup (0s)
INFO: Report finished

INFO: Extraction Run Summary
      Total CPU Time      : 1(s)
      Total Real Time     : 2(s)
      Peak Memory Used    : 31.00 (M)

INFO: NVN Run Summary
      Total CPU Time      : 0(s)
      Total Real Time     : 0(s)
      Peak Memory Used    : 243.91 (M)

INFO: LVS Summary
      Total CPU Time      : 1(s)
      Total Real Time     : 2(s)
      Peak Memory Used    : 243.91 (M)

=====
# Run Result          : MATCH
# Run Summary         : [INFO] ERC Results: Empty
#                     : [INFO] Extraction Clean
# ERC Summary File    : and2.sum
# Extraction Report File : and2.rep
# Comparison Report File : and2.rep.cls
#
=====

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 1 Peak memory(MB): 69.55
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:27:52 2025

Pegasus finished normally. 2025-11-23 22:27:52

```

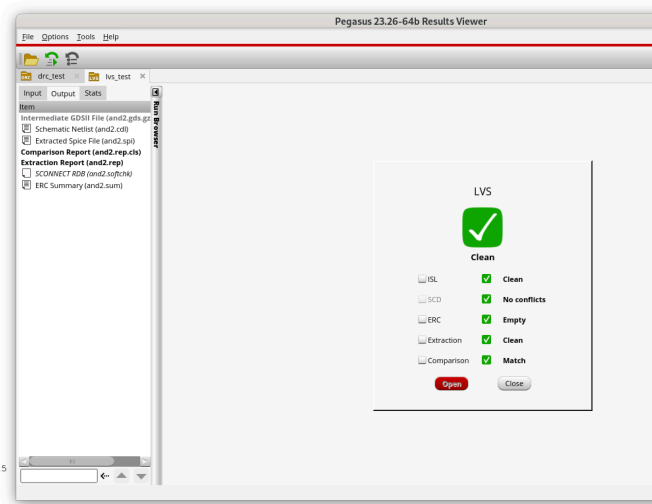


Figure 18: AND 2-input LVS Test

Figure 18 shows the LVS test for the 2 input AND layout and it was able to pass.

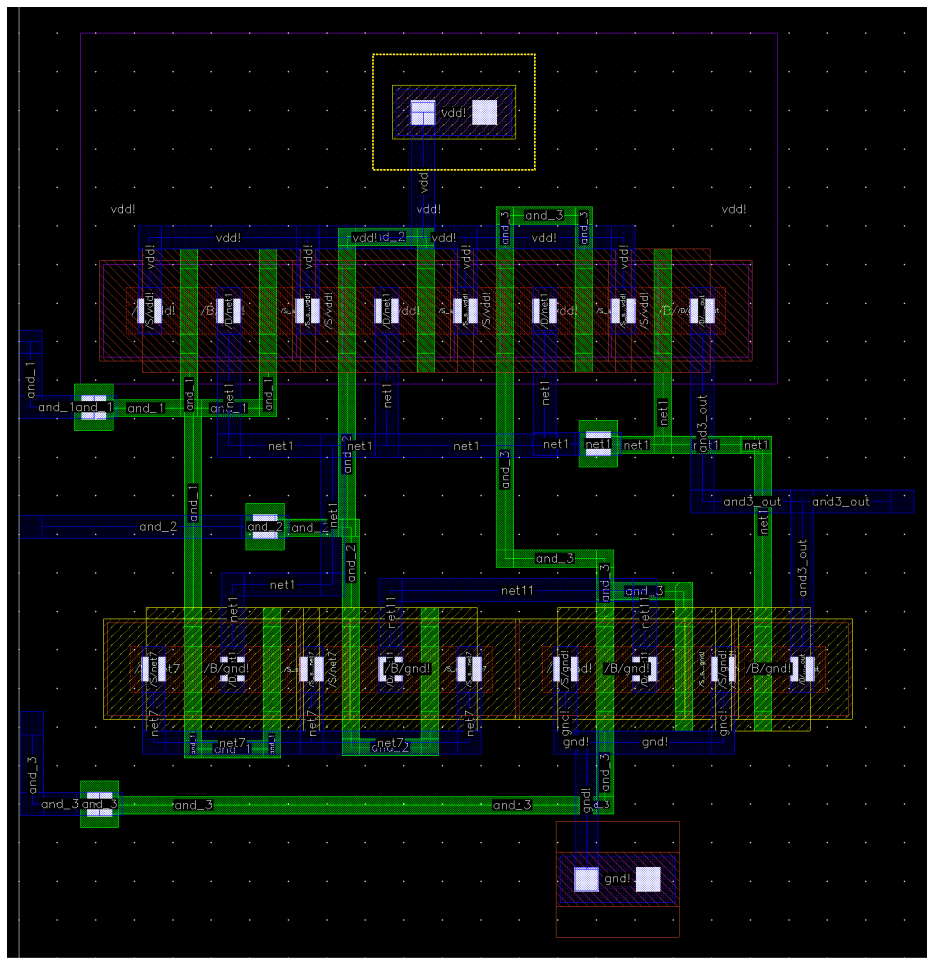


Figure 19: AND 3-input Layout

Figure 20 shows the layout for the 2 input AND gate.

```

Remaining rules: ESD.8_2
INFO: Rule ESD.8_2 completed with no violations, 562/562
INFO: Status: Runset 100% complete as of 2025-11-23 22:28:49 Elapsed: 00:00:02
Worker: Active CPUs Total Mem(MB) Active Mem(MB) CPU time(s)
      0: 1 / 1 22 12 1
INFO: Status: Finishing as of 2025-11-23 22:28:49 Elapsed: 00:00:02 CPU time[s]: 2
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
      0: 1 2 1 51.12 22 12
Worker resource usage summary: Total CPU[s]: 1 Run duration[s]: 2 Max peak memory[MB]: 22
Worker CPU utilization summary: 1 of 2 available CPU seconds (50.00%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/and3.sum
INFO: All selected rules completed
Total CPU Time : 1(s)
Total Real Time : 3(s)
Total Original Geometry : 237(268)
Total DRC RuleChecks : 562
Total DRC Results : 0 (0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/and3.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:28:49

```

Figure 20: AND 3-input DRC Test

Figure 20 shows the DRC Test for the 3 input AND layout and it was able to run with 0 errors.

```

Report Init (0s)
Load common queries (0s)
Create report (1s)
Cleanup (0s)
INFO: Report finished
INFO: Extraction Run Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 31.00 (M)
INFO: NVN Run Summary
      Total CPU Time : 0(s)
      Total Real Time : 1(s)
      Peak Memory Used : 243.91 (M)
INFO: LVS Summary
      Total CPU Time : 1(s)
      Total Real Time : 3(s)
      Peak Memory Used : 243.91 (M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
# ERC Summary File : and3.sum #
# Extraction Report File : and3.rep #
# Comparison Report File : and3.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 69.55
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:31:20 2025

Pegasus finished normally. 2025-11-23 22:31:20

```

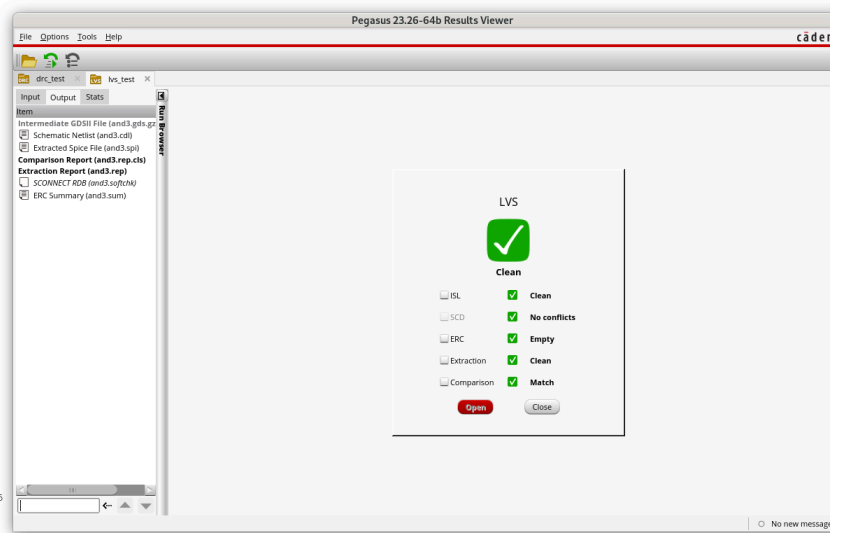


Figure 21: AND 3-input LVS Test

Figure 21 shows the LVS test for the 3 input AND layout and it was able to pass.

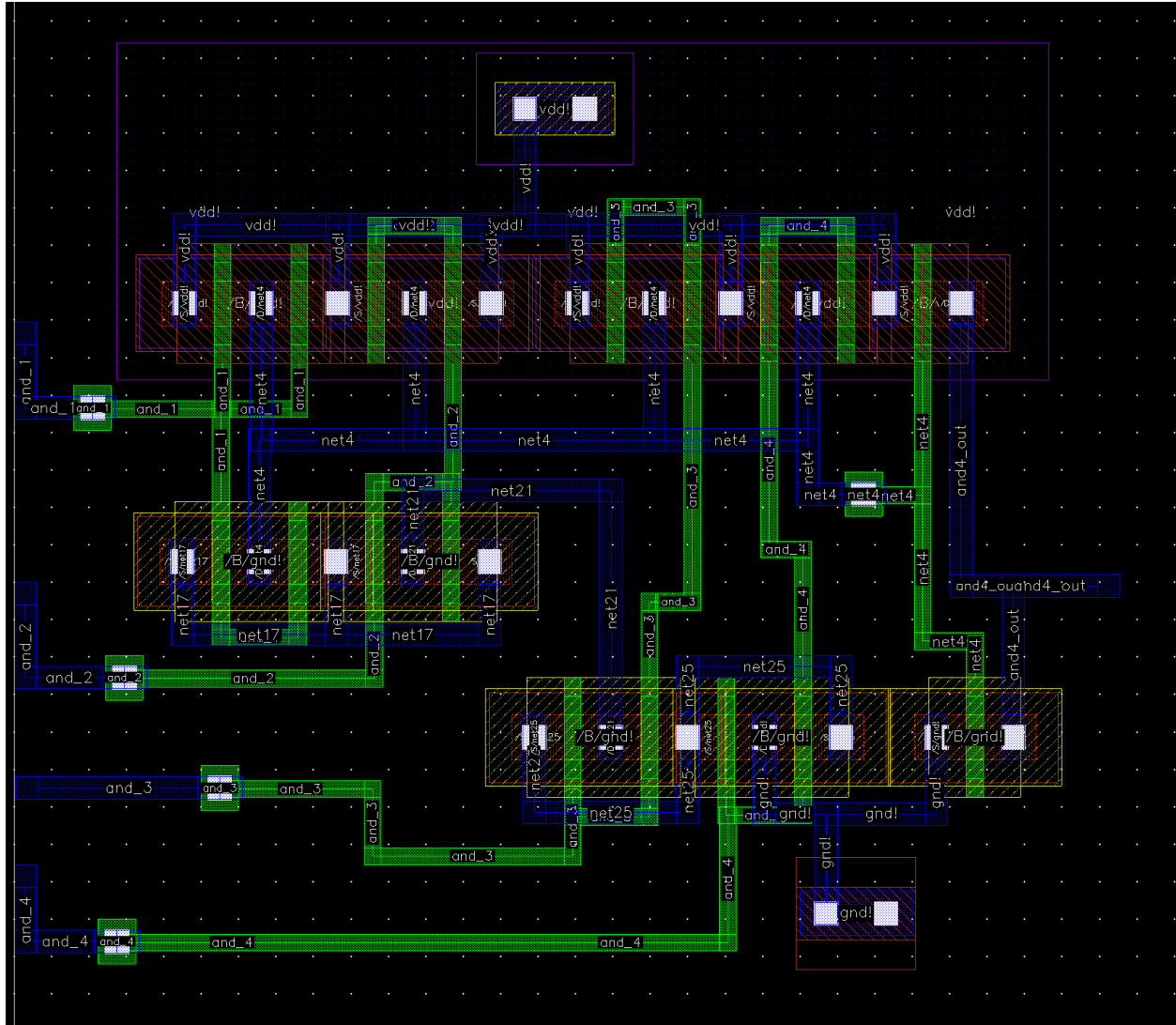


Figure 22: AND 4-input Layout

Figure 22 shows the 4 input AND gate layout.

```

0:      1 / 1      22      12      2
INFO:    Status: Finishing as of 2025-11-23 22:33:58 Elapsed: 00:00:04 CPU time[s]: 4
INFO:    Resource usage by worker at exit:
          CPU      Elapsed CPUs      CPU      Peak      Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0:      2      4      1      51.67      22      12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 4 Max peak memory[MB]: 22
Worker CPU utilization summary: 2 of 4 available CPU seconds (50.00%) with 1 average available CPU(s)
INFO:    Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/and4.sum
INFO:    All selected rules completed
          Total CPU Time      : 2 (s)
          Total Real Time     : 5 (s)
          Total Original Geometry : 264 (344)
          Total DRC RuleChecks  : 562
          Total DRC Results    : 0 (0)
          ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/and4.drc_errors.ascii
INFO:    Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:33:58

```

Figure 23: AND 4-input DRC Test

Figure 23 shows the DRC Test for the 4 input AND layout and it was able to run with 0 errors.

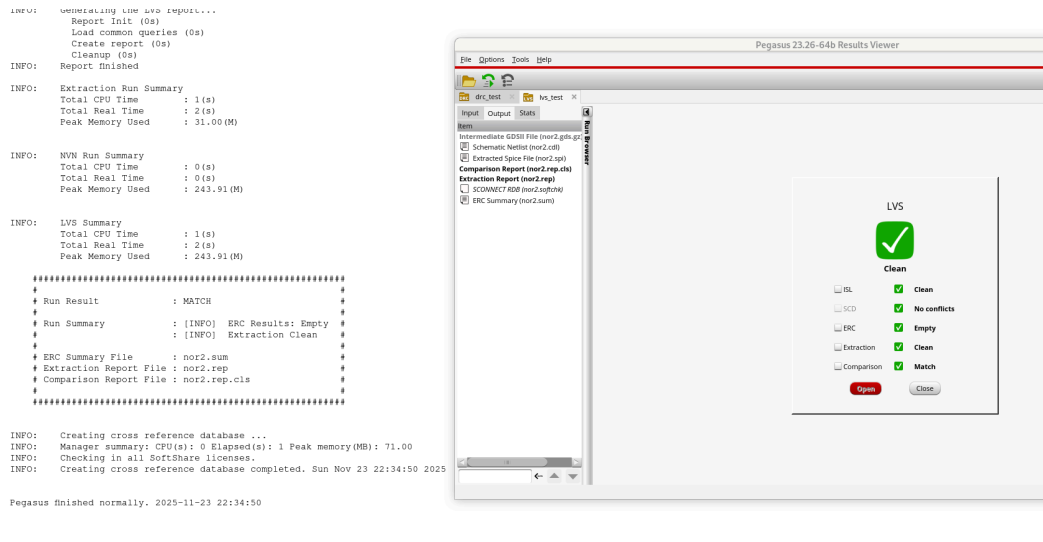


Figure 24: AND 4-input LVS Test

Figure 24 shows the LVS test for the 4 input AND layout and it was able to pass.

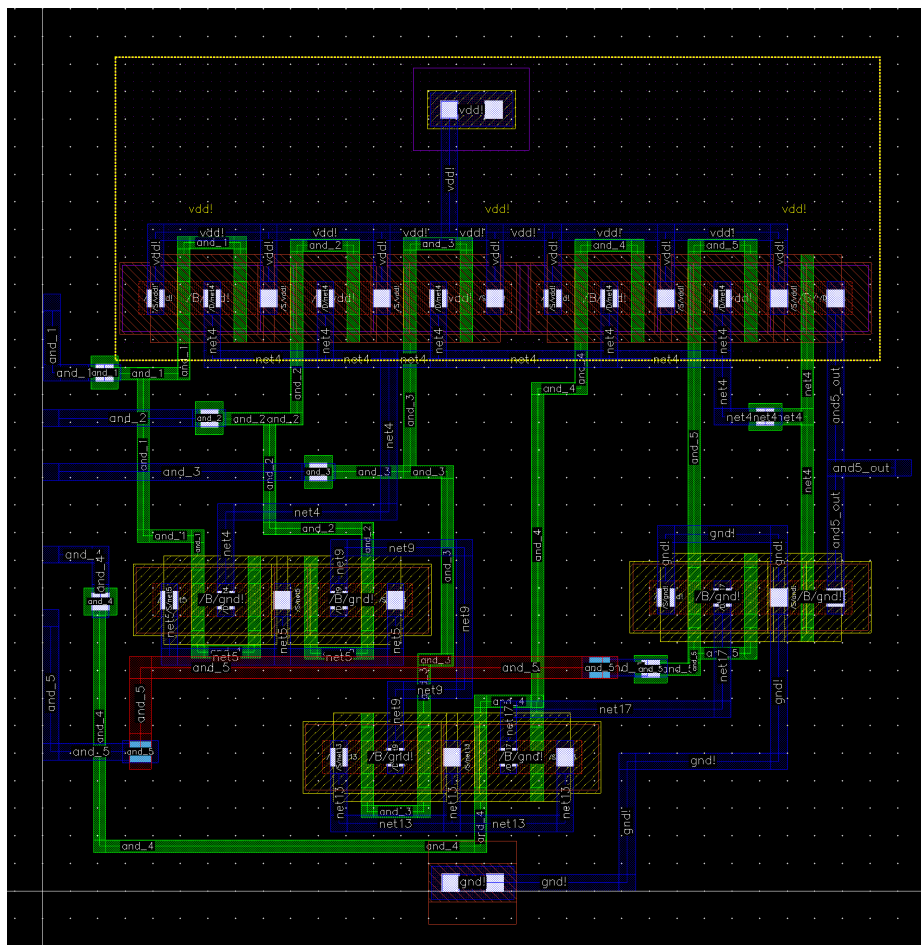


Figure 25: AND 5-input Layout

Figure 25 shows the layout for the 5 input AND gate.

```

INFO:      Status: Runset 100% complete as of 2025-11-23 22:36:20   Elapsed: 00:00:02
Worker: Active CPUs   Total Mem(MB)  Active Mem(MB)  CPU time(s)
      0:      1 /      1           22           12           1
INFO:      Status: Finishing as of 2025-11-23 22:36:20   Elapsed: 00:00:02   CPU time[s]: 2
INFO:      Resource usage by worker at exit:
      CPU      Elapsed  CPUs      CPU      Peak      Peak engine
Worker: time[s] time[s]  Allocated usage[%] Memory[MB] Memory[MB]
      0:      1      2           1      49.82      22      12
Worker resource usage summary: Total CPU[s]: 1 Run duration[s]: 2 Max peak memory[MB]: 22
Worker CPU utilization summary: 1 of 2 available CPU seconds (50.00%) with 1 average available CPU(s)
INFO:      Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/and5.sum
INFO:      All selected rules completed
      Total CPU Time           : 1(s)
      Total Real Time          : 3(s)
      Total Original Geometry  : 304(423)
      Total DRC RuleChecks     : 562
      Total DRC Results        : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/and5.drc_errors.ascii
INFO:      Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:36:20

```

Figure 26: AND 5-input DRC Test

Figure 26 shows the DRC Test for the 5 input AND layout and it was able to run with 0 errors.

```

INFO:      Generating the LVS report...
      Report Init (1s)
      Load common queries (0s)
      Create report (0s)
      Cleanup (0s)
INFO:      Report finished
INFO:      Extraction Run Summary
      Total CPU Time           : 1(s)
      Total Real Time          : 2(s)
      Peak Memory Used         : 31.00 (M)
INFO:      NVN Run Summary
      Total CPU Time           : 0(s)
      Total Real Time          : 1(s)
      Peak Memory Used         : 243.91 (M)
INFO:      LVS Summary
      Total CPU Time           : 1(s)
      Total Real Time          : 3(s)
      Peak Memory Used         : 243.91 (M)

#####
#
# Run Result           : MATCH
#
# Run Summary          : [INFO] ERC Results: Empty
#                      : [INFO] Extraction Clean
#
# ERC Summary File     : and5.sum
# Extraction Report File : and5.rep
# Comparison Report File : and5.rep.cls
#
#####

INFO:      Creating cross reference database ...
INFO:      Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 69.55
INFO:      Checking in all SoftShare licenses.
INFO:      Creating cross reference database completed. Sun Nov 23 22:37:37 2025

Pegasus finished normally. 2025-11-23 22:37:37

```

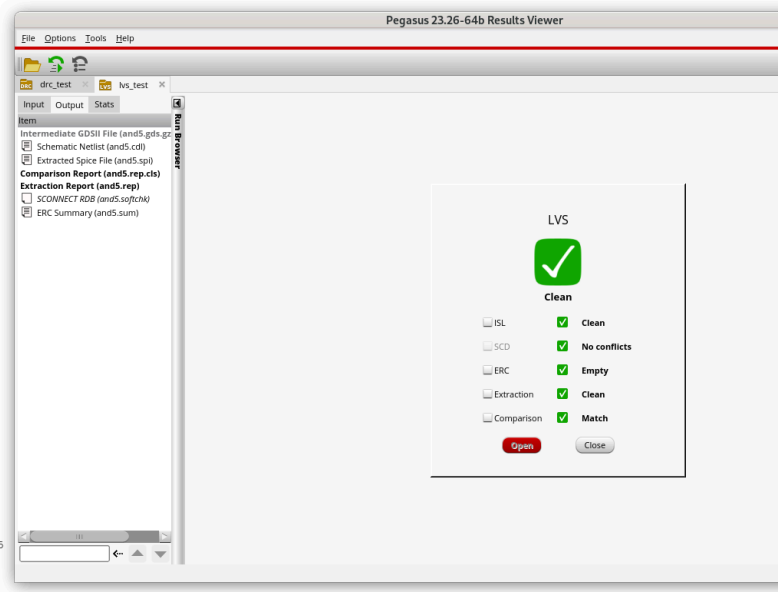


Figure 27: AND 5-input LVS Test

Figure 27 shows the LVS test for the 5 input AND layout and it was able to pass.

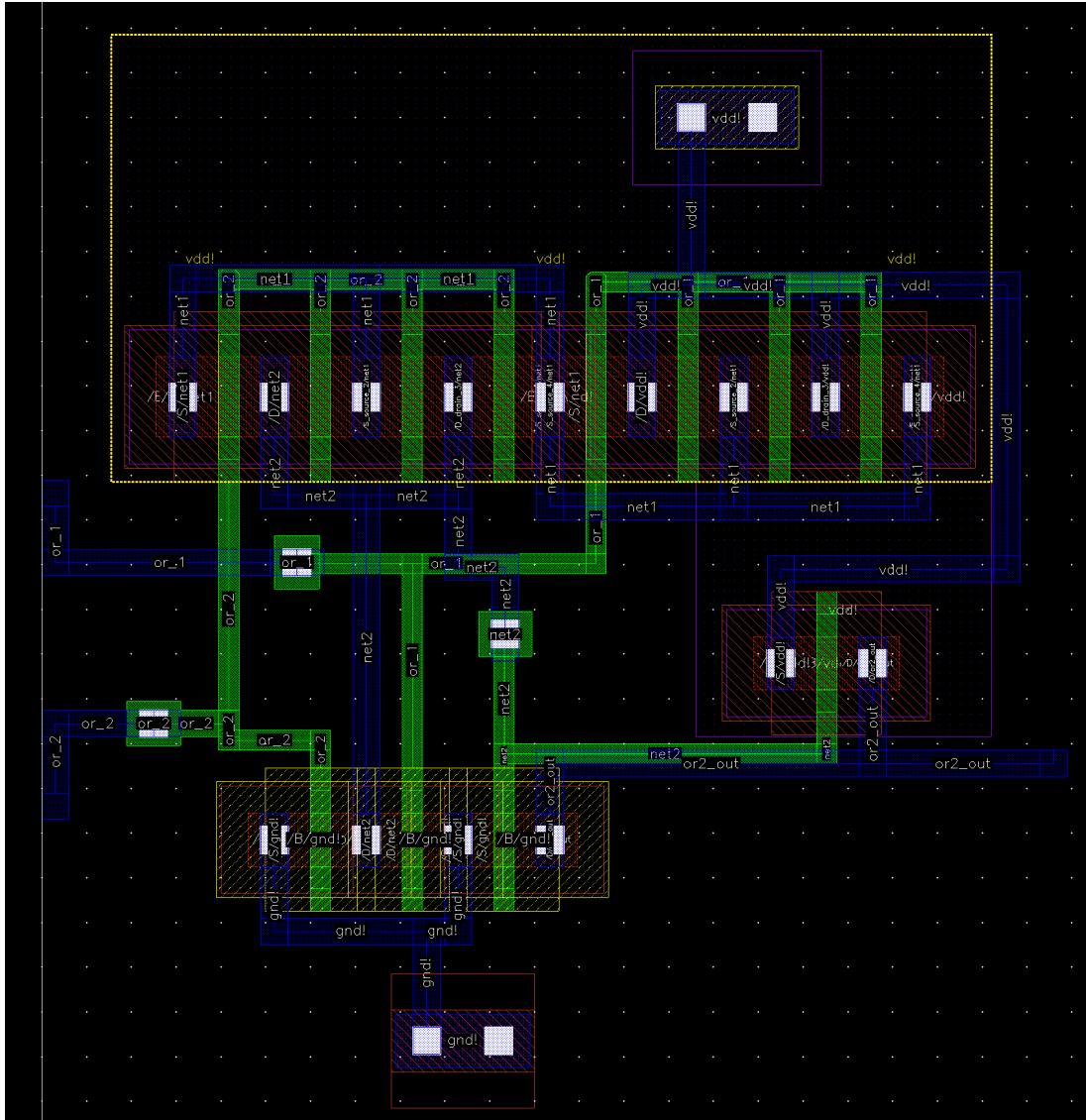


Figure 28: OR 2-input Layout

Figure 28 shows the 2 input OR gate layout.

```

0: 1 / 1 22 12 1
INFO: Status: Finishing as of 2025-11-23 22:38:24 Elapsed: 00:00:02 CPU time[s]: 2
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 1 2 1 47.86 22 12
Worker resource usage summary: Total CPU[s]: 1 Run duration[s]: 2 Max peak memory[MB]: 22
Worker CPU utilization summary: 1 of 2 available CPU seconds (50.00%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/or2.sum
INFO: All selected rules completed
      Total CPU Time : 1(s)
      Total Real Time : 3(s)
      Total Original Geometry : 227(227)
      Total DRC RuleChecks : 562
      Total DRC Results : 0(0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/or2.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:38:24

```

Figure 29: OR 2-input DRC Test

Figure 29 shows the DRC Test for the 2 input OR layout and it was able to run with 0 errors.

```

report_lutil (1s)
Load common queries (0s)
Create report (0s)
Cleanup (0s)
INFO: Report finished

INFO: Extraction Run Summary
Total CPU Time : 1(s)
Total Real Time : 2(s)
Peak Memory Used : 31.00 (M)

INFO: NVN Run Summary
Total CPU Time : 0(s)
Total Real Time : 2(s)
Peak Memory Used : 241.97 (M)

INFO: LVS Summary
Total CPU Time : 1(s)
Total Real Time : 4(s)
Peak Memory Used : 241.97 (M)

#####
# Run Result : MATCH
# Run Summary : ([INFO] ERC Results: Empty
# : ([INFO] Extraction Clean
#
# ERC Summary File : or2.sum
# Extraction Report File : or2.rep
# Comparison Report File : or2.rep.cls
#
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory (MB): 68.86
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:39:04 2025

Pegasus finished normally. 2025-11-23 22:39:04

```

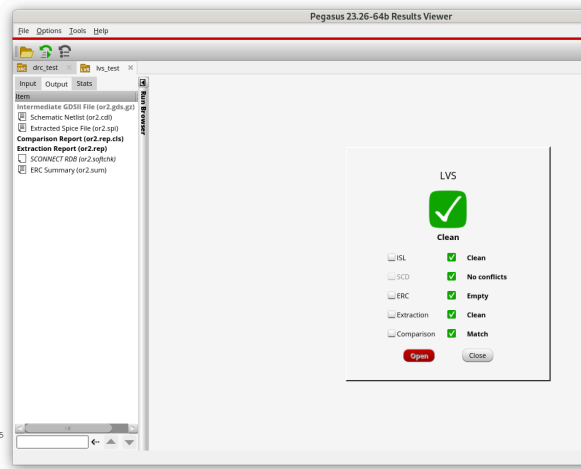


Figure 30: OR 2-input LVS Test

Figure 30 shows the LVS test for the 2 input OR layout and it was able to pass.

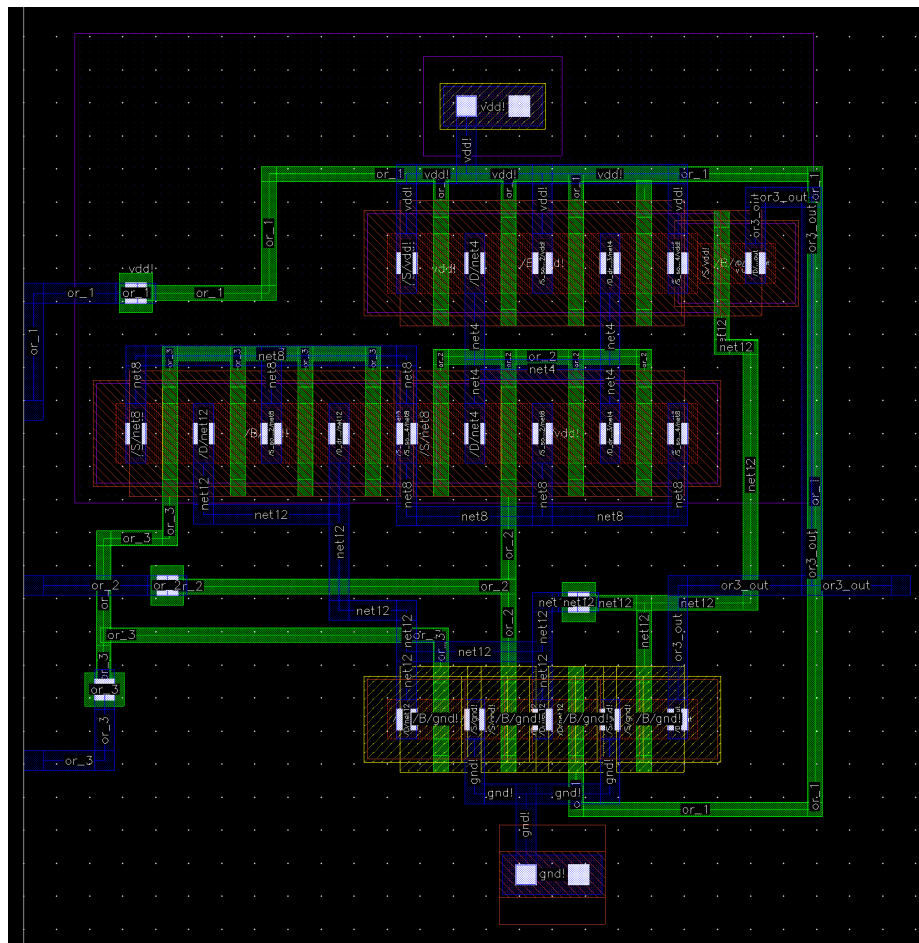


Figure 31: OR 3-input Layout

Figure 31 shows the 3 input OR gate layout.

```

0: 1 / 1 22 12 1
INFO: Status: Finishing as of 2025-11-23 22:39:45 Elapsed: 00:00:02 CPU time[s]: 2
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
      0: 1 2 1 48.04 22 12
Worker resource usage summary: Total CPU[s]: 1 Run duration[s]: 2 Max peak memory[MB]: 22
Worker CPU utilization summary: 1 of 2 available CPU seconds (50.00%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/or3.sum
INFO: All selected rules completed
      Total CPU Time : 1(s)
      Total Real Time : 3(s)
      Total Original Geometry : 294(300)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/or3.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:39:45

```

Figure 32: OR 3-input DRC Test

Figure 32 shows the DRC Test for the 3 input OR layout and it was able to run with 0 errors.

```

INFO: Report finished
INFO: Extraction Run Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 31.00 (M)

INFO: NVN Run Summary
      Total CPU Time : 0(s)
      Total Real Time : 0(s)
      Peak Memory Used : 243.97 (M)

INFO: LVS Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 243.97 (M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
# ERC Summary File : or3.sum #
# Extraction Report File : or3.rep #
# Comparison Report File : or3.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 1 Peak memory (MB): 70.75
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:40:18 2025

Pegasus finished normally. 2025-11-23 22:40:18

```

The screenshot shows the LVS (Logic Verification System) test results. A large green checkmark indicates a 'Clean' status. Below it, a list of checks is shown with their results: ISL (Clean), SCD (No conflicts), ERC (Empty), Extraction (Clean), and Comparison (Match). The 'Open' button is highlighted in red.

Figure 33: OR 3-input LVS Test

Figure 33 shows the LVS test for the 3 input OR layout and it was able to pass.

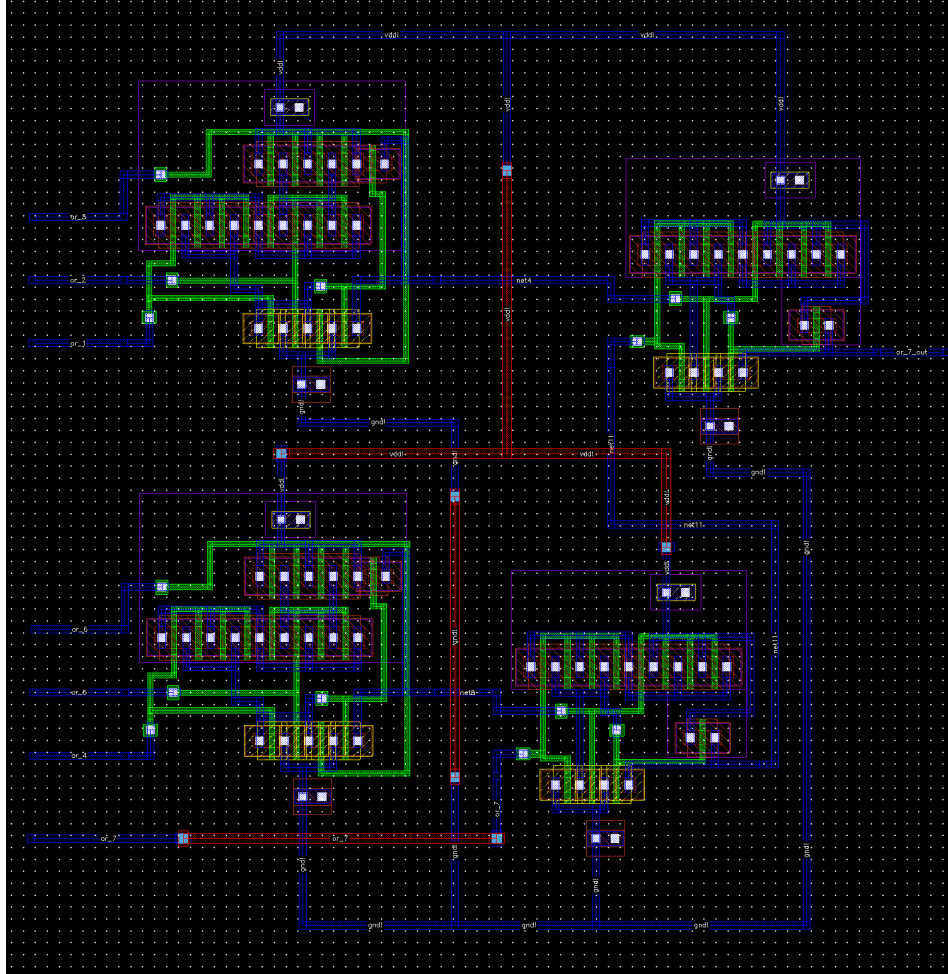


Figure 34: OR 7-input Layout

Figure 34 shows the 7 input OR gate layout.

```

0: 1 / 1 26 12 2
INFO: Status: Finishing as of 2025-11-23 22:41:01 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 2 3 1 63.05 26 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/or7.sum
INFO: All selected rules completed
Total CPU Time : 2(s)
Total Real Time : 4(s)
Total Original Geometry : 458(1126)
Total DRC RuleChecks : 562
Total DRC Results : 0 (0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/or7.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:41:01

```

Figure 35: OR 7-input DRC Test

Figure 35 shows the DRC Test for the 7 input OR layout and it was able to run with 0 errors.

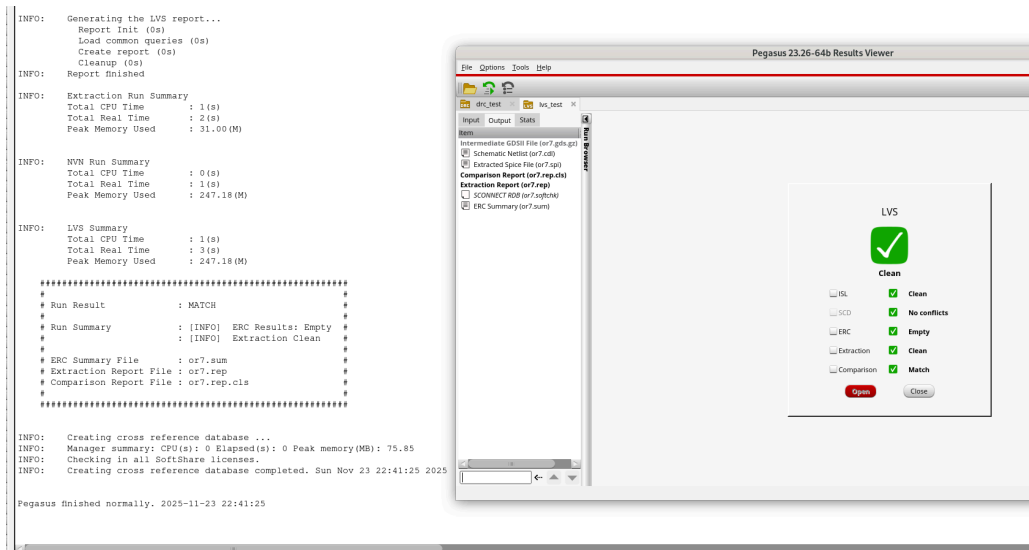


Figure 36: OR 7-input LVS Test

Figure 36 shows the LVS test for the 7 input OR layout and it was able to pass.

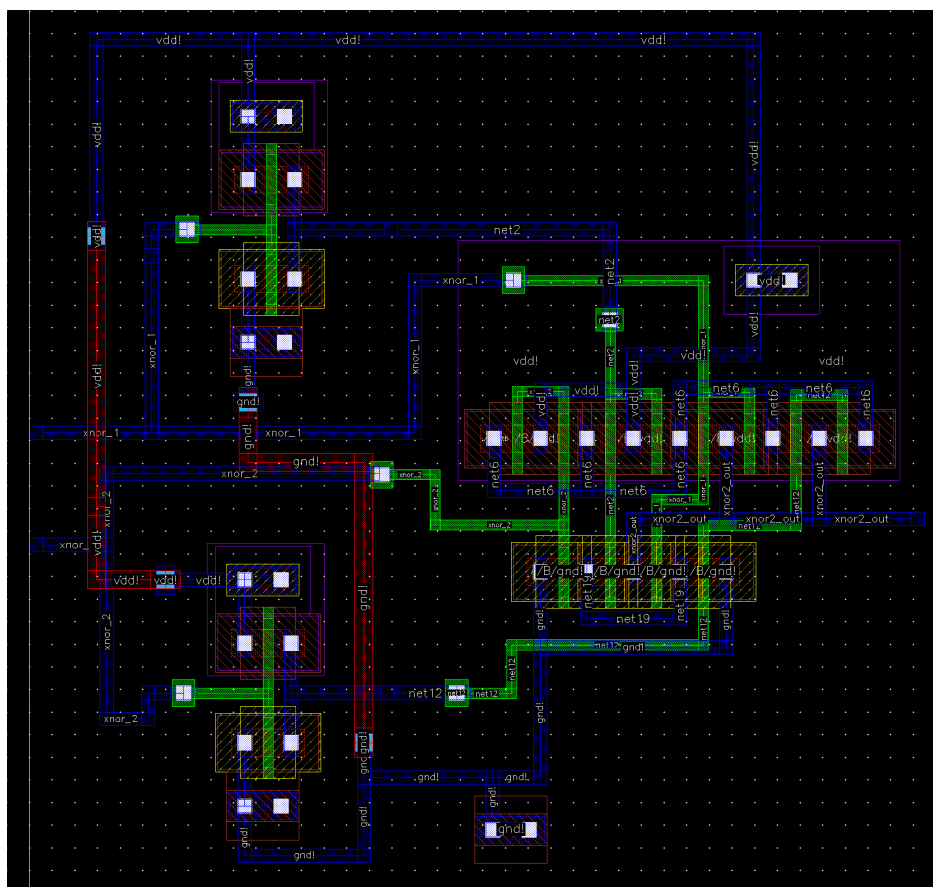


Figure 37: XNOR 2-input Layout

Figure 37 shows the 2 input xnor gate layout.


```

Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 25
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/xnor2.sum
INFO: All selected rules completed
Total CPU Time : 2(s)
Total Real Time : 4(s)
Total Original Geometry : 291(394)
Total DRC RuleChecks : 562
Total DRC Results : 0(0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/xnor2.drc_errors.asci
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:42:57

```

Figure 38: XNOR 2-input DRC Test

Figure 38 shows the DRC Test for the 2 input XNOR gate layout and it was able to run with 0 errors.

```

INFO: Generating the LVS report...
Report Init (0s)
Load common queries (0s)
Create report (1s)
Cleanup (0s)
INFO: Report finished

INFO: Extraction Run Summary
Total CPU Time : 1(s)
Total Real Time : 2(s)
Peak Memory Used : 31.00(M)

INFO: NVN Run Summary
Total CPU Time : 1(s)
Total Real Time : 1(s)
Peak Memory Used : 249.73(M)

INFO: LVS Summary
Total CPU Time : 2(s)
Total Real Time : 3(s)
Peak Memory Used : 249.73(M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
# ERC Summary File : xnor2.sum #
# Extraction Report File : xnor2.rep #
# Comparison Report File : xnor2.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 74.52
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:43:33 2025

Pegasus finished normally. 2025-11-23 22:43:33

```

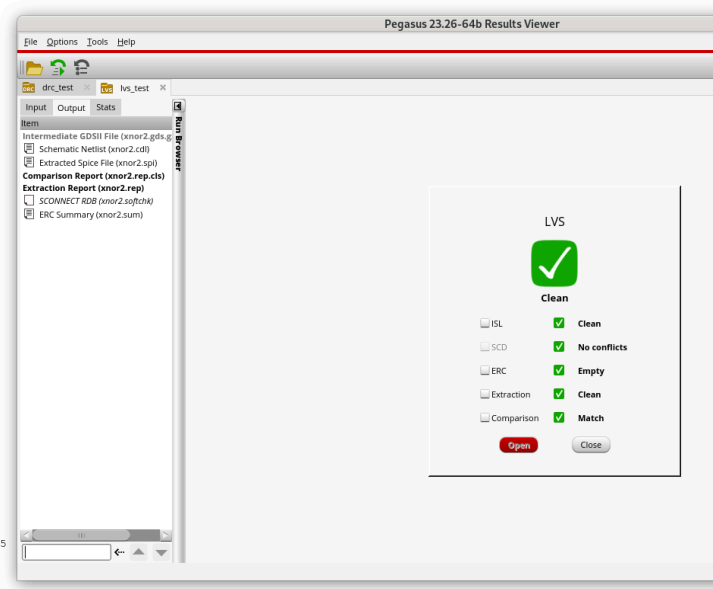


Figure 39: XNOR 2-input LVS Test

Figure 39 shows the LVS test for the 2 input XNOR layout and it was able to pass.

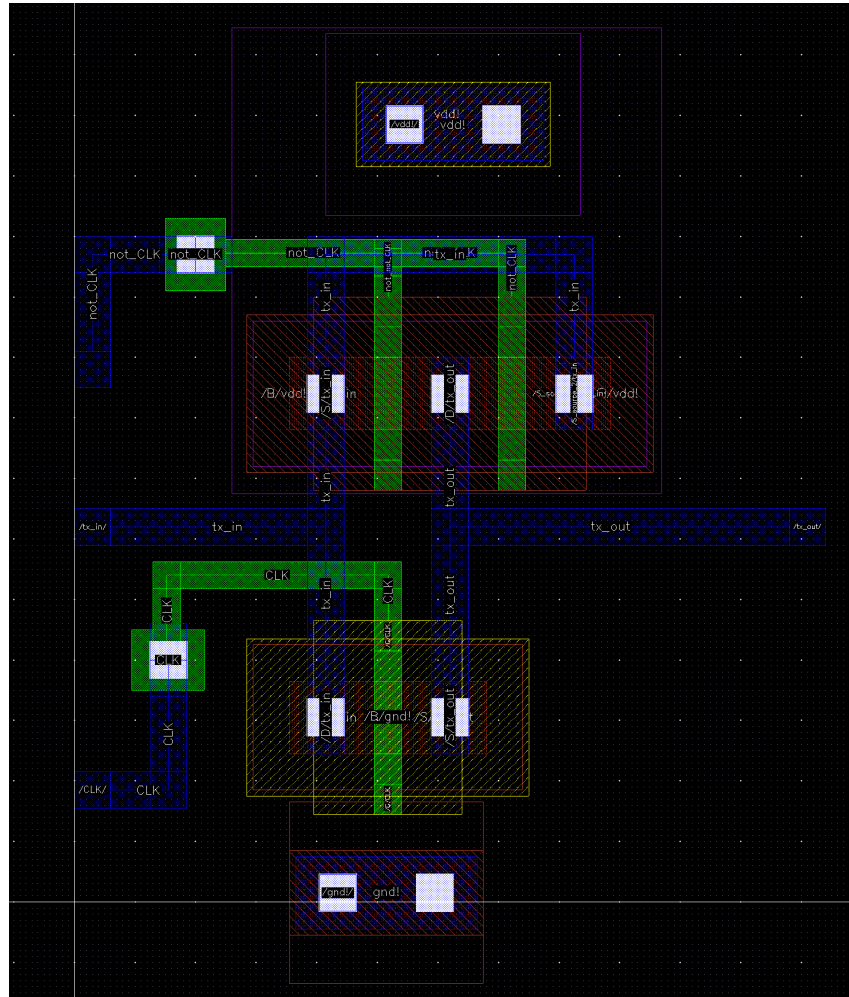


Figure 40: Transmission Gate Layout

Figure 40 shows the transmission gate layout.

```

0:      1 /      1      22      12      1
INFO:  Status: Finishing as of 2025-11-23 22:45:30 Elapsed: 00:00:02 CPU time[s]: 2
INFO:  Resource usage by worker at exit:
      CPU Elapsed CPUs      CPU      Peak      Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
      0:      1      2      1      49.59      22      12
Worker resource usage summary: Total CPU[s]: 1 Run duration[s]: 2 Max peak memory[MB]: 22
Worker CPU utilization summary: 1 of 2 available CPU seconds (50.00%) with 1 average available CPU(s)
INFO:  Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/tx.sum
INFO:  All selected rules completed
      Total CPU Time      : 1(s)
      Total Real Time      : 3(s)
      Total Original Geometry      : 83(83)
      Total DRC RuleChecks      : 562
      Total DRC Results      : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/tx.drc_errors.ascii
INFO:  Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:45:30

```

Figure 41: Transmission Gate DRC Test

Figure 41 shows the DRC Test for the transmission gate layout and it was able to run with 0 errors.

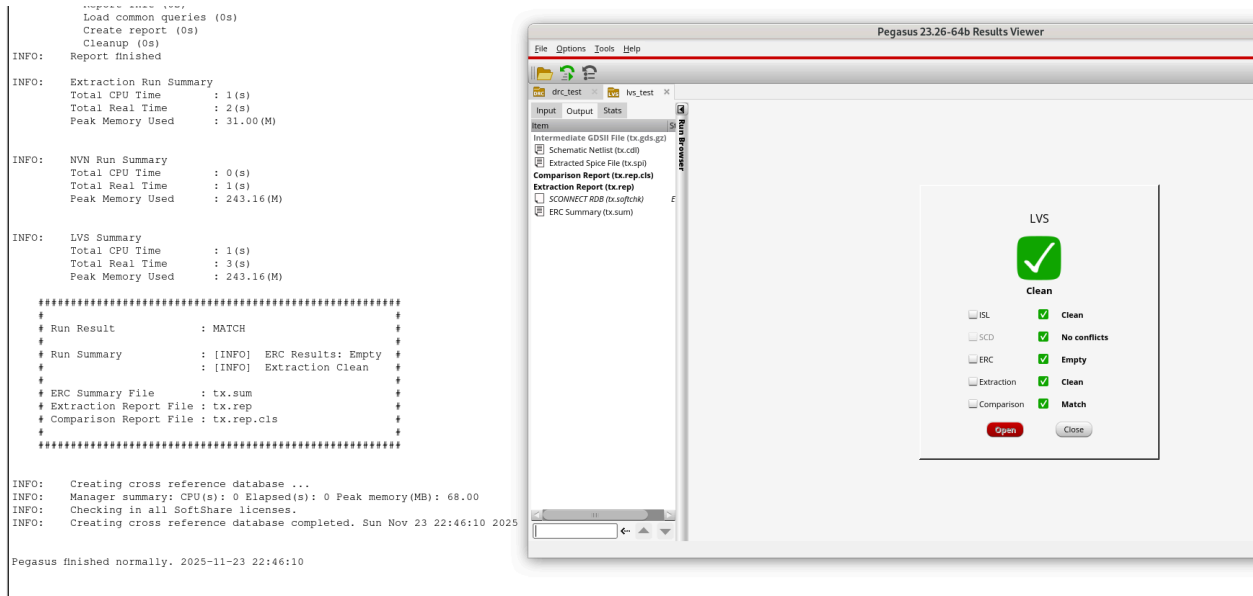


Figure 42: Transmission Gate LVS Test

Figure 42 shows the LVS test for the transmission gate layout and it was able to pass.

4.2 Lower Level Modules

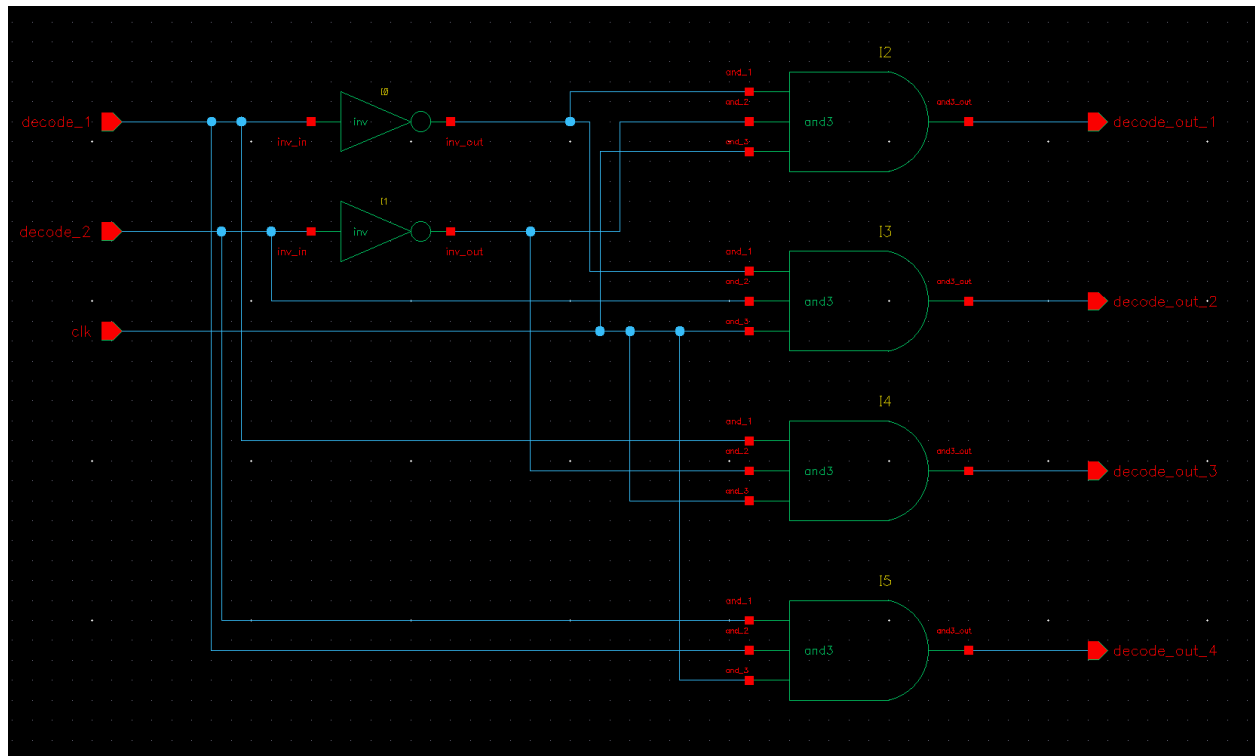


Figure 43: Decoder 2 to 4 Schematic

Figure 43 shows the schematic for the 2 to 4 decoder.

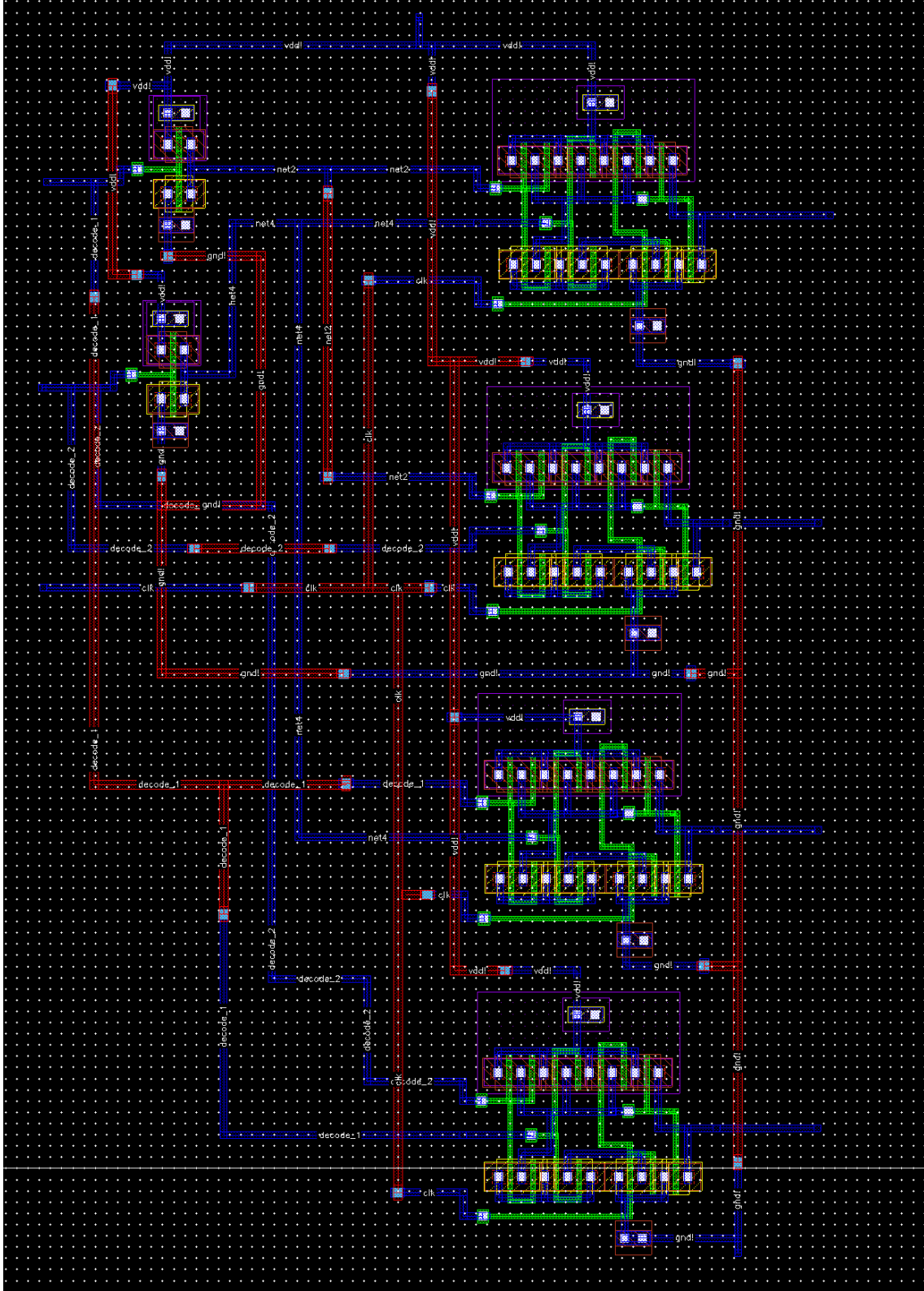


Figure 44: Decoder 2 to 4 Layout

Figure 44 shows the layout for the 2 to 4 decoder.

```

0: 1 / 1 25 12 2
INFO: Status: Finishing as of 2025-11-23 22:47:25 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 2 3 1 63.45 25 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 25
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/decoder2to4.sum
INFO: All selected rules completed
      Total CPU Time : 2(s)
      Total Real Time : 4(s)
      Total Original Geometry : 397(1367)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/decoder2to4.drc_errors.ascii
INFO: Checking in all SoftShare licenses.
Pegasus finished normally. 2025-11-23 22:47:25

```

Figure 45: Decoder 2 to 4 DRC Test

Figure 45 shows the DRC Test for the 2 to 4 decoder layout and it was able to run with 0 errors.

```

INFO: Report Init (1s)
      Load common queries (0s)
      Create report (0s)
      Cleanup (0s)
INFO: Report finished
INFO: Extraction Run Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 31.00 (M)
INFO: NVN Run Summary
      Total CPU Time : 0(s)
      Total Real Time : 1(s)
      Peak Memory Used : 253.57 (M)
INFO: LVS Summary
      Total CPU Time : 1(s)
      Total Real Time : 3(s)
      Peak Memory Used : 253.57 (M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
# ERC Summary File : decoder2to4.sum #
# Extraction Report File : decoder2to4.rep #
# Comparison Report File : decoder2to4.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 77.58
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:47:59 2025
Pegasus finished normally. 2025-11-23 22:47:59

```

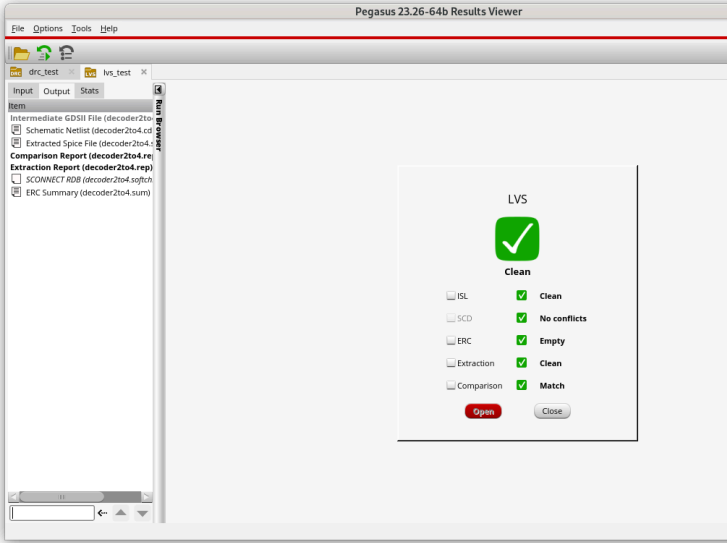


Figure 46: Decoder 2 to 4 LVS Test

Figure 46 shows the LVS test for the 2 to 4 decoder layout and it was able to pass.

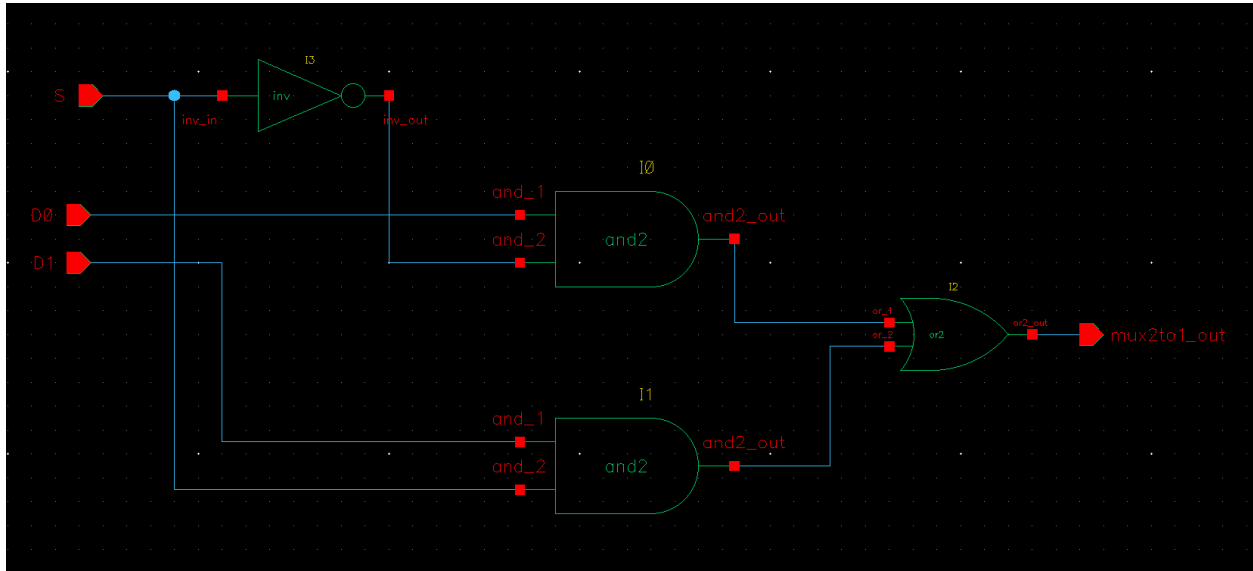


Figure 47: Multiplexer 2 to 1 Schematic

Figure 47 shows the schematic for the 2 to 1 multiplexer.

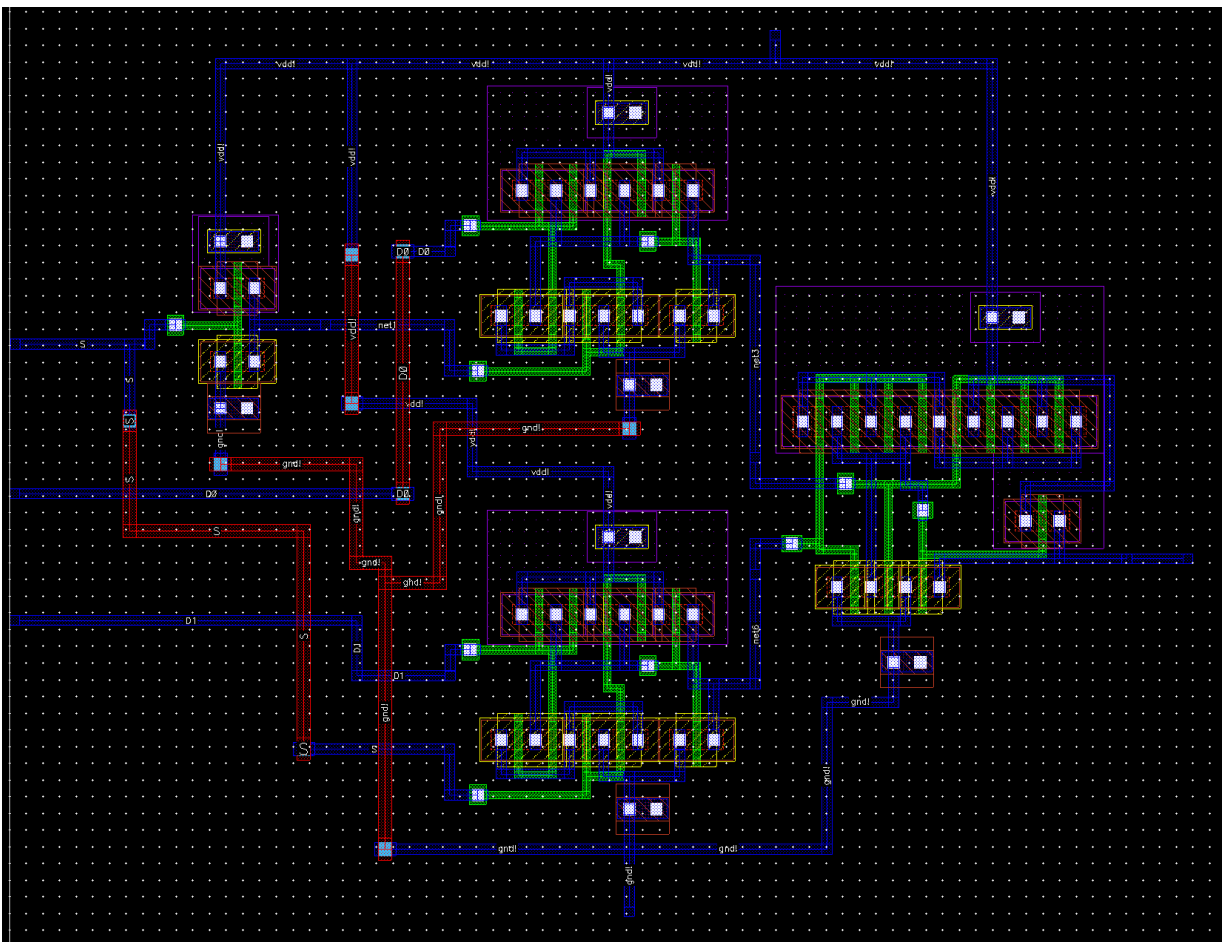


Figure 48: Multiplexer 2 to 1 Layout

Figure 48 shows the layout for the 2 to 1 multiplexer.

```

INFO:      Status: Finishing as of 2025-11-23 22:49:26 Elapsed: 00:00:03 CPU time[s]: 3
INFO:      Resource usage by worker at exit:
          CPU      Elapsed CPUs      CPU      Peak      Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
          0:      2      3          1      61.61      25      12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 25
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO:      Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/mux2to1.sum
INFO:      All selected rules completed
          Total CPU Time      : 2 (s)
          Total Real Time     : 4 (s)
          Total Original Geometry : 483 (763)
          Total DRC RuleChecks : 562
          Total DRC Results    : 0 (0)
          ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/mux2to1.drc_errors.ascii
INFO:      Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:49:26

```

Figure 49: Multiplexer 2 to 1 DRC Test

Figure 49 shows the DRC Test for the 2 to 1 multiplexer layout and it was able to run with 0 errors.

```

INFO:      Load common queries (0s)
INFO:      Create report (0s)
INFO:      Cleanup (0s)
INFO:      Report finished

INFO:      Extraction Run Summary
          Total CPU Time      : 1 (s)
          Total Real Time     : 2 (s)
          Peak Memory Used    : 31.00 (M)

INFO:      NVN Run Summary
          Total CPU Time      : 0 (s)
          Total Real Time     : 1 (s)
          Peak Memory Used    : 250.62 (M)

INFO:      LVS Summary
          Total CPU Time      : 1 (s)
          Total Real Time     : 3 (s)
          Peak Memory Used    : 250.62 (M)

#####
# Run Result      : MATCH
#
# Run Summary     : [INFO] ERC Results: Empty
#                 : [INFO] Extraction Clean
#
# ERC Summary File : mux2to1.sum
# Extraction Report File : mux2to1.rep
# Comparison Report File : mux2to1.rep.cls
#
#####

INFO:      Creating cross reference database ...
INFO:      Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 79.14
INFO:      Checking in all SoftShare licenses.
INFO:      Creating cross reference database completed. Sun Nov 23 22:50:05 2025

Pegasus finished normally. 2025-11-23 22:50:05

```

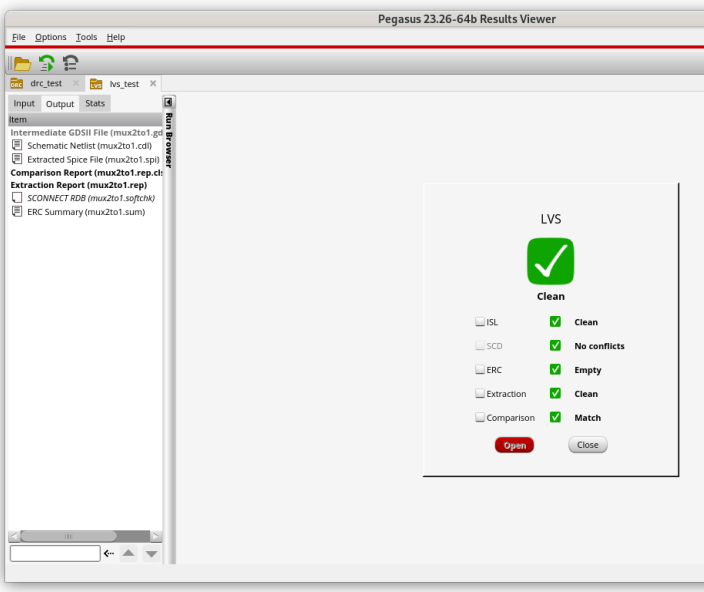


Figure 50: Multiplexer 2 to 1 LVS Test

Figure 50 shows the LVS test for the 2 to 1 multiplexer layout and it was able to pass.

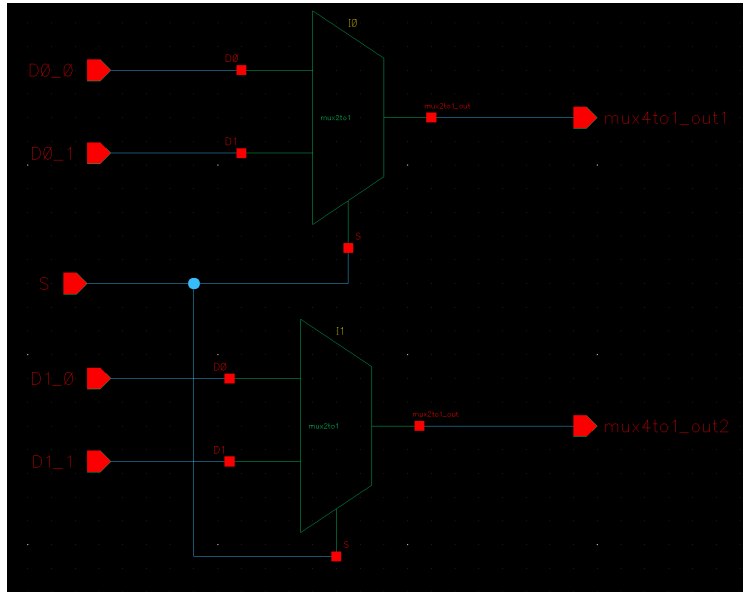


Figure 51: 2- bit Multiplexer 2 to 1 Schematic

Figure 51 shows the schematic for a 2-bit 2 to 1 multiplexer.

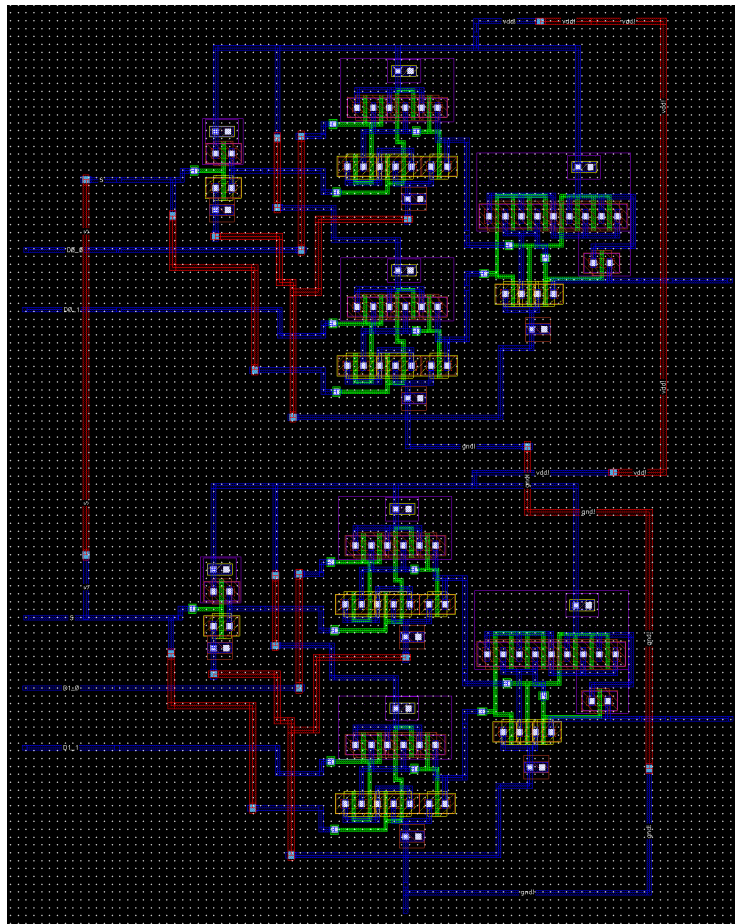


Figure 52: 2-bit Multiplexer 2 to 1 Layout

Figure 52 shows the layout for the 2-bit 2 to 1 multiplexer.


```

0: 1 / 1 25 12 2
INFO: Status: Finishing as of 2025-11-23 22:50:58 Elapsed: 00:00:04 CPU time[s]: 4
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 2 4 1 50.00 25 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 4 Max peak memory[MB]: 25
Worker CPU utilization summary: 2 of 4 available CPU seconds (50.00%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/mux2to1_2bit.sum
INFO: All selected rules completed
      Total CPU Time : 2(s)
      Total Real Time : 5(s)
      Total Original Geometry : 521(1579)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/mux2to1_2bit.drc_errors.asci
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:50:58

```

Figure 53: 2-bit Multiplexer 2 to 1 DRC Test

Figure 53 shows the DRC Test for the 2-bit 2 to 1 multiplexer layout and it was able to run with 0 errors.

```

INFO: Cleanup (0s)
      Report finished
INFO: Extraction Run Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 31.00 (M)
INFO: NVN Run Summary
      Total CPU Time : 0(s)
      Total Real Time : 1(s)
      Peak Memory Used : 254.00 (M)
INFO: LVS Summary
      Total CPU Time : 1(s)
      Total Real Time : 3(s)
      Peak Memory Used : 254.00 (M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
# ERC Summary File : mux2to1_2bit.sum #
# Extraction Report File : mux2to1_2bit.rep #
# Comparison Report File : mux2to1_2bit.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 84.54
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:51:29 2025

Pegasus finished normally. 2025-11-23 22:51:29

```

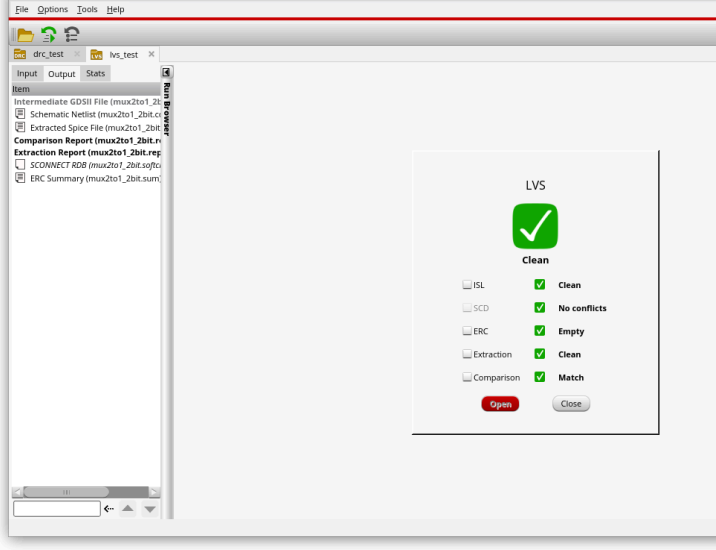


Figure 54: 2-bit Multiplexer 2 to 1 LVS Test

Figure 54 shows the LVS test for the 2-bit 2 to 1 multiplexer layout and it was able to pass.

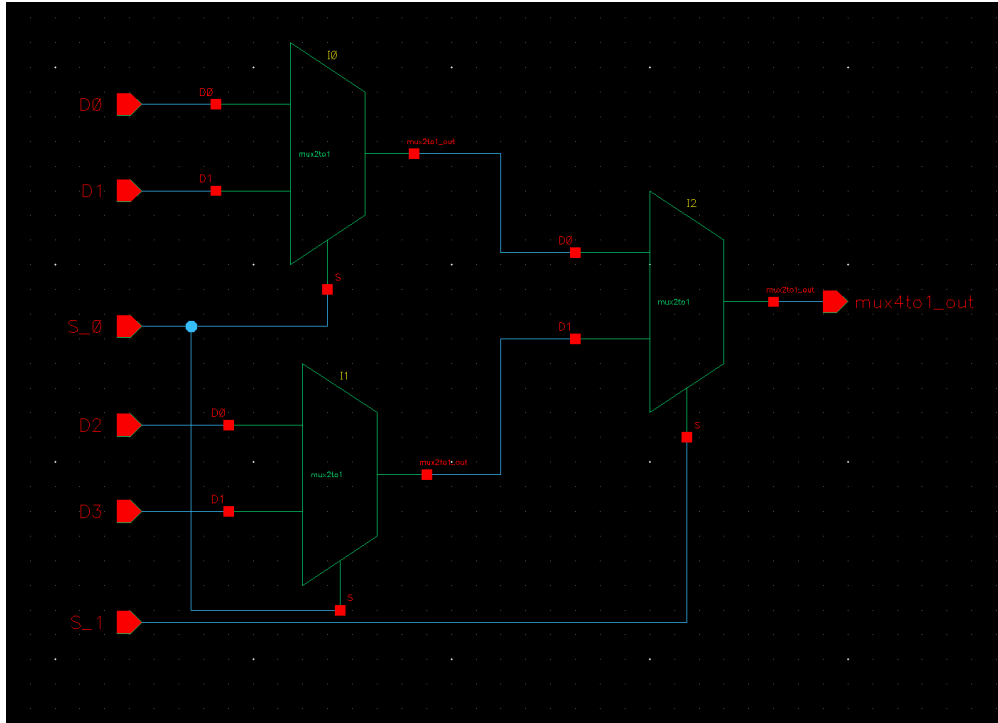


Figure 55: Multiplexer 4 to 1 Schematic

Figure 55 shows the schematic for the 4 to 1 multiplexer.

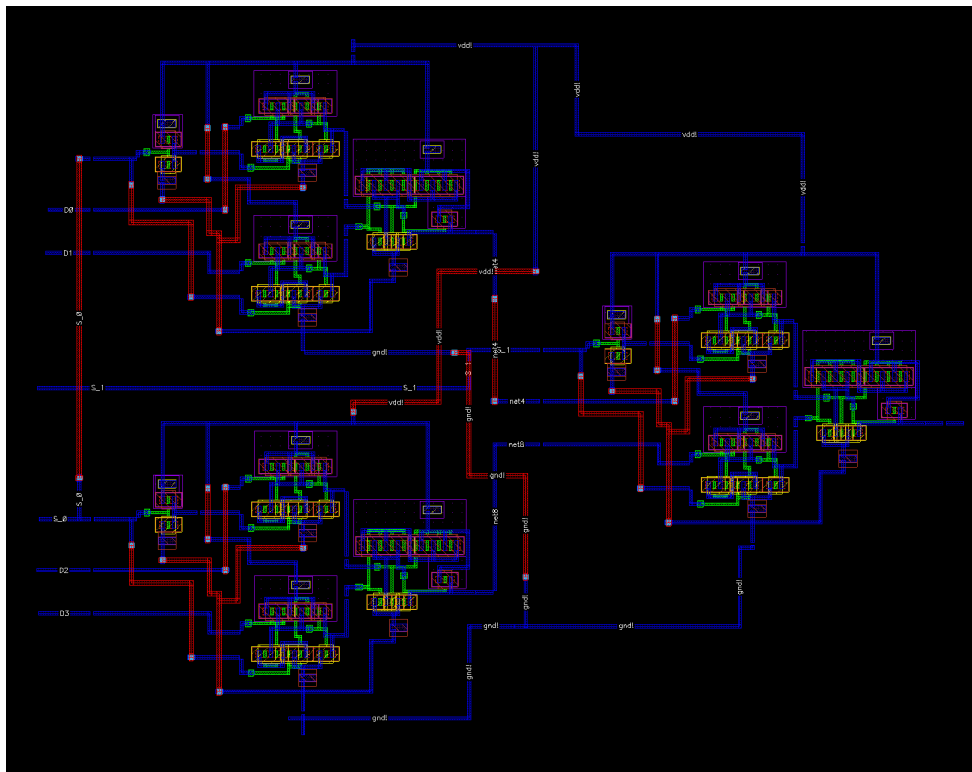


Figure 56: Multiplexer 4 to 1 Layout

Figure 56 shows the layout for the 4 to 1 multiplexer.

```

INFO: Status: Finishing as of 2025-11-23 22:52:32 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
      0: 2 3 1 64.87 25 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 25
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/mux4to1.sum
INFO: All selected rules completed
      Total CPU Time : 2(s)
      Total Real Time : 4(s)
      Total Original Geometry : 538(2365)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/mux4to1.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:52:32

```

Figure 57: Multiplexer 4 to 1 DRC Test

Figure 57 shows the DRC Test for the 4 to 1 multiplexer layout and it was able to run with 0 errors.

```

      Load common queries (0s)
      Create report (0s)
      Cleanup (0s)
INFO: Report finished

INFO: Extraction Run Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 31.00 (M)

INFO: NVN Run Summary
      Total CPU Time : 0(s)
      Total Real Time : 0(s)
      Peak Memory Used : 250.62 (M)

INFO: LVS Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 250.62 (M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
#
# ERC Summary File : mux4to1.sum #
# Extraction Report File : mux4to1.rep #
# Comparison Report File : mux4to1.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 83.11
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:52:56 2025

Pegasus finished normally. 2025-11-23 22:52:57

```

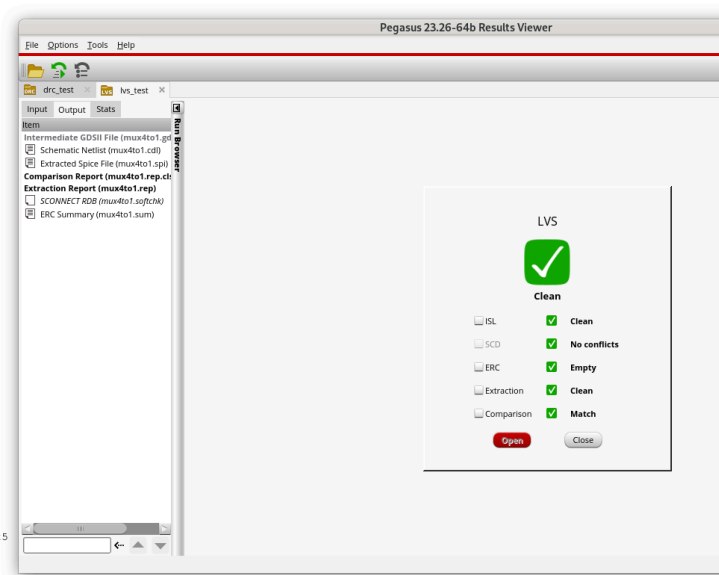


Figure 58: Multiplexer 4 to 1 LVS Test

Figure 58 shows the LVS test for the 4 to 1 multiplexer layout and it was able to pass.

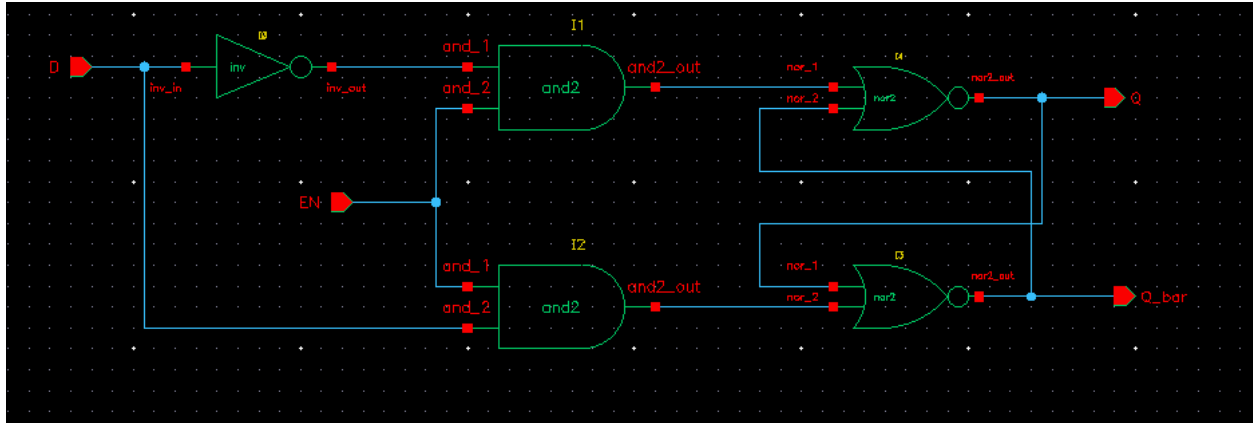


Figure 59: D-Latch Schematic

Figure 59 shows the schematic for a D-Latch.

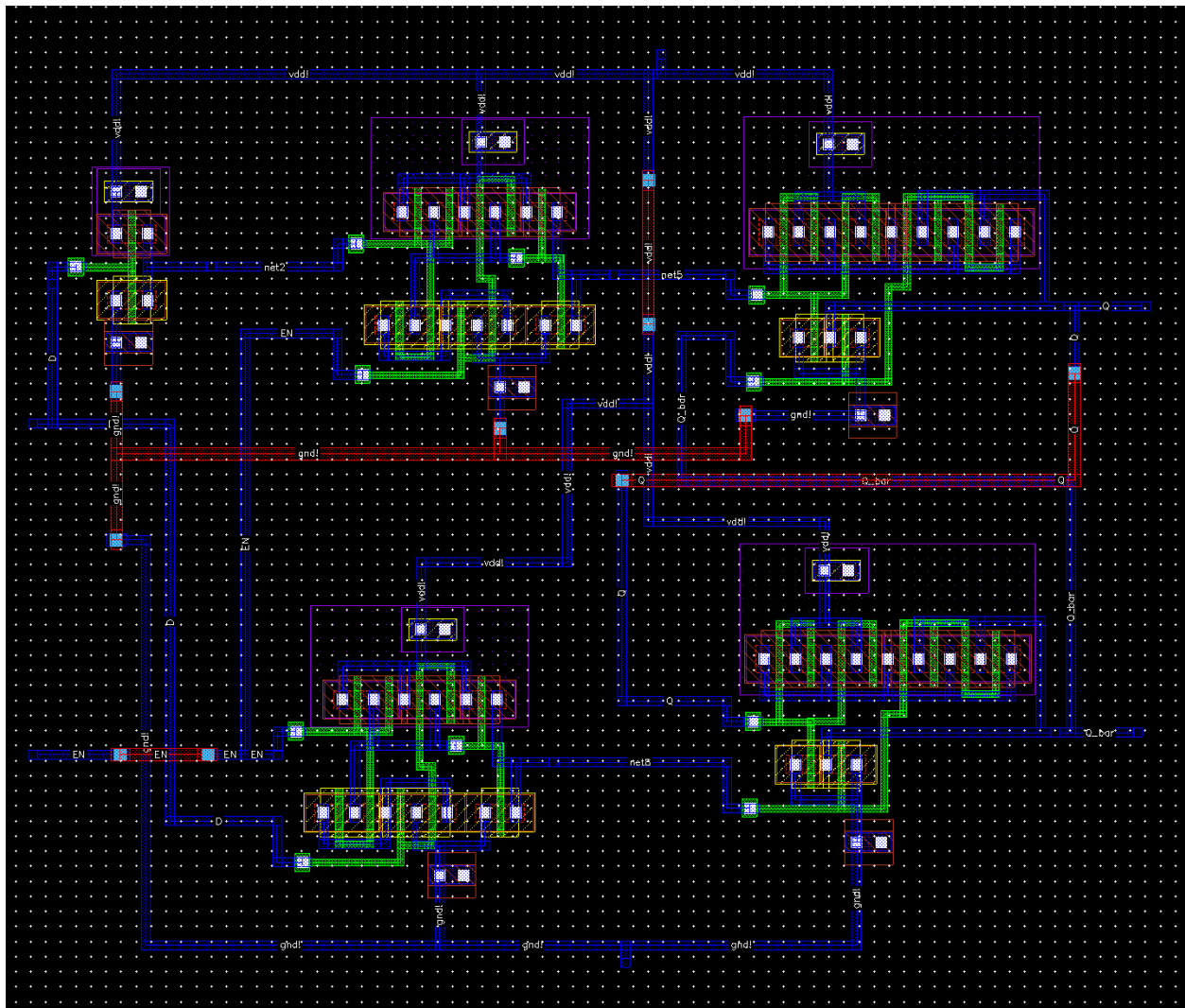


Figure 60: D-Latch Layout

Figure 60 shows the layout for the D-Latch.

```

0: 1 / 1 25 12 2
INFO: Status: Finishing as of 2025-11-23 22:53:56 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 2 3 1 62.67 25 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 25
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/dlatch.sum
INFO: All selected rules completed
      Total CPU Time : 2(s)
      Total Real Time : 4(s)
      Total Original Geometry : 473(926)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/dlatch.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:53:56

```

Figure 61: D-Latch DRC Test

Figure 61 shows the DRC Test for the D-Latch layout and it was able to run with 0 errors.

```

INFO: Cleanup (usr)
INFO: Report finished
INFO: Extraction Run Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 31.00 (M)
INFO: NVN Run Summary
      Total CPU Time : 0(s)
      Total Real Time : 0(s)
      Peak Memory Used : 248.88 (M)
INFO: LVS Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 248.88 (M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
# ERC Summary File : dlatch.sum #
# Extraction Report File : dlatch.rep #
# Comparison Report File : dlatch.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 78.26
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:54:18 2025

Pegasus finished normally. 2025-11-23 22:54:18

```

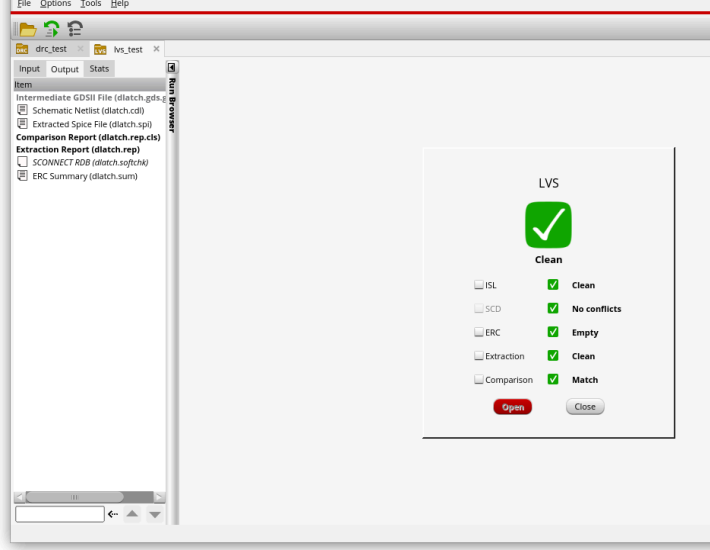


Figure 62: D-Latch LVS Test

Figure 62 shows the LVS test for the D-Latch layout and it was able to pass.

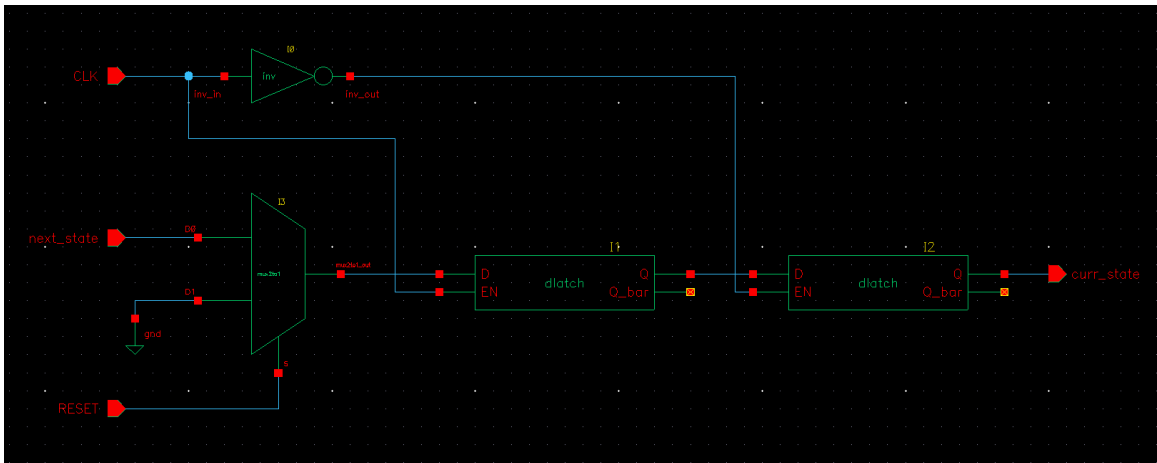


Figure 63: Negative Edge Flip Flop Schematic

Figure 63 shows the schematic for a flip flop that updates at negedge.

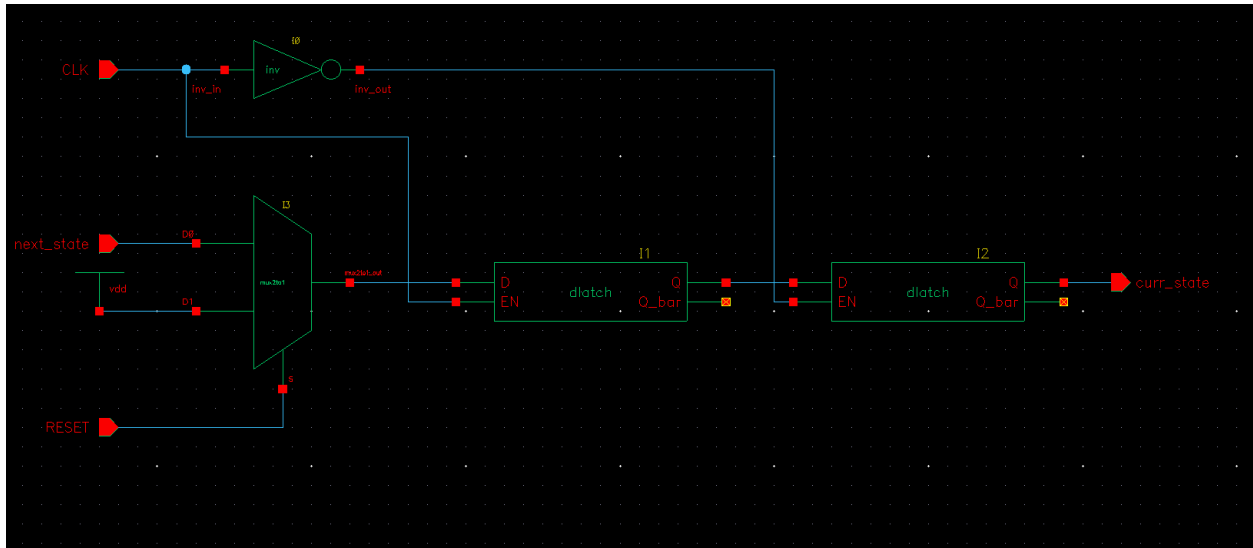


Figure 64: N-Edge Flip Flip with Reset Value 1 Schematic

Figure 64 shows the schematic for a flip flop that updates at negedge. This flip flop has a reset value of 1, which is used for the state zero (IDLE).

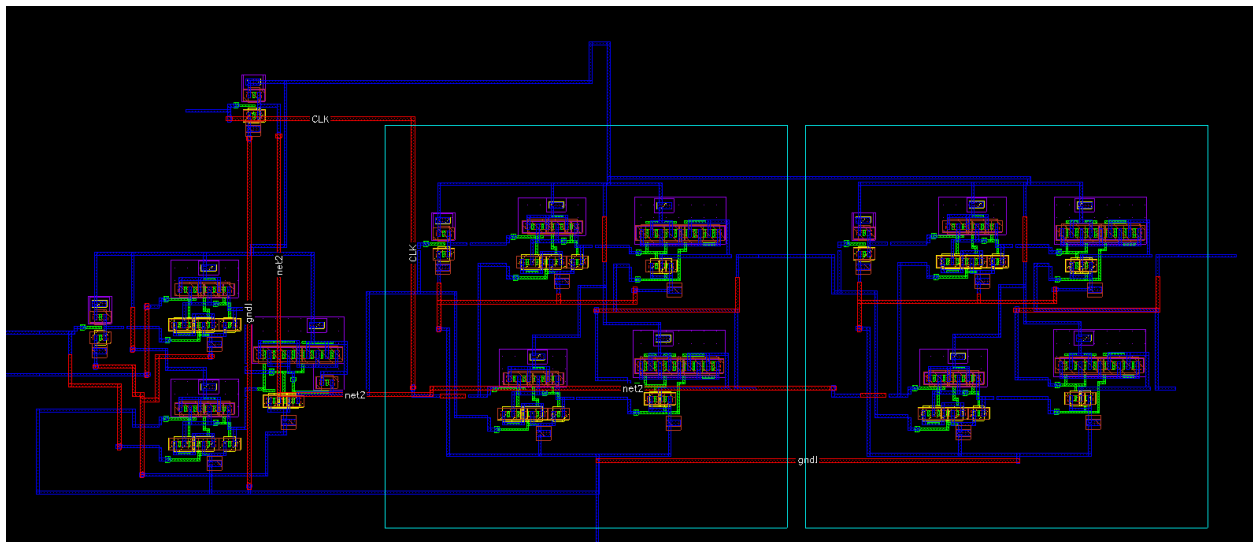


Figure 65: N-Edge Flip Flop Layout

Figure 65 shows the layout for the negative edge flip flop. The layout for a reset value for 1 is the same as this but has the input connected to vdd instead of gnd so it was not shown.

```

0: 1 / 1 26 12 2
INFO: Status: Finishing as of 2025-11-23 22:56:04 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 2 3 1 60.55 26 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/set_curr_state.sum
INFO: All selected rules completed
      Total CPU Time : 2(s)
      Total Real Time : 4(s)
      Total Original Geometry : 683 (2746)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/set_curr_state.drc_errors.asci
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 22:56:04

```

Figure 66: N-Edge Flip Flop DRC Test

Figure 66 shows the DRC Test for the negative edge flip flop layout and it was able to run with 0 errors.

```

INFO: Cleanup (0s)
      Report finished

INFO: Extraction Run Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 31.00 (M)

INFO: NVN Run Summary
      Total CPU Time : 0(s)
      Total Real Time : 1(s)
      Peak Memory Used : 254.51 (M)

INFO: LVS Summary
      Total CPU Time : 1(s)
      Total Real Time : 3(s)
      Peak Memory Used : 254.51 (M)

#####
# Run Result : MATCH #
# #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
# #
# ERC Summary File : set_curr_state.sum #
# Extraction Report File : set_curr_state.rep #
# Comparison Report File : set_curr_state.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 81.42
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Sun Nov 23 22:56:32 2025

Pegasus finished normally. 2025-11-23 22:56:32

```

Figure 67: N-Edge Flip Flop LVS Test

Figure 67 shows the LVS test for the negative edge flip flop layout and it was able to pass.

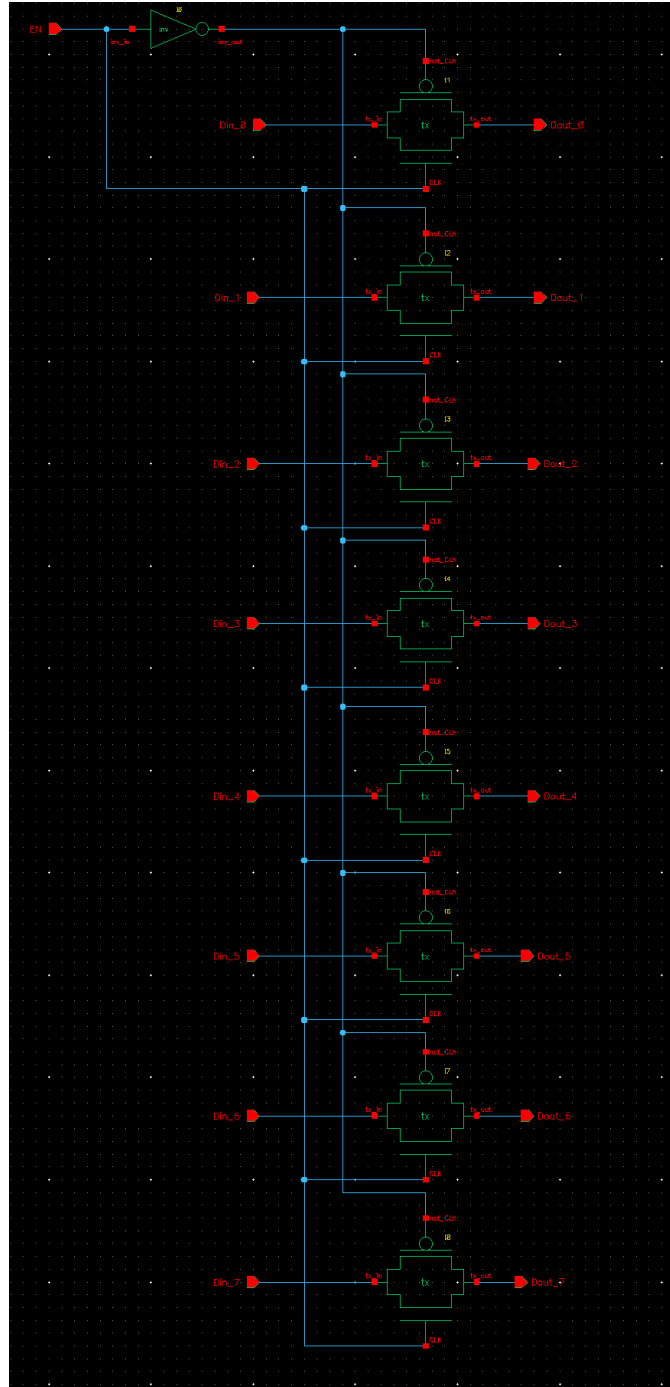


Figure 68: Tristate Buffer Schematic

Figure 68 shows the schematic for a 8-bit tri state buffer.

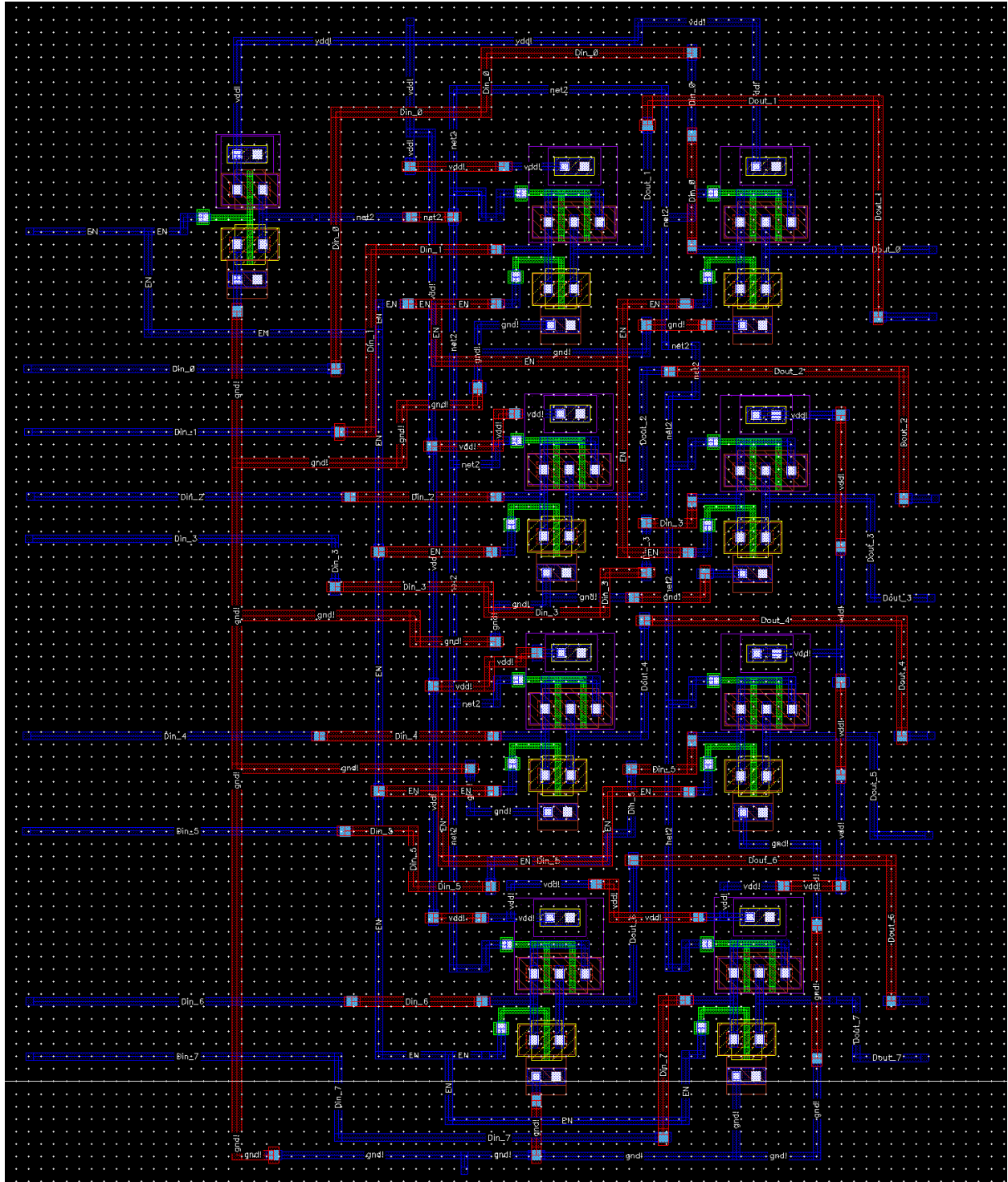


Figure 69: Tristate Buffer Layout

Figure 69 shows the layout for the 8-bit tristate buffer.

```

INFO:      Status: Finishing as of 2025-11-23 23:00:42 Elapsed: 00:00:03 CPU time[s]: 3
INFO:      Resource usage by worker at exit:
          CPU      Elapsed  CPUs      CPU      Peak      Peak engine
Worker:  time[s]  time[s]  Allocated  usage[%]  Memory[MB]  Memory[MB]
          0:      2      3      1      64.13      26      12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO:      Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/tri_buffer.sum
INFO:      All selected rules completed
          Total CPU Time      : 2(s)
          Total Real Time     : 4(s)
          Total Original Geometry : 357(1164)
          Total DRC RuleChecks  : 562
          Total DRC Results    : 0 (0)
          ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/tri_buffer.drc_e
INFO:      Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 23:00:42

```

Figure 70: Tristate Buffer DRC Test

Figure 70 shows the DRC Test for the 8-bit tristate buffer layout and it was able to run with 0 errors.

```

INFO:      Generating the LVS report...
          Report Init (0s)
          Load common queries (0s)
          Create report (0s)
          Cleanup (0s)
INFO:      Report finished

INFO:      Extraction Run Summary
          Total CPU Time      : 1(s)
          Total Real Time     : 2(s)
          Peak Memory Used    : 31.00 (M)

INFO:      NVN Run Summary
          Total CPU Time      : 0(s)
          Total Real Time     : 0(s)
          Peak Memory Used    : 251.18 (M)

INFO:      LVS Summary
          Total CPU Time      : 1(s)
          Total Real Time     : 2(s)
          Peak Memory Used    : 251.18 (M)

#####
# Run Result      : MATCH #
# Run Summary    : [INFO] ERC Results: Empty #
#               : [INFO] Extraction Clean #
# ERC Summary File : tri_buffer.sum #
# Extraction Report File : tri_buffer.rep #
# Comparison Report File : tri_buffer.rep.cls #
#####

INFO:      Creating cross reference database ...
INFO:      Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 75.69
INFO:      Checking in all SoftShare licenses.
INFO:      Creating cross reference database completed. Sun Nov 23 23:01:06 2025

Pegasus finished normally. 2025-11-23 23:01:06

```

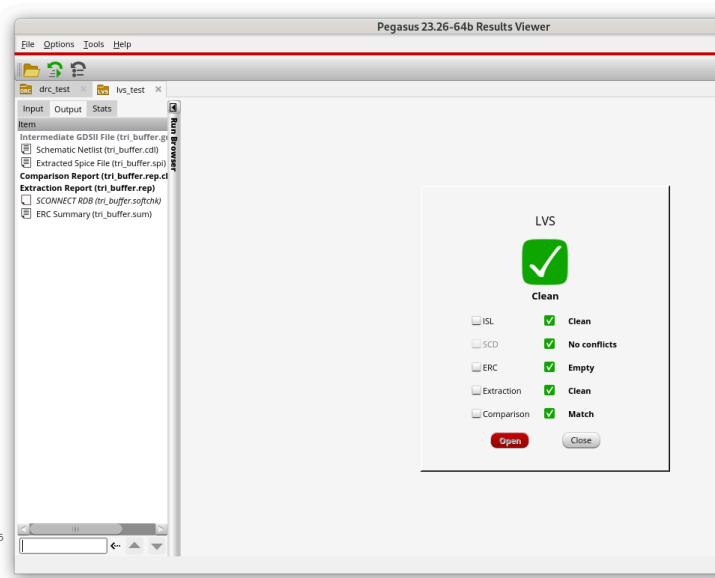


Figure 71: Tristate Buffer LVS Test

Figure 71 shows the LVS test for the 8-bit tristate buffer layout and it was able to pass.

4.3 Cache Cell

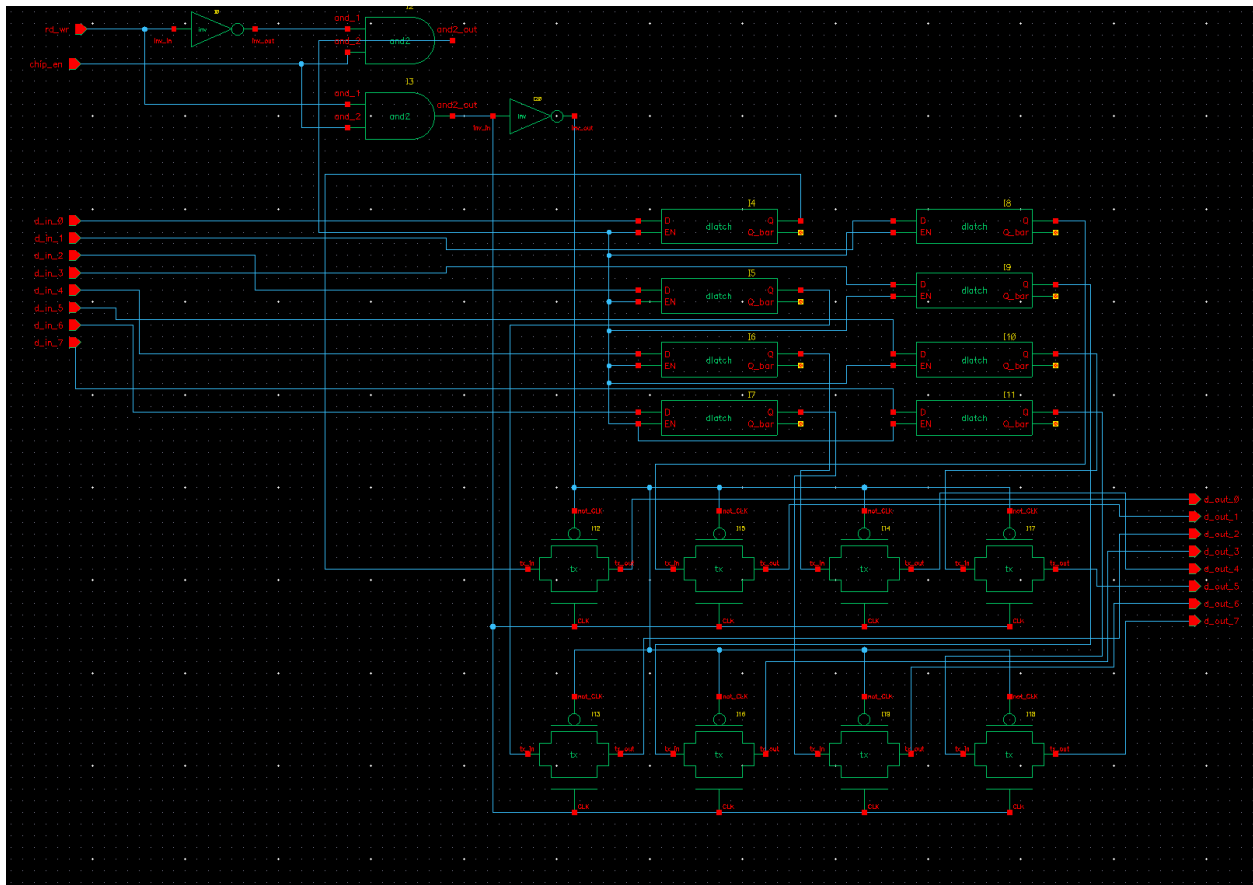


Figure 72: Cache Cell Schematic

Figure 72 shows the schematic for the cache cell.

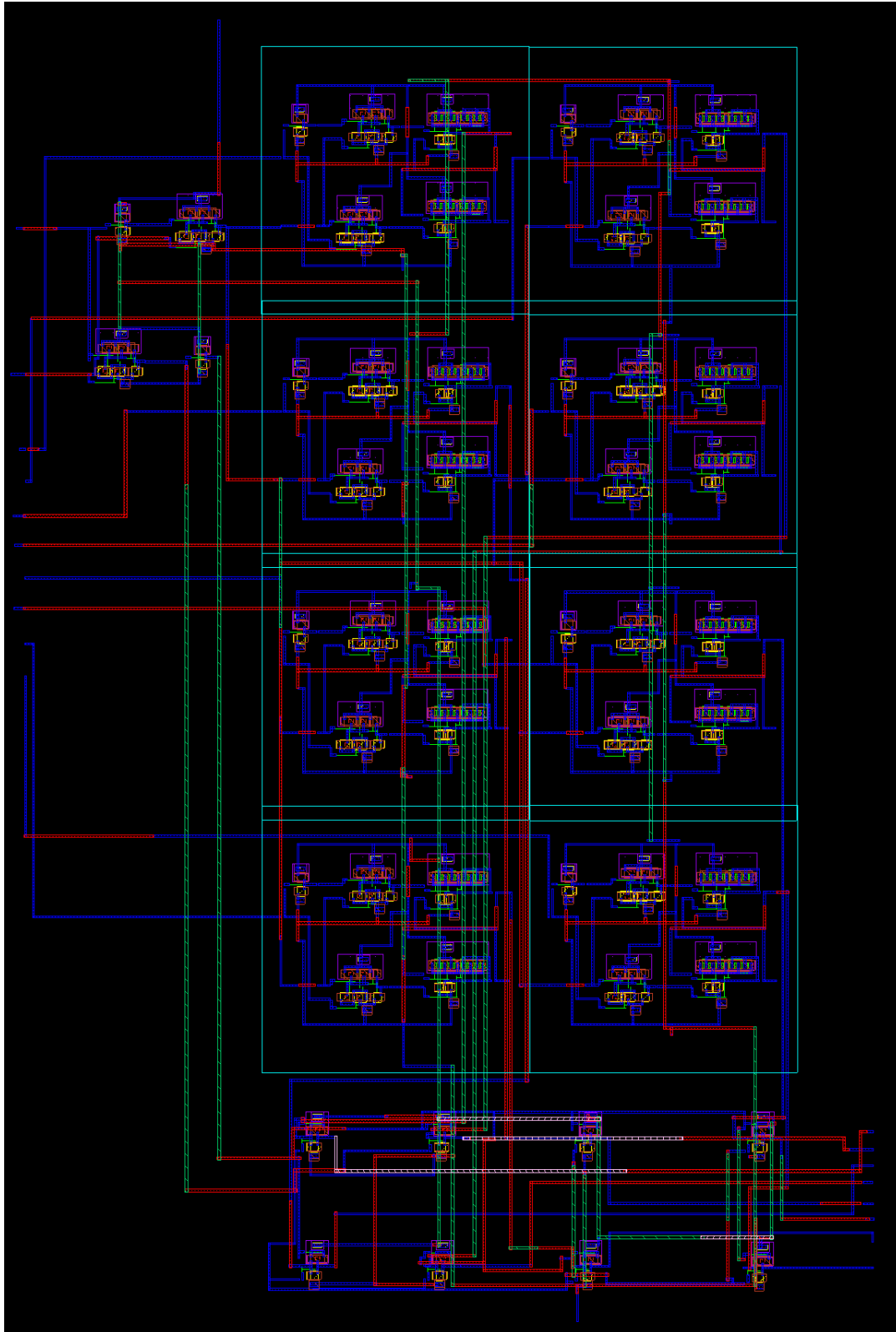


Figure 73: Cache Cell Layout

Figure 73 shows the layout for the cache cell.

```

INFO:      Status: Finishing as of 2025-11-23 23:02:45   Elapsed: 00:00:03   CPU time[s]: 3
INFO:      Resource usage by worker at exit:
           CPU      Elapsed  CPUs      CPU      Peak      Peak engine
Worker:    time[s]  time[s]  Allocated  usage[%]  Memory[MB]  Memory[MB]
0:         2       3       1       61.40     26         12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO:      Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache_cell.sum
INFO:      All selected rules completed
           Total CPU Time      : 2(s)
           Total Real Time     : 4(s)
           Total Original Geometry : 997(9595)
           Total DRC RuleChecks : 562
           Total DRC Results    : 0 (0)
           ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache_cell.drc_errors.ascii
INFO:      Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-23 23:02:45

```

Figure 74: Cache Cell DRC Test

Figure 74 shows the DRC Test for the cache cell layout and it was able to run with 0 errors.

```

Load common queries (0s)
Create report (0s)
Cleanup (0s)
INFO:      Report finished

INFO:      Extraction Run Summary
           Total CPU Time      : 1(s)
           Total Real Time     : 2(s)
           Peak Memory Used    : 31.00 (M)

INFO:      NVN Run Summary
           Total CPU Time      : 1(s)
           Total Real Time     : 1(s)
           Peak Memory Used    : 252.76 (M)

INFO:      LVS Summary
           Total CPU Time      : 2(s)
           Total Real Time     : 3(s)
           Peak Memory Used    : 252.76 (M)

#####
# Run Result      : MATCH #
# Run Summary    : [INFO] ERC Results: Empty #
#                : [INFO] Extraction Clean #
# ERC Summary File : cache_cell.sum #
# Extraction Report File : cache_cell.rep #
# Comparison Report File : cache_cell.rep.cis #
#####

INFO:      Creating cross reference database ...
INFO:      Manager summary: CPU(s): 0 Elapsed(s): 1 Peak memory(MB): 80.30
INFO:      Checking in all SoftShare licenses.
INFO:      Creating cross reference database completed. Mon Nov 24 09:51:05 2025

Pegasus finished normally. 2025-11-24 09:51:05

```

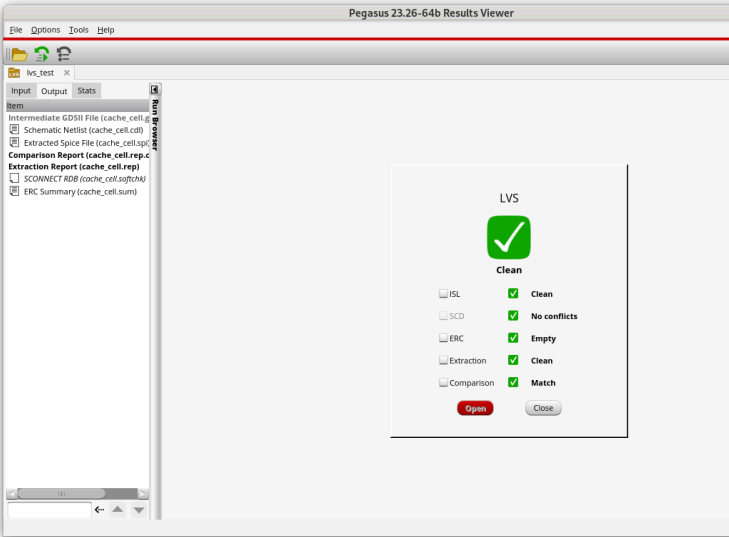


Figure 75: Cache Cell LVS Test

Figure 75 shows the LVS test for the cache cell layout and it was able to pass.

4.4 Cache Block

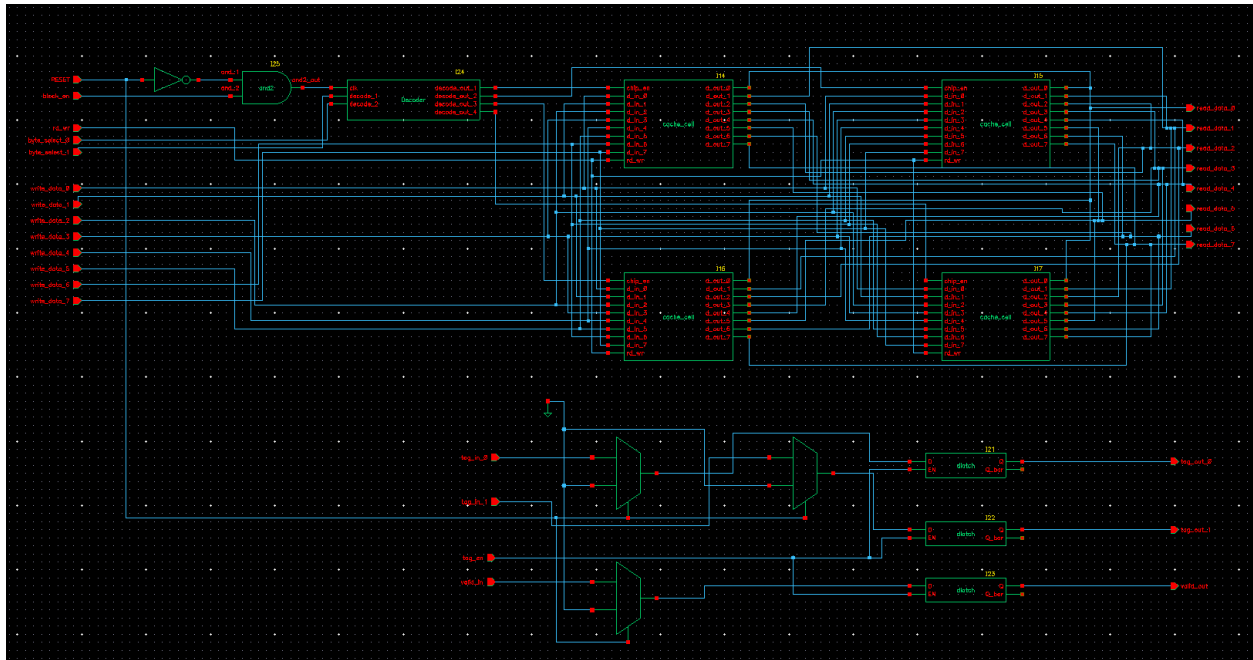


Figure 76: Cache Block Schematic

Figure 76 shows the schematic for the cache block.

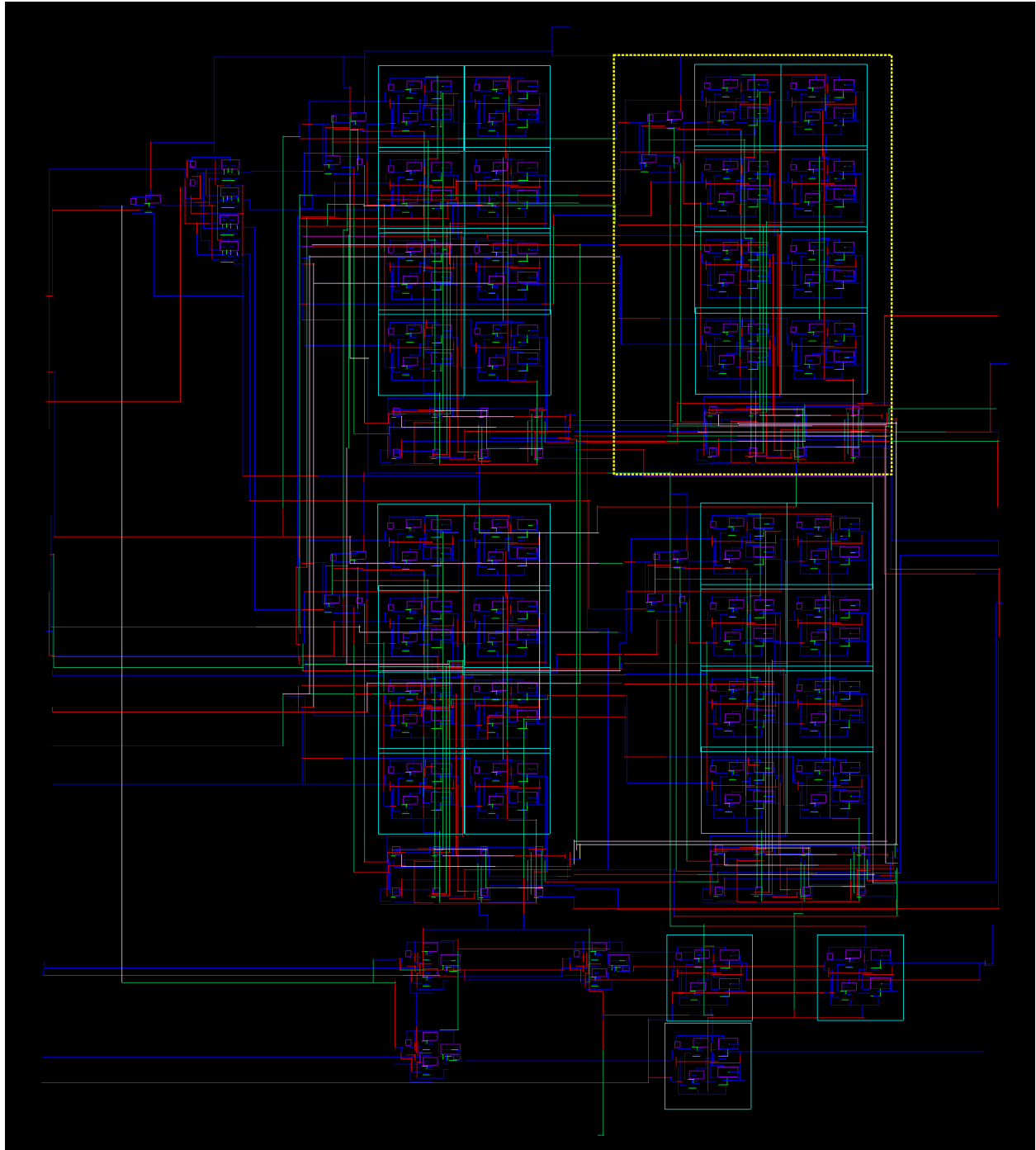


Figure 77: Cache Block Layout

Figure 77 shows the cache block layout.

```

U: 1 / 1 26 12 2
INFO: Status: Finishing as of 2025-11-24 09:53:05 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 2 3 1 59.43 26 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache_block.sum
INFO: All selected rules completed
      Total CPU Time : 2(s)
      Total Real Time : 4(s)
      Total Original Geometry : 2074(46768)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache_block.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 09:53:05

```

Figure 78: Cache Block DRC Test

Figure 78 shows the DRC Test for the cache block layout and it was able to run with 0 errors.

```

      Create report (0s)
      Cleanup (0s)
INFO: Report finished
INFO: Extraction Run Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 31.00 (M)
INFO: NVN Run Summary
      Total CPU Time : 0(s)
      Total Real Time : 1(s)
      Peak Memory Used : 260.32 (M)
INFO: LVS Summary
      Total CPU Time : 1(s)
      Total Real Time : 3(s)
      Peak Memory Used : 260.32 (M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
#
# ERC Summary File : cache_block.sum #
# Extraction Report File : cache_block.rep #
# Comparison Report File : cache_block.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 80.98
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Mon Nov 24 09:53:36 2025

Pegasus finished normally. 2025-11-24 09:53:36

```

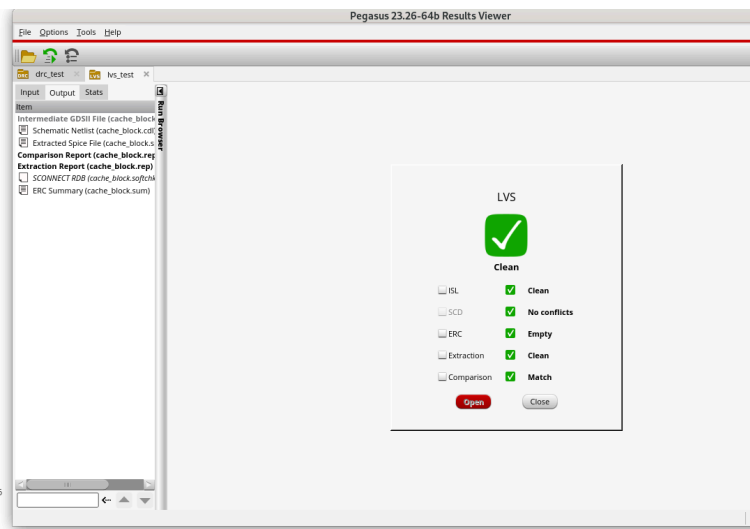


Figure 79: Cache Block LVS Test

Figure 79 shows the LVS test for the cache block layout and it was able to pass.

4.5 Cache Array

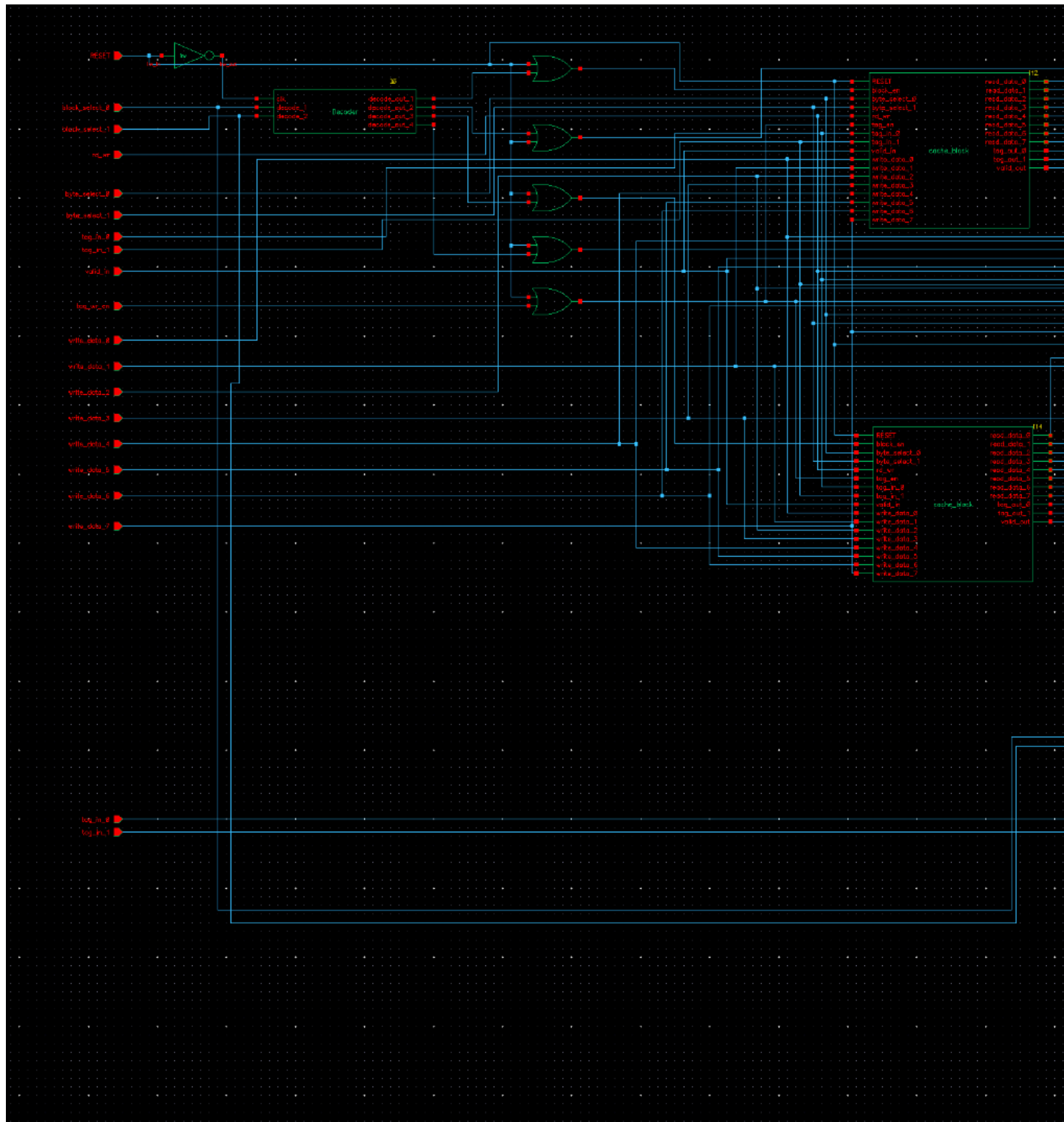


Figure 80: Cache Array Schematic

Figure 80 shows the schematic for the cache array.

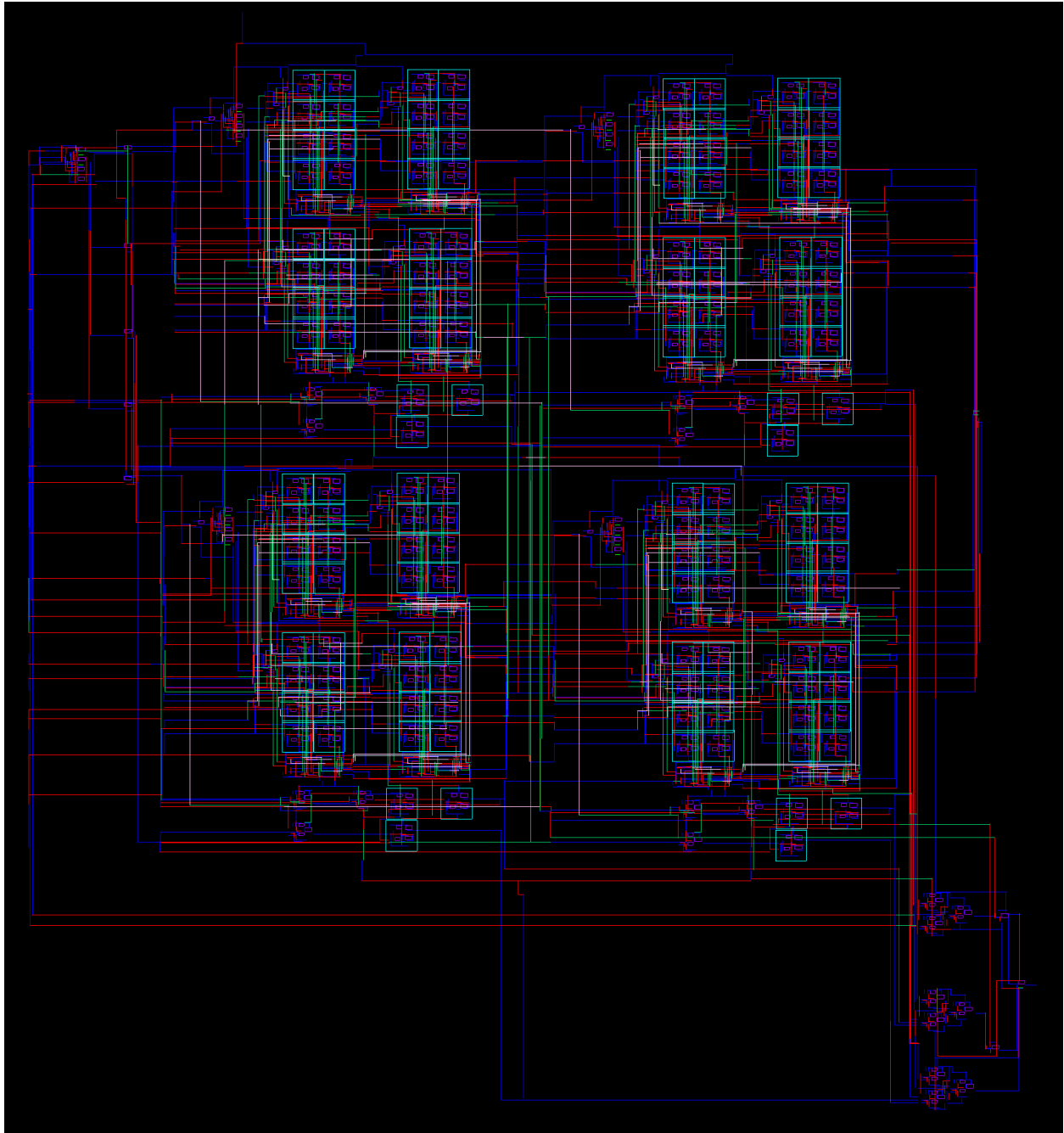


Figure 81: Cache Array Layout

Figure 81 shows the layout for the cache array.

```

INFO:      Status: Finishing as of 2025-11-24 09:55:51   Elapsed: 00:00:04   CPU time[s]: 4
INFO:      Resource usage by worker at exit:
          CPU      Elapsed  CPUs      CPU      Peak      Peak engine
Worker:  time[s]  time[s]  Allocated  usage[%]  Memory[MB]  Memory[MB]
          0:         2         4         1      55.26      116         16
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 4 Max peak memory[MB]: 116
Worker CPU utilization summary: 2 of 4 available CPU seconds (50.00%) with 1 average available CPU(s)
INFO:      Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache_array.sum
INFO:      All selected rules completed
          Total CPU Time      : 3(s)
          Total Real Time     : 5(s)
          Total Original Geometry : 3456(200792)
          Total DRC RuleChecks  : 562
          Total DRC Results    : 0 (0)
          ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache_array.drc_errors.ascii
INFO:      Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 09:55:51

```

Figure 82: Cache Array DRC Test

Figure 82 shows the DRC Test for the cache array layout and it was able to run with 0 errors.

```

INFO:      Create report (0s)
INFO:      Cleanup (0s)
INFO:      Report finished
INFO:      Extraction Run Summary
          Total CPU Time      : 1(s)
          Total Real Time     : 2(s)
          Peak Memory Used    : 31.00 (M)
INFO:      NVN Run Summary
          Total CPU Time      : 1(s)
          Total Real Time     : 1(s)
          Peak Memory Used    : 261.12 (M)
INFO:      LVS Summary
          Total CPU Time      : 2(s)
          Total Real Time     : 3(s)
          Peak Memory Used    : 261.12 (M)

#####
# Run Result      : MATCH #
# Run Summary    : [INFO] ERC Results: Empty #
#                : [INFO] Extraction Clean #
# ERC Summary File : cache_array.sum #
# Extraction Report File : cache_array.rep #
# Comparison Report File : cache_array.rep.cls #
#####

INFO:      Creating cross reference database ...
INFO:      Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 80.97
INFO:      Checking in all SoftShare licenses.
INFO:      Creating cross reference database completed. Mon Nov 24 09:56:26 2025

Pegasus finished normally. 2025-11-24 09:56:26

```

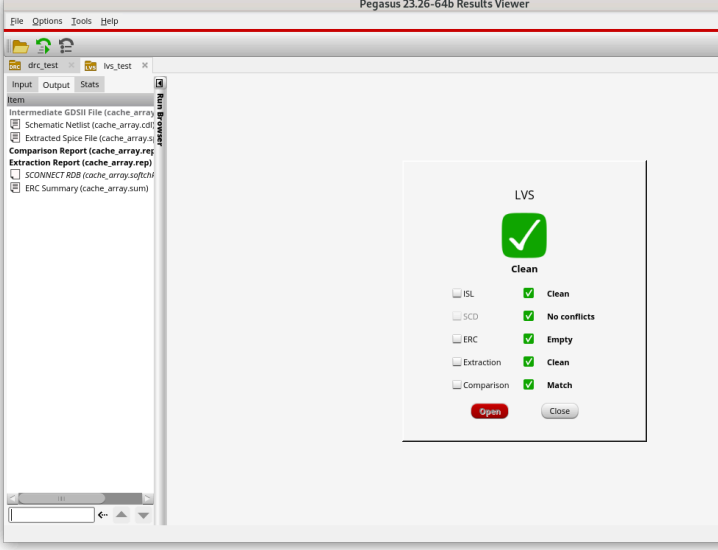


Figure 83: Cache Array LVS Test

Figure 83 shows the LVS test for the cache array layout and it was able to pass.

4.6 Cache FSM

4.6.1 Cache FSM Helper Modules:

The figures in 4.6.1 are all part of FSM but they are just split up into different modules so the design can be easily debugged.

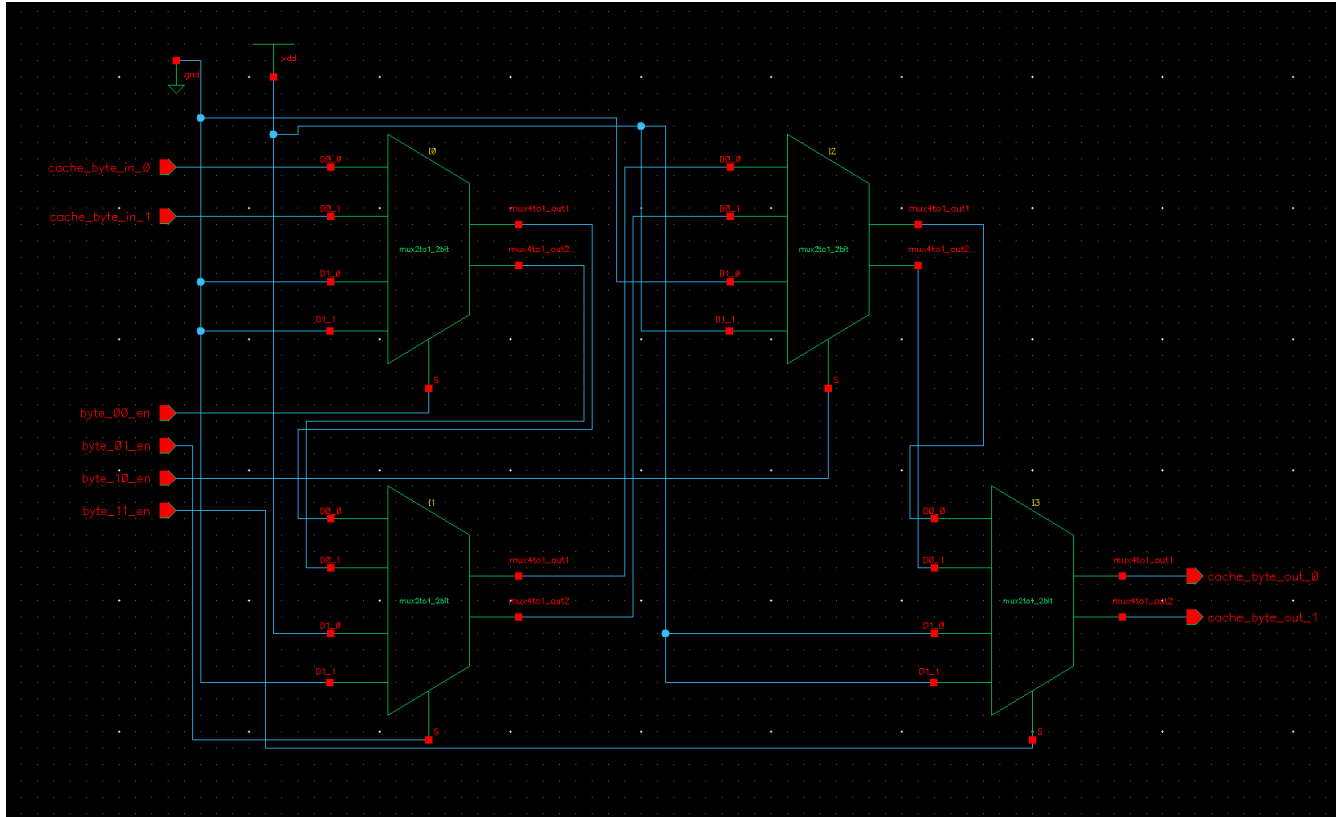


Figure 84: Byte Selector Schematic

Figure 84 shows the byte selector schematic.



Figure 85: Byte Selector Layout

Figure 85 shows the byte selector layout.

```

0: 1 / 1 26 12 2
INFO: Status: Finishing as of 2025-11-24 09:57:26 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
      0: 2 3 1 62.84 26 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/byte_selector.sum
INFO: All selected rules completed
      Total CPU Time : 2(s)
      Total Real Time : 4(s)
      Total Original Geometry : 646(6534)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/byte_selector.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 09:57:26

```

Figure 86: Byte Selector DRC Test

Figure 86 shows the DRC Test for the byte selector layout and it was able to run with 0 errors.

```

INFO: Cleanup report (1s)
INFO: Cleanup (0s)
INFO: Report finished

INFO: Extraction Run Summary
      Total CPU Time : 1(s)
      Total Real Time : 2(s)
      Peak Memory Used : 31.00 (M)

INFO: NVN Run Summary
      Total CPU Time : 0(s)
      Total Real Time : 1(s)
      Peak Memory Used : 265.39 (M)

INFO: LVS Summary
      Total CPU Time : 1(s)
      Total Real Time : 3(s)
      Peak Memory Used : 265.39 (M)

#####
# Run Result : MATCH
# Run Summary : [INFO] ERC Results: Empty
# : [INFO] Extraction Clean
#
# ERC Summary File : byte_selector.sum
# Extraction Report File : byte_selector.rep
# Comparison Report File : byte_selector.rep.cls
#
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 91.09
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Mon Nov 24 09:59:30 2025

Pegasus finished normally. 2025-11-24 09:59:30

```

Figure 87: Byte Selector LVS Test

Figure 87 shows the LVS test for the byte selector layout and it was able to pass.

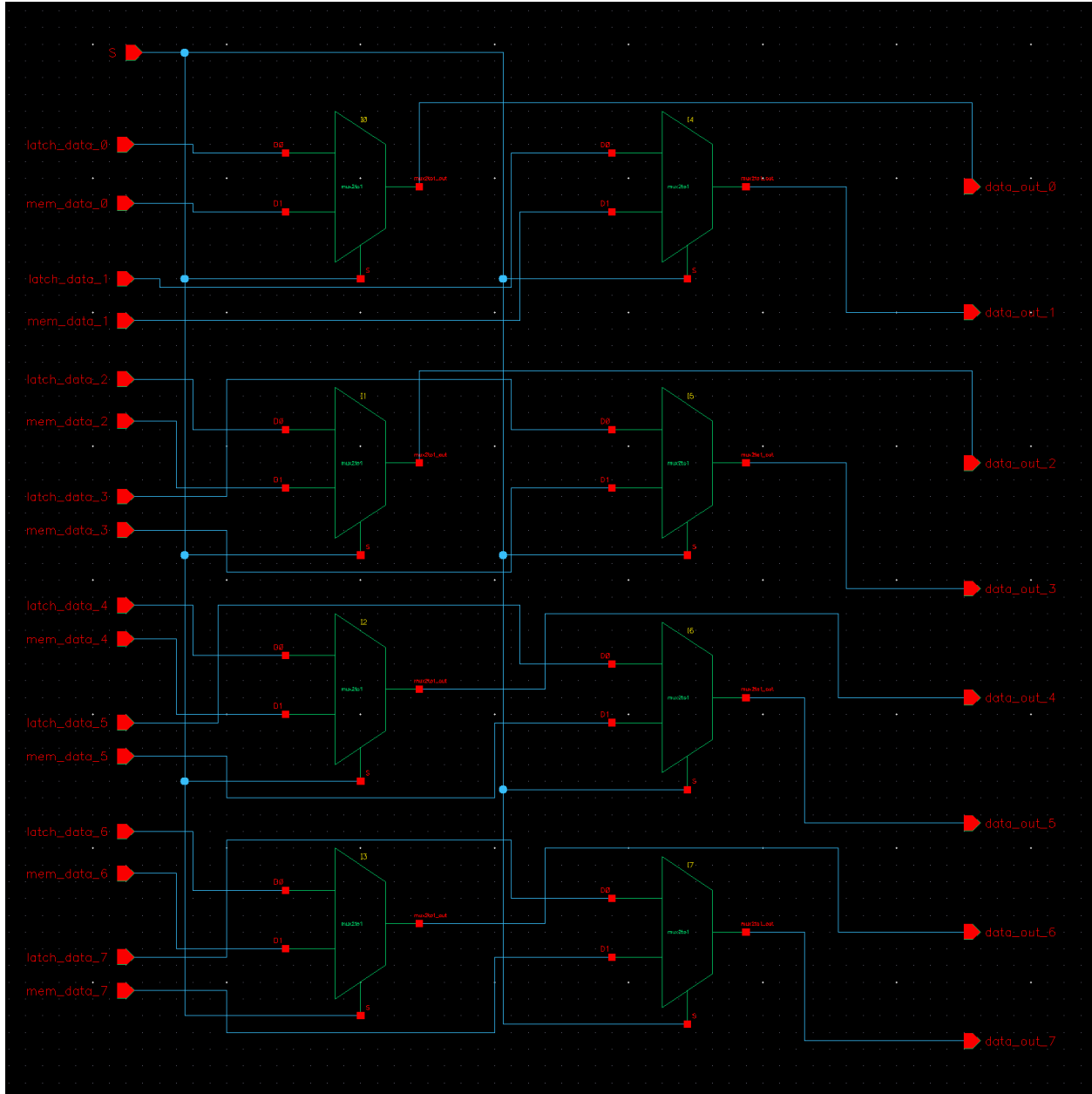


Figure 88: Select Write Data Schematic

Figure 88 shows the select write data schematic. This decides whether to write from cache data or memory data.

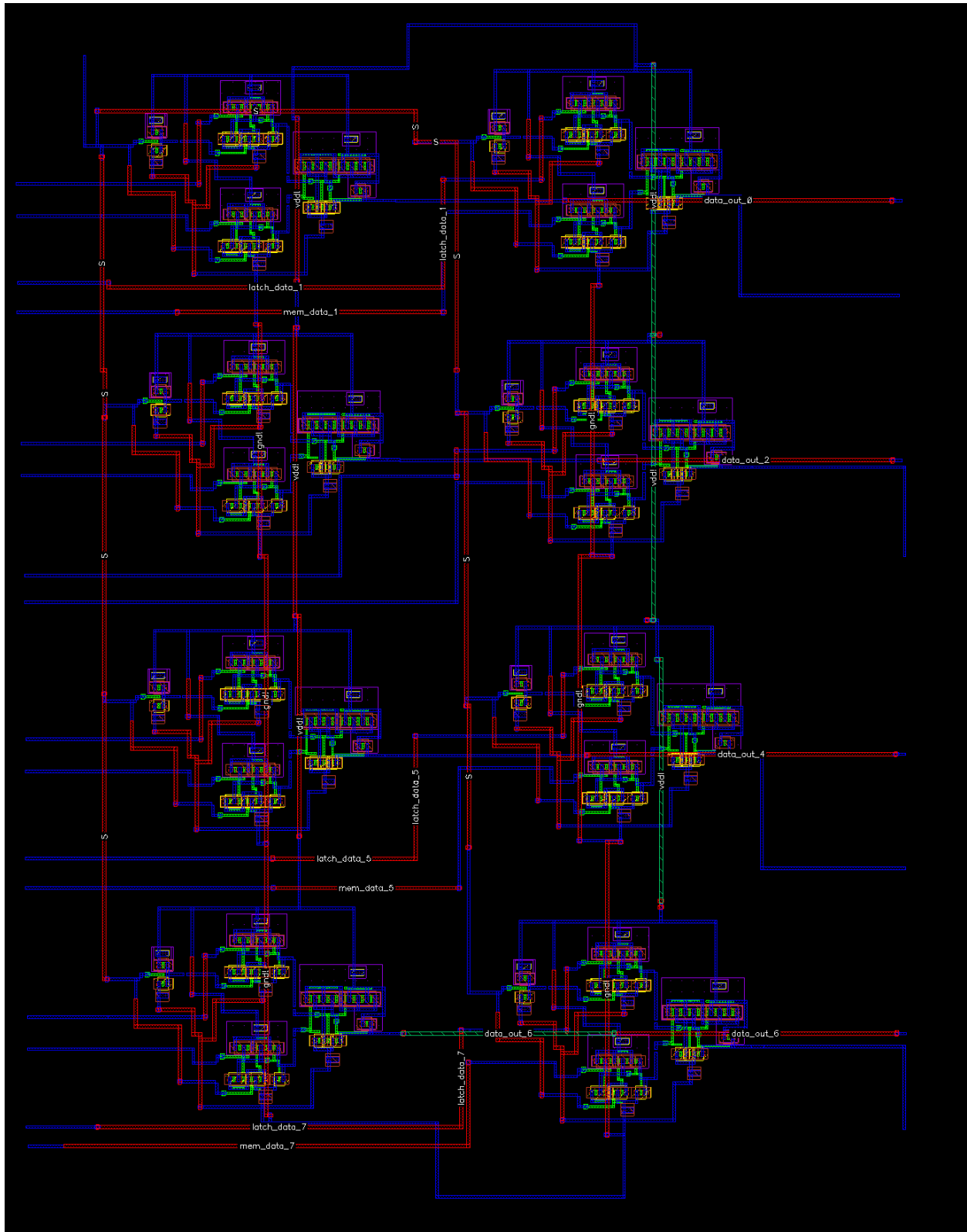


Figure 89: Select Write Data Layout

Figure 89 shows the layout for the select write data.


```

0: 1 / 1 26 12 2
INFO: Status: Finishing as of 2025-11-24 10:28:17 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
Worker: CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 2 3 1 62.15 26 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/select_write_data.sum
INFO: All selected rules completed
Total CPU Time : 2(s)
Total Real Time : 4(s)
Total Original Geometry : 669(6452)
Total DRC RuleChecks : 562
Total DRC Results : 0 (0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/select_write_data.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 10:28:17

```

Figure 90: Select Write Data DRC Test

Figure 90 shows the DRC Test for the select write data layout and it was able to run with 0 errors.

```

Create report (0s)
Cleanup (0s)
INFO: Report finished

INFO: Extraction Run Summary
Total CPU Time : 1(s)
Total Real Time : 2(s)
Peak Memory Used : 31.00(M)

INFO: NVN Run Summary
Total CPU Time : 1(s)
Total Real Time : 1(s)
Peak Memory Used : 259.76(M)

INFO: LVS Summary
Total CPU Time : 2(s)
Total Real Time : 3(s)
Peak Memory Used : 259.76(M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
# ERC Summary File : select_write_data.sum #
# Extraction Report File : select_write_data.rep #
# Comparison Report File : select_write_data.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 87.25
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Mon Nov 24 10:28:59 2025

Pegasus finished normally. 2025-11-24 10:28:59

```

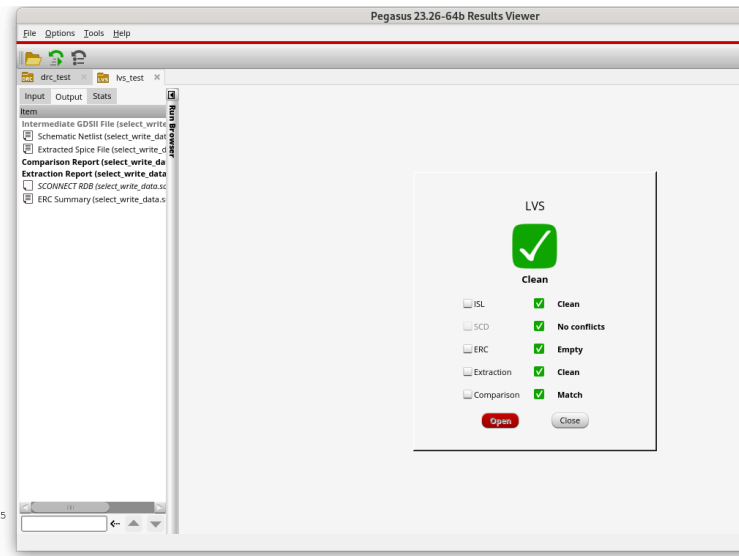


Figure 91: Select Write Data LVS Test

Figure 91 shows the LVS test for the select write data layout and it was able to pass.

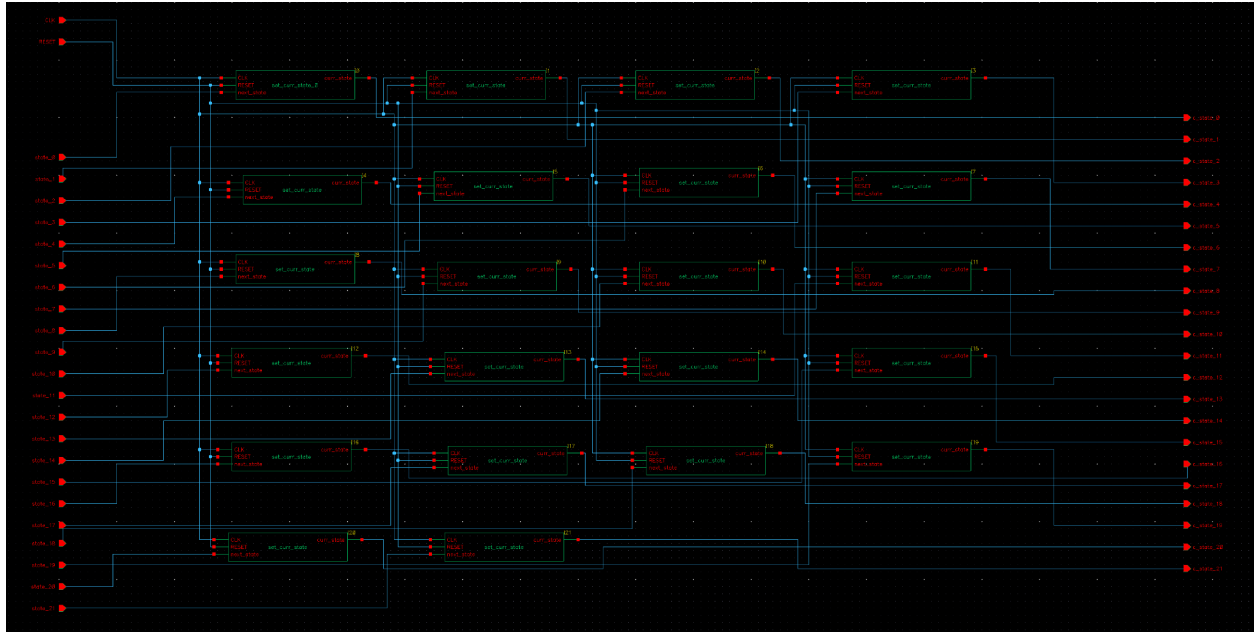


Figure 92: Set Current States Schematic

Figure 92 is the schematic for the set current states. This sets the current state to the next state at the negative edge.

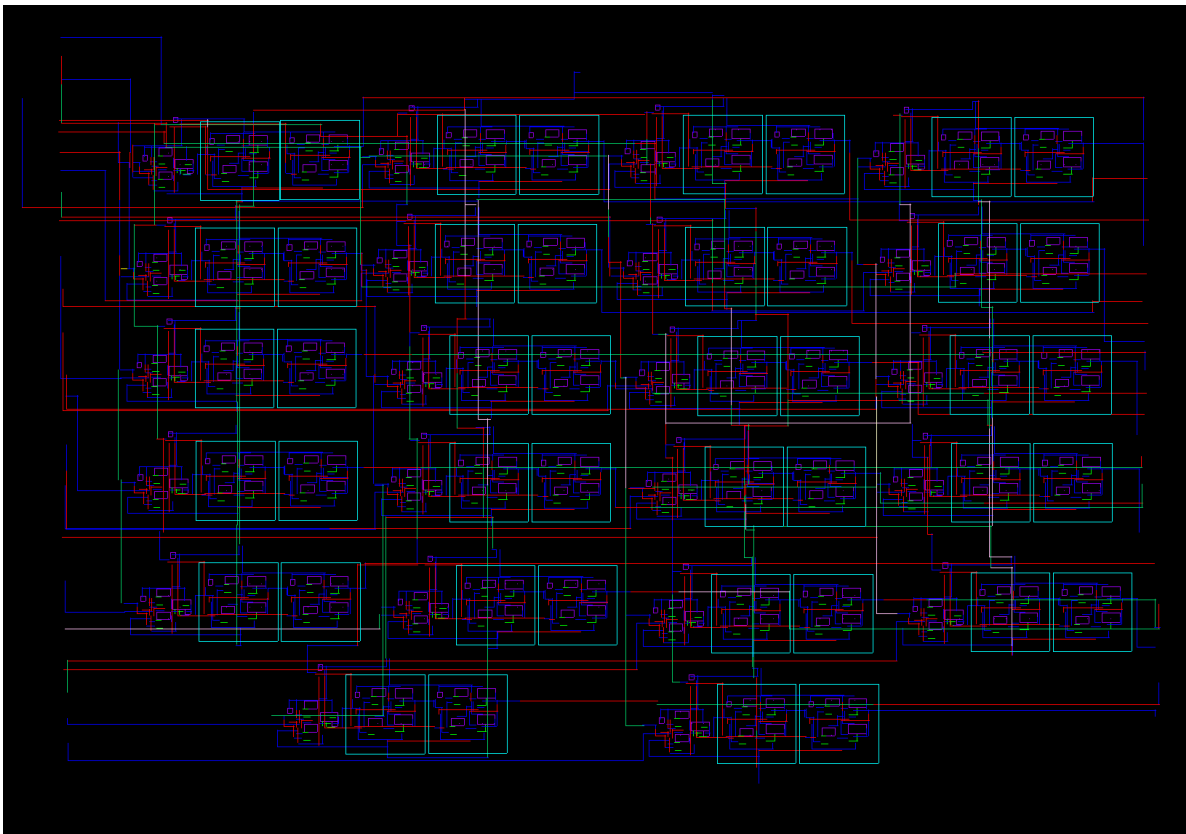


Figure 93: Set Current States Layout

Figure 93 shows the layout for the set current states.

```

0:      1 / 1      26      12      2
INFO:   Status: Finishing as of 2025-11-24 10:30:49 Elapsed: 00:00:03 CPU time[s]: 3
INFO:   Resource usage by worker at exit:
      CPU      Elapsed CPUs      CPU      Peak      Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
      0:      2      3      1      61.32      26      12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO:   Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/set_curr_states.sum
INFO:   All selected rules completed
      Total CPU Time      : 2(s)
      Total Real Time      : 4(s)
      Total Original Geometry      : 1393(62129)
      Total DRC RuleChecks      : 562
      Total DRC Results      : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/set_curr_states.drc_errors.ascii
INFO:   Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 10:30:49

```

Figure 94: Set Current States DRC Test

Figure 94 shows the DRC Test for the set current states layout and it was able to run with 0 errors.

```

Create report (0s)
Cleanup (0s)
Report finished
INFO:   Extraction Run Summary
      Total CPU Time      : 1(s)
      Total Real Time      : 2(s)
      Peak Memory Used      : 31.00 (M)

INFO:   NVN Run Summary
      Total CPU Time      : 0(s)
      Total Real Time      : 0(s)
      Peak Memory Used      : 264.13 (M)

INFO:   LVS Summary
      Total CPU Time      : 1(s)
      Total Real Time      : 2(s)
      Peak Memory Used      : 264.13 (M)

#####
# Run Result      : MATCH #
# Run Summary      : [INFO] ERC Results: Empty #
#                  : [INFO] Extraction Clean #
# ERC Summary File : set_curr_states.sum #
# Extraction Report File : set_curr_states.rep #
# Comparison Report File : set_curr_states.rep.cis #
#####

INFO:   Creating cross reference database ...
INFO:   Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory (MB): 90.25
INFO:   Checking in all SoftShare licenses.
INFO:   Creating cross reference database completed. Mon Nov 24 10:31:29 2025

Pegasus finished normally. 2025-11-24 10:31:29

```

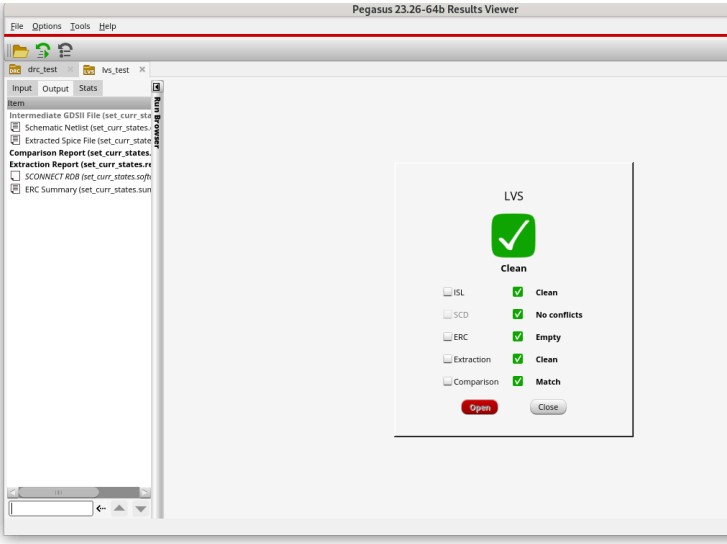


Figure 95: Set Current States LVS Test

Figure 95 shows the LVS test for the set current states layout and it was able to pass.

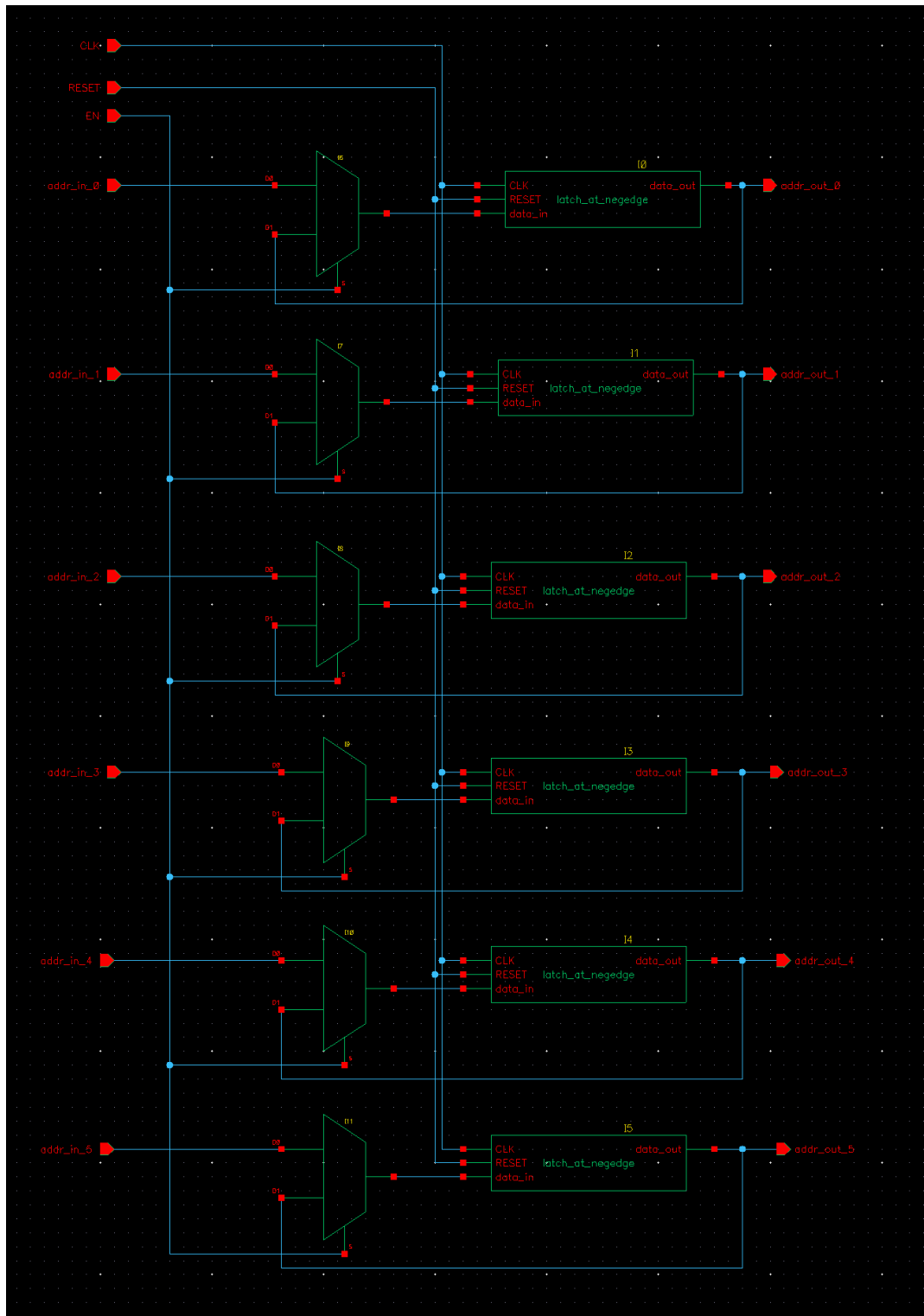


Figure 96: Latch Cache Address Schematic

Figure 96 shows the latch cache address schematic. This module saves the address passed in.

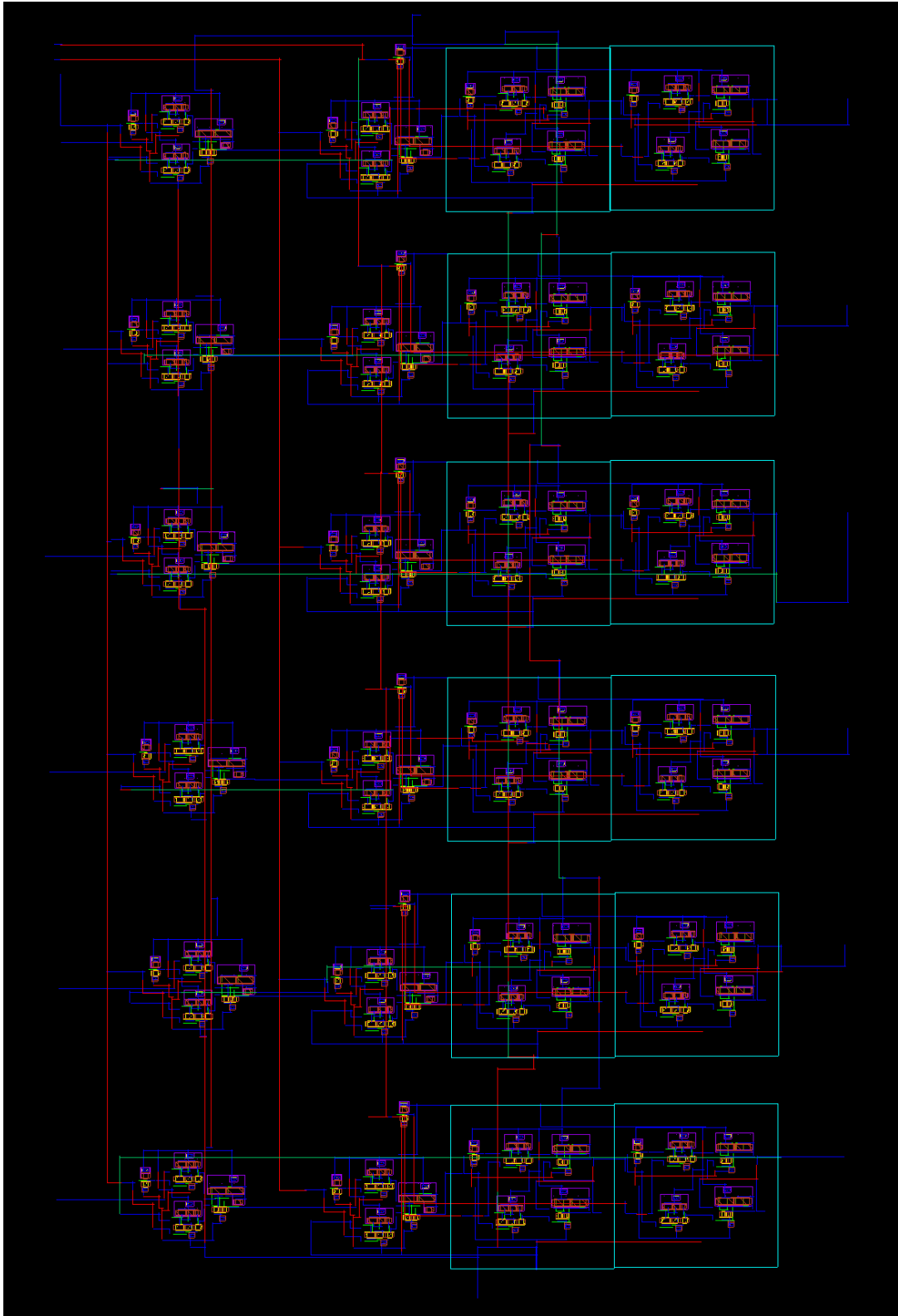


Figure 97: Latch Cache Address Layout

Figure 97 shows the layout for the latch cache address.

```

INFO:      Status: Finishing as of 2025-11-24 10:32:44 Elapsed: 00:00:03 CPU time[s]: 3
INFO:      Resource usage by worker at exit:
Worker:   CPU      Elapsed  CPUs      CPU      Peak      Peak engine
time[s]   time[s]   Allocated  usage[%]  Memory[MB]  Memory[MB]
0:         2         3         1        62.45      26         12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO:      Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/latch_cache_address.sum
INFO:      All selected rules completed
Total CPU Time           : 2(s)
Total Real Time          : 4(s)
Total Original Geometry  : 957(21632)
Total DRC RuleChecks     : 562
Total DRC Results        : 0 (0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/latch_cache_address.drc_errors.ascii
INFO:      Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 10:32:44

```

Figure 98: Latch Cache Address DRC Test

Figure 98 shows the DRC Test for the latch cache address layout and it was able to run with 0 errors.

```

INFO:      Create report (us)
Cleanup (0s)
Report finished

INFO:      Extraction Run Summary
Total CPU Time       : 1(s)
Total Real Time      : 2(s)
Peak Memory Used     : 31.00 (M)

INFO:      NVN Run Summary
Total CPU Time       : 0(s)
Total Real Time      : 1(s)
Peak Memory Used     : 264.20 (M)

INFO:      LVS Summary
Total CPU Time       : 1(s)
Total Real Time      : 3(s)
Peak Memory Used     : 264.20 (M)

#####
# Run Result          : MATCH
# Run Summary         : [INFO] ERC Results: Empty
#                     : [INFO] Extraction Clean
# ERC Summary File    : latch_cache_address.sum
# Extraction Report File : latch_cache_address.rep
# Comparison Report File : latch_cache_address.rep.cls
#####

INFO:      Creating cross reference database ...
INFO:      Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 91.00
INFO:      Checking in all SoftShare licenses.
INFO:      Creating cross reference database completed. Mon Nov 24 10:33:16 2025

Pegasus finished normally. 2025-11-24 10:33:16

```

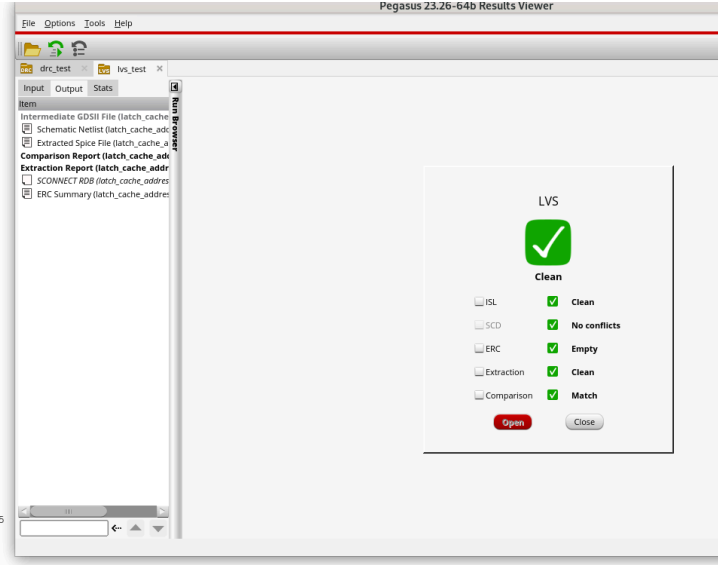


Figure 99: Latch Cache Address LVS Test

Figure 99 shows the LVS test for the latch cache address layout and it was able to pass.

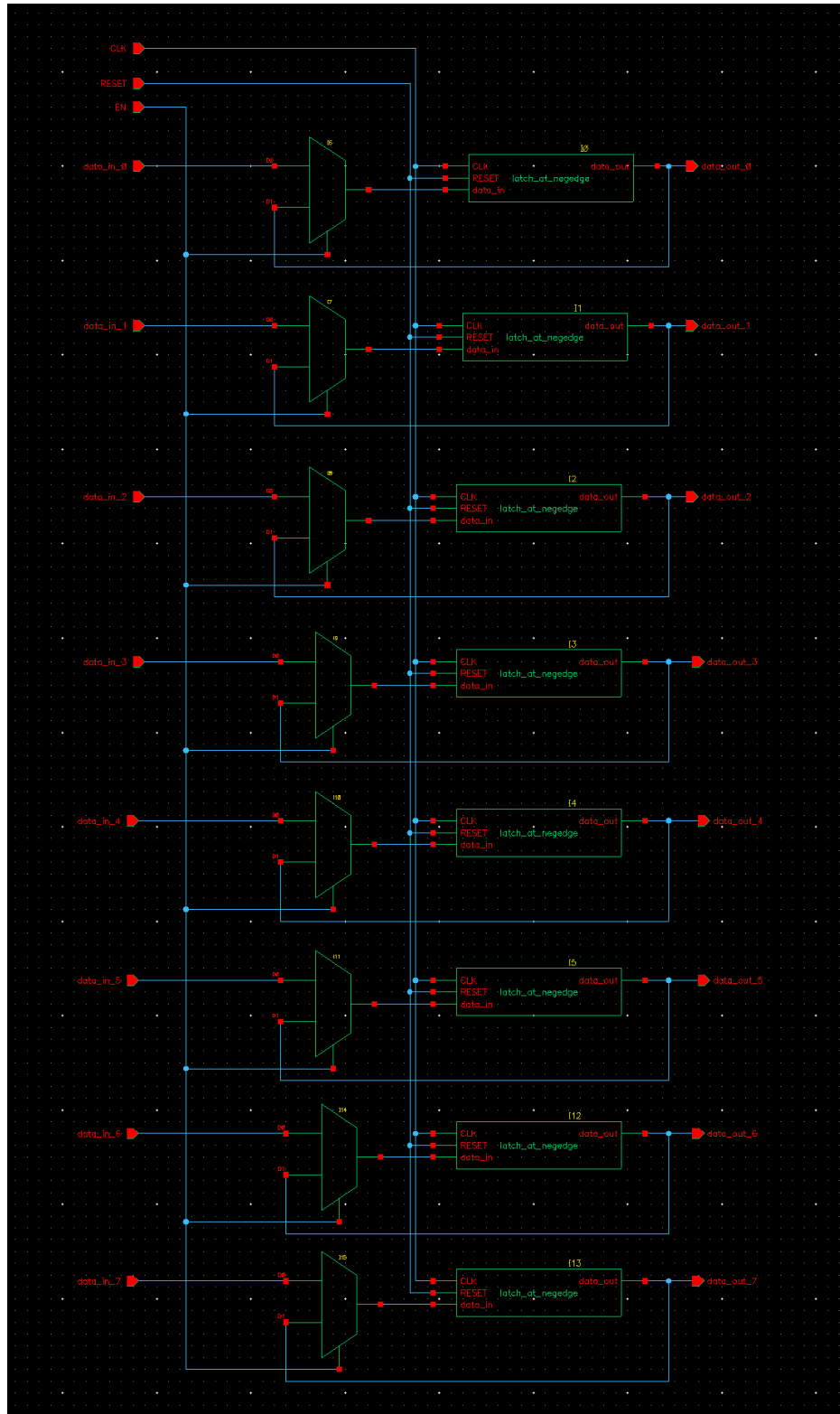


Figure 100: Latch Cache Data Schematic

Figure 100 shows the latch cache data schematic. This module is used to store the data passed in.

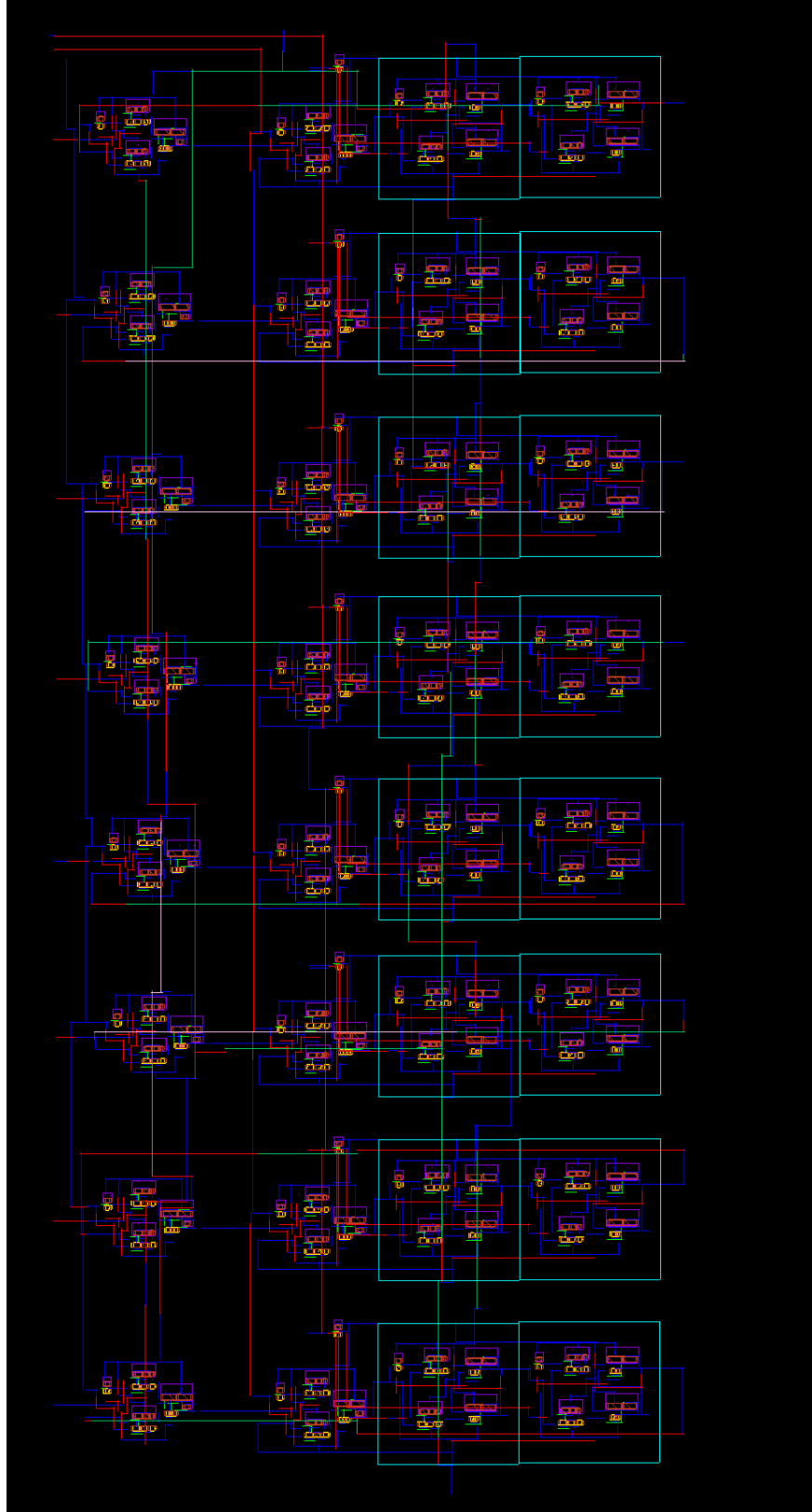


Figure 101: Latch Cache Data Layout

Figure 101 shows the layout for the latch cache data.


```

0: 1 / 1 26 12 2
INFO: Status: Finishing as of 2025-11-24 10:33:56 Elapsed: 00:00:03 CPU time[s]: 3
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 2 3 1 63.13 26 12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/latch_cache_data.sum
INFO: All selected rules completed
      Total CPU Time : 2 (s)
      Total Real Time : 4 (s)
      Total Original Geometry : 1040 (28881)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/latch_cache_data.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 10:33:56

```

Figure 102: Latch Cache Data DRC Test

Figure 102 shows the DRC Test for the latch cache data layout and it was able to run with 0 errors.

```

Load common queries (0s)
Create report (0s)
Cleanup (0s)
INFO: Report finished
INFO: Extraction Run Summary
      Total CPU Time : 1 (s)
      Total Real Time : 2 (s)
      Peak Memory Used : 31.00 (M)
INFO: NVN Run Summary
      Total CPU Time : 0 (s)
      Total Real Time : 1 (s)
      Peak Memory Used : 259.89 (M)
INFO: LVS Summary
      Total CPU Time : 1 (s)
      Total Real Time : 3 (s)
      Peak Memory Used : 259.89 (M)

#####
# Run Result : MATCH #
# Run Summary : [INFO] ERC Results: Empty #
# : [INFO] Extraction Clean #
# ERC Summary File : latch_cache_data.sum #
# Extraction Report File : latch_cache_data.rep #
# Comparison Report File : latch_cache_data.rep.cls #
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 88.13
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Mon Nov 24 10:34:33 2025

Pegasus finished normally. 2025-11-24 10:34:33

```

Figure 103: Latch Cache Data LVS Test

Figure 103 shows the LVS test for the latch cache data layout and it was able to pass.

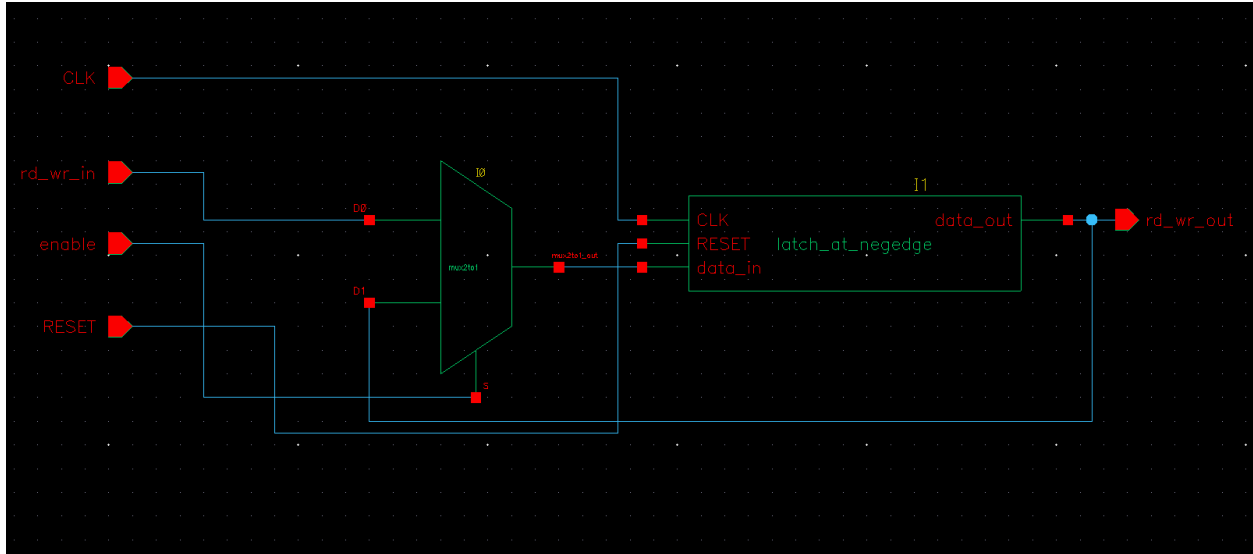


Figure 104: Latch Read Write Schematic

Figure 104 shows the latch read write schematic. This module is used to store the read write bit.

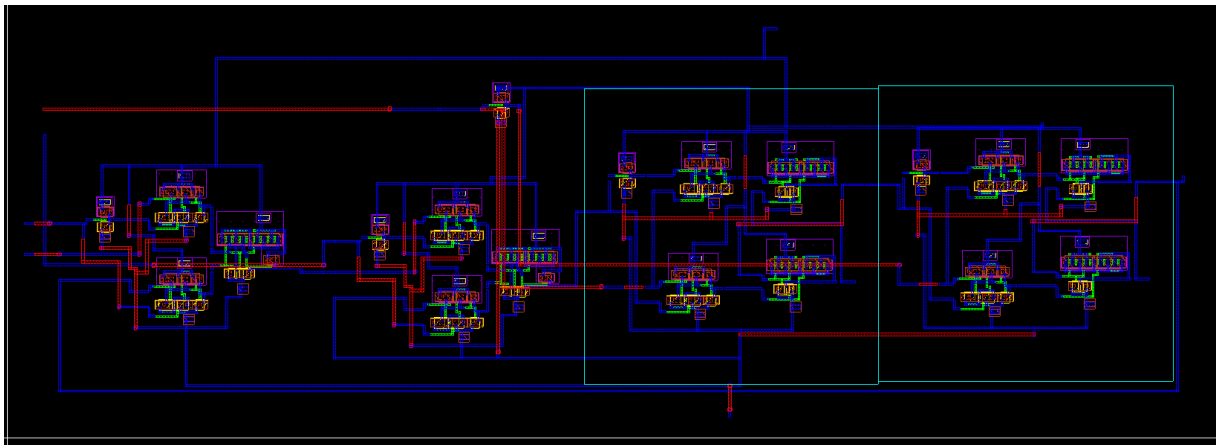


Figure 105: Latch Read Write Layout

Figure 105 shows the latch read and write layout.

```

U:      1 / 1      2b      12      2
INFO:    Status: Finishing as of 2025-11-24 10:35:25 Elapsed: 00:00:03 CPU time[s]: 3
INFO:    Resource usage by worker at exit:
Worker:  CPU Elapsed CPUs CPU Peak Peak
         time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0:        2      3      1      62.94      26      12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 26
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO:    Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/latch_rd_wr.sum
INFO:    All selected rules completed
Total CPU Time           : 2(s)
Total Real Time          : 4(s)
Total Original Geometry   : 717(3573)
Total DRC RuleChecks      : 562
Total DRC Results         : 0 (0)
ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/latch_rd_wr.drc_errors.ascii
INFO:    Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 10:35:25

```

Figure 106: Latch Read Write DRC Test

Figure 106 shows the DRC Test for the latch read write layout and it was able to run with 0 errors.

```

Load common queries (0s)
Create report (0s)
Cleanup (0s)
INFO: Report finished

INFO: Extraction Run Summary
Total CPU Time      : 1(s)
Total Real Time     : 2(s)
Peak Memory Used    : 31.00 (M)

INFO: NVN Run Summary
Total CPU Time      : 0(s)
Total Real Time     : 0(s)
Peak Memory Used    : 255.88 (M)

INFO: LVS Summary
Total CPU Time      : 1(s)
Total Real Time     : 2(s)
Peak Memory Used    : 255.88 (M)

#####
# Run Result          : MATCH
#
# Run Summary         : [INFO] ERC Results: Empty
#                     : [INFO] Extraction Clean
#
# ERC Summary File    : latch_rd_wr.sum
# Extraction Report File : latch_rd_wr.rep
# Comparison Report File : latch_rd_wr.rep.cls
#
#####

INFO: Creating cross reference database ...
INFO: Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 87.14
INFO: Checking in all SoftShare licenses.
INFO: Creating cross reference database completed. Mon Nov 24 10:36:06 2025

Pegasus finished normally. 2025-11-24 10:36:06

```

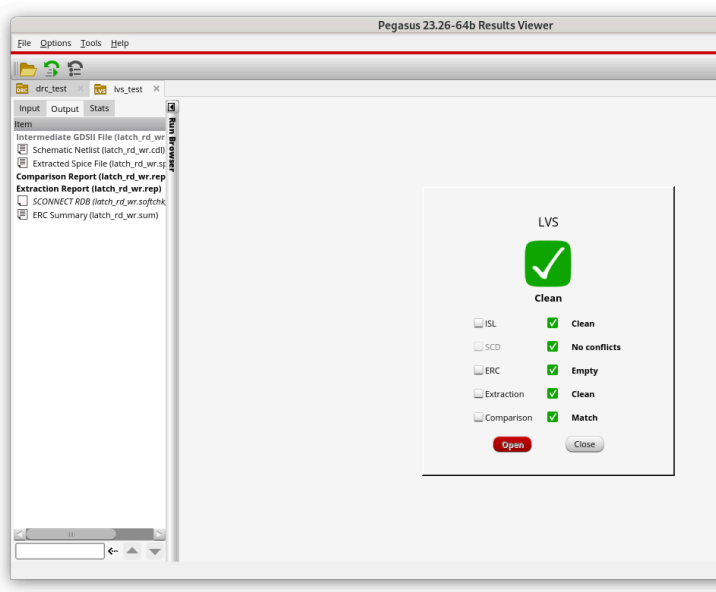


Figure 107: Latch Read Write LVS Test

Figure 107 shows the LVS test for the 107 latch read write layout and it was able to pass.

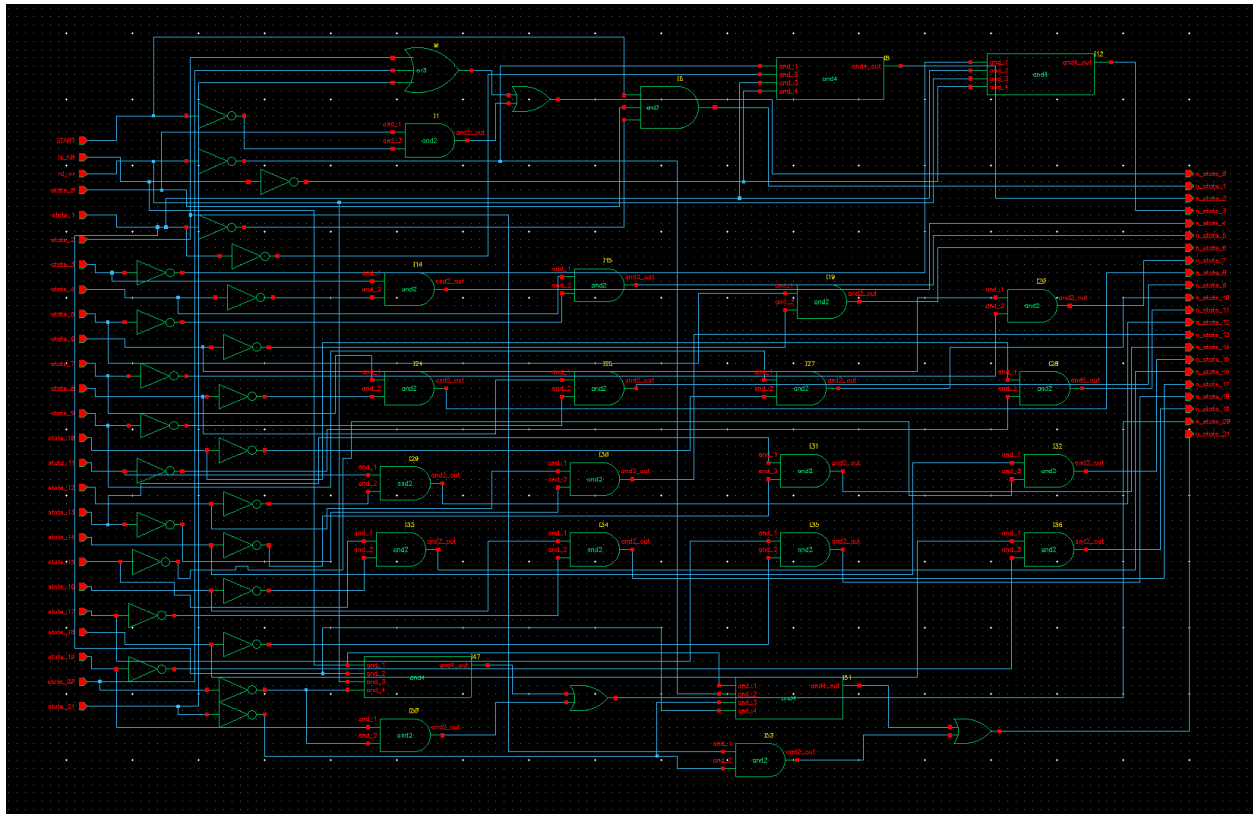


Figure 108: FSM States Schematic

Figure 108 shows the schematic for the FSM States. This module is where the next state logic is determined.

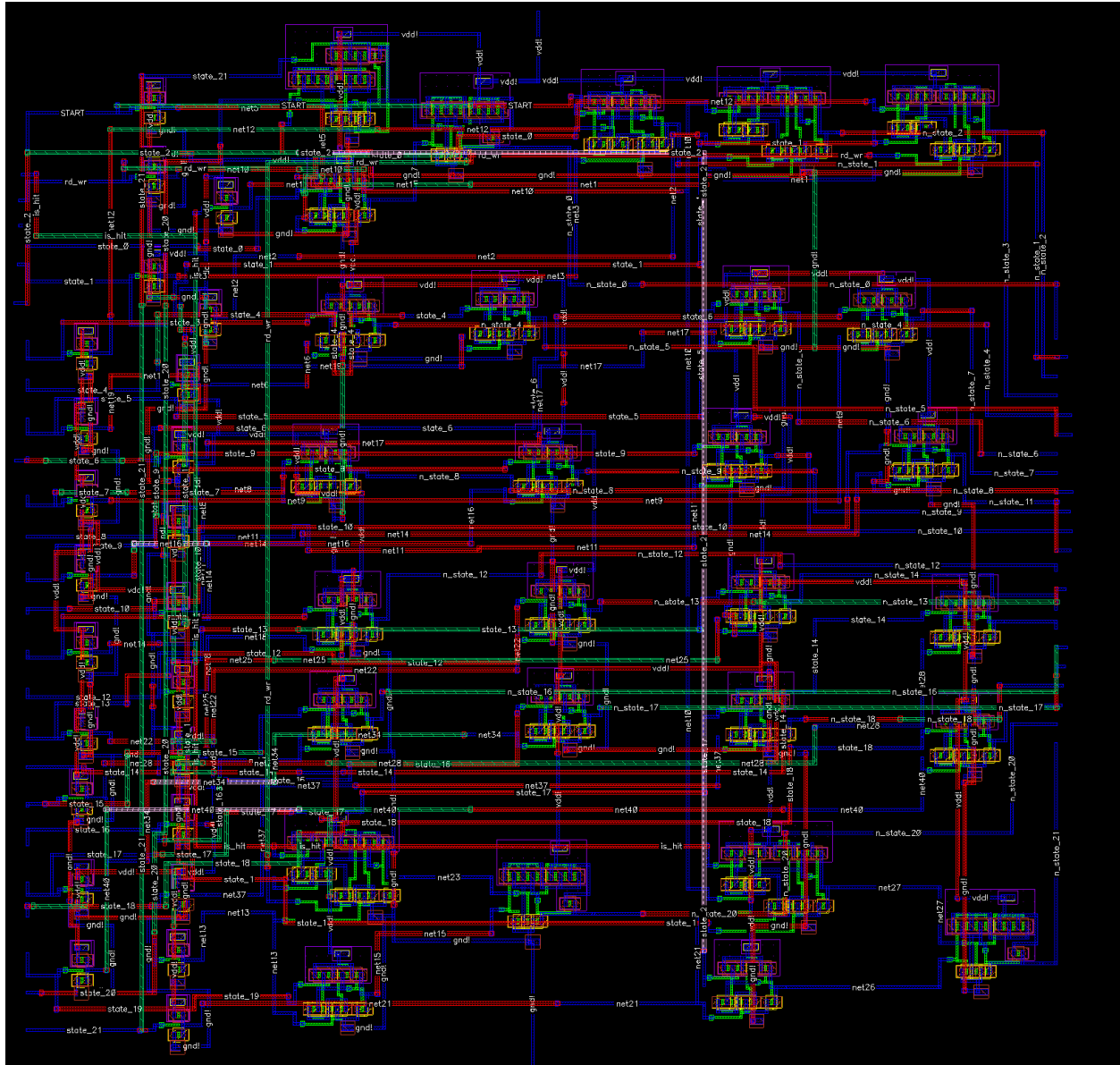


Figure 109: FSM States Layout

Figure 109 shows the layout for the FSM states.

```

INFO:      Status: Finishing as of 2025-11-24 10:37:15 Elapsed: 00:00:03 CPU time[s]: 3
INFO:      Resource usage by worker at exit:
          CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
          0:      2      3      1      62.87      112      12
Worker resource usage summary: Total CPU[s]: 2 Run duration[s]: 3 Max peak memory[MB]: 112
Worker CPU utilization summary: 2 of 3 available CPU seconds (66.67%) with 1 average available CPU(s)
INFO:      Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/fsm_states.sum
INFO:      All selected rules completed
          Total CPU Time      : 2 (s)
          Total Real Time     : 4 (s)
          Total Original Geometry : 1873 (9899)
          Total DRC RuleChecks  : 562
          Total DRC Results     : 0 (0)
          ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/fsm_states.drc_errors.ascii
INFO:      Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 10:37:15

```

Figure 110: FSM States DRC Test

Figure 110 shows the DRC Test for the FSM states layout and it was able to run with 0 errors.

```

INFO:      Cleanup (0s)
INFO:      Report finished
INFO:      Extraction Run Summary
          Total CPU Time      : 1 (s)
          Total Real Time     : 2 (s)
          Peak Memory Used    : 31.00 (M)
INFO:      NVN Run Summary
          Total CPU Time      : 0 (s)
          Total Real Time     : 1 (s)
          Peak Memory Used    : 254.15 (M)
INFO:      LVS Summary
          Total CPU Time      : 1 (s)
          Total Real Time     : 3 (s)
          Peak Memory Used    : 254.15 (M)

#####
# Run Result      : MATCH
#
# Run Summary     : [INFO] ERC Results: Empty
#                  : [INFO] Extraction Clean
#
# ERC Summary File : fsm_states.sum
# Extraction Report File : fsm_states.rep
# Comparison Report File : fsm_states.rep.cls
#
#####

INFO:      Creating cross reference database ...
INFO:      Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 80.12
INFO:      Checking in all SoftShare licenses.
INFO:      Creating cross reference database completed. Mon Nov 24 10:37:46 2025

Pegasus finished normally. 2025-11-24 10:37:46

```

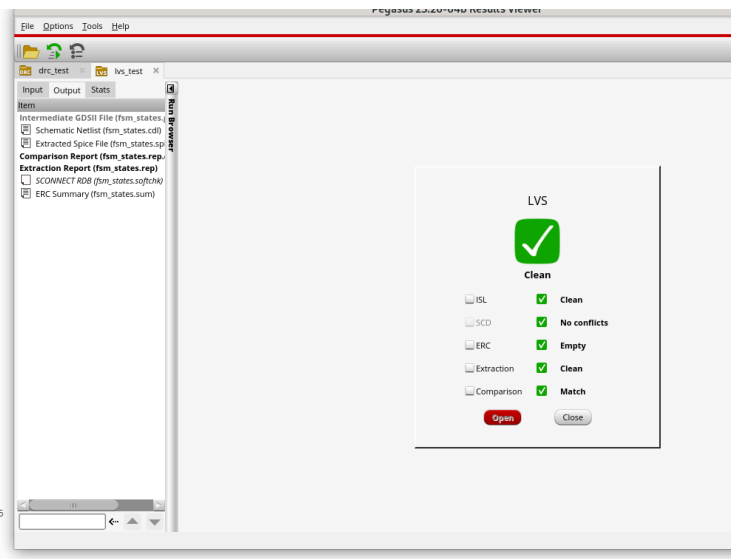


Figure 111: FSM States LVS Test

Figure 111 shows the LVS test for the FSM states layout and it was able to pass.

Figure 112 shows the cache FSM schematic.



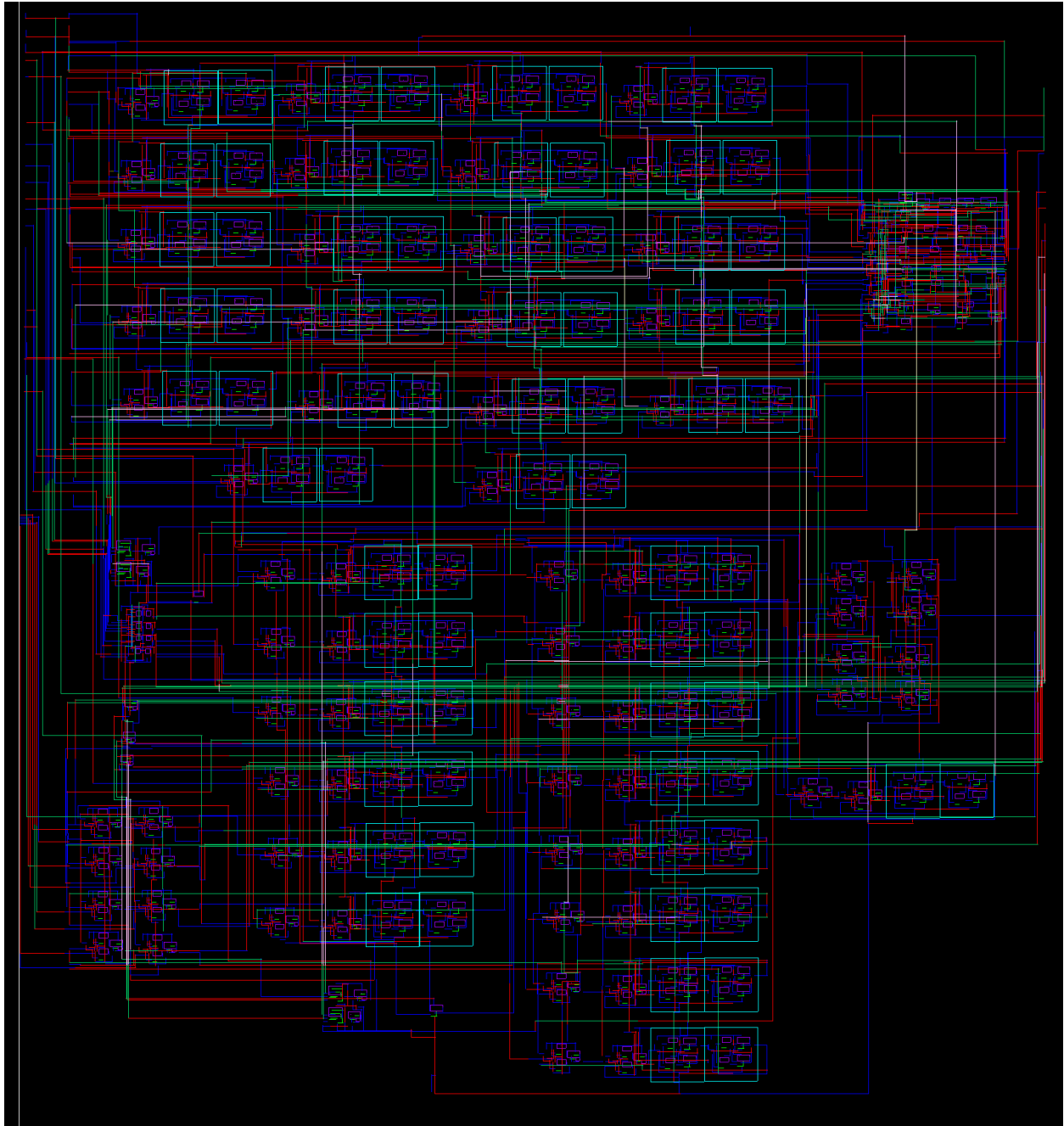


Figure 113: Cache FSM Layout

Figure 113 shows the cache FSM layout.

```

INFO:      Status: Finishing as of 2025-11-24 10:38:43 Elapsed: 00:00:04 CPU time[s]: 4
INFO:      Resource usage by worker at exit:
          CPU      Elapsed  CPUs      CPU      Peak      Peak engine
Worker:  time[s]  time[s]  Allocated  usage[%]  Memory[MB]  Memory[MB]
          0:        3        4          1      69.31      121        16
Worker resource usage summary: Total CPU[s]: 3 Run duration[s]: 4 Max peak memory[MB]: 121
Worker CPU utilization summary: 3 of 4 available CPU seconds (75.00%) with 1 average available CPU(s)
INFO:      Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache_fsm.sum
INFO:      All selected rules completed
          Total CPU Time      : 3 (s)
          Total Real Time     : 5 (s)
          Total Original Geometry : 5558 (147456)
          Total DRC RuleChecks  : 562
          Total DRC Results    : 0 (0)
          ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache_fsm.drc_errors.asci
INFO:      Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 10:38:43

```

Figure 114: Cache FSM DRC Test

Figure 114 shows the DRC Test for the cache FSM layout and it was able to run with 0 errors.

```

INFO:      Create report (0s)
INFO:      Cleanup (0s)
INFO:      Report finished
INFO:      Extraction Run Summary
          Total CPU Time      : 1 (s)
          Total Real Time     : 2 (s)
          Peak Memory Used    : 31.00 (M)
INFO:      NVN Run Summary
          Total CPU Time      : 1 (s)
          Total Real Time     : 1 (s)
          Peak Memory Used    : 274.46 (M)
INFO:      LVS Summary
          Total CPU Time      : 2 (s)
          Total Real Time     : 3 (s)
          Peak Memory Used    : 274.46 (M)

#####
# Run Result      : MATCH
#
# Run Summary     : [INFO] ERC Results: Empty
#                  : [INFO] Extraction Clean
#                  : [INFO] Some Sections Have Been Truncated
#
# ERC Summary File : cache_fsm.sum
# Extraction Report File : cache_fsm.rep
# Comparison Report File : cache_fsm.rep.cls
#
#####

INFO:      Creating cross reference database ...
INFO:      Manager summary: CPU(s): 0 Elapsed(s): 0 Peak memory(MB): 90.43
INFO:      Checking in all SoftShare licenses.
INFO:      Creating cross reference database completed. Mon Nov 24 10:39:19 2025

Pegasus finished normally. 2025-11-24 10:39:19

```

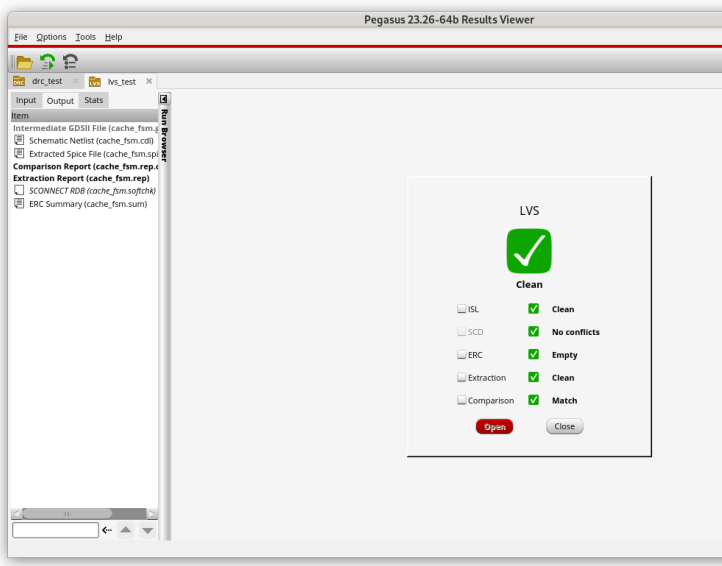


Figure 115: Cache FSM LVS Test

Figure 115 shows the LVS test for the Cache FSM layout and it was able to pass.

4.7 Full Cache

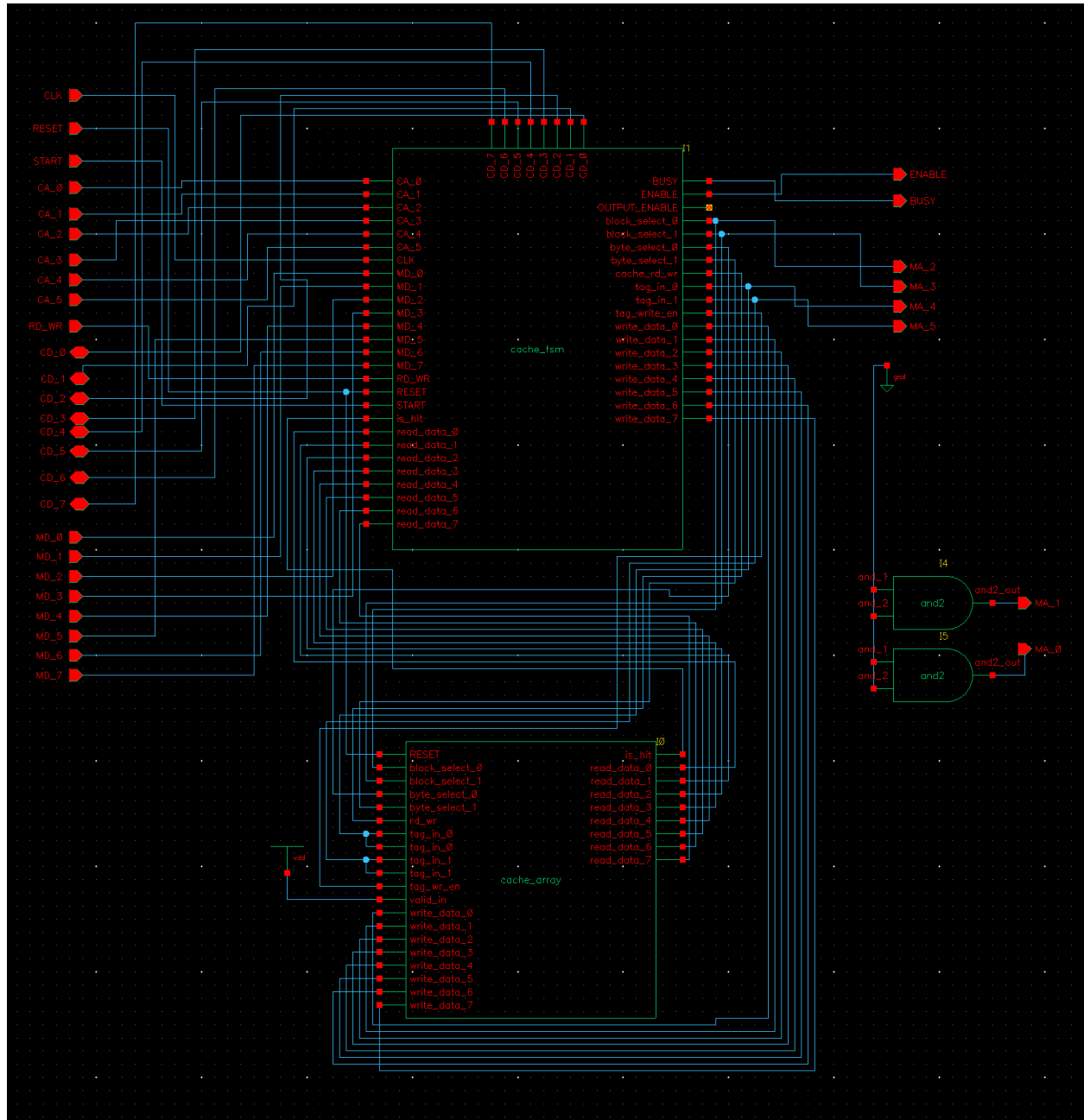


Figure 116: Cache Schematic

Figure 116 shows the full cache implementation schematic. The two AND gates are used to just get the memory bits 1 and 0 to be logic 0.

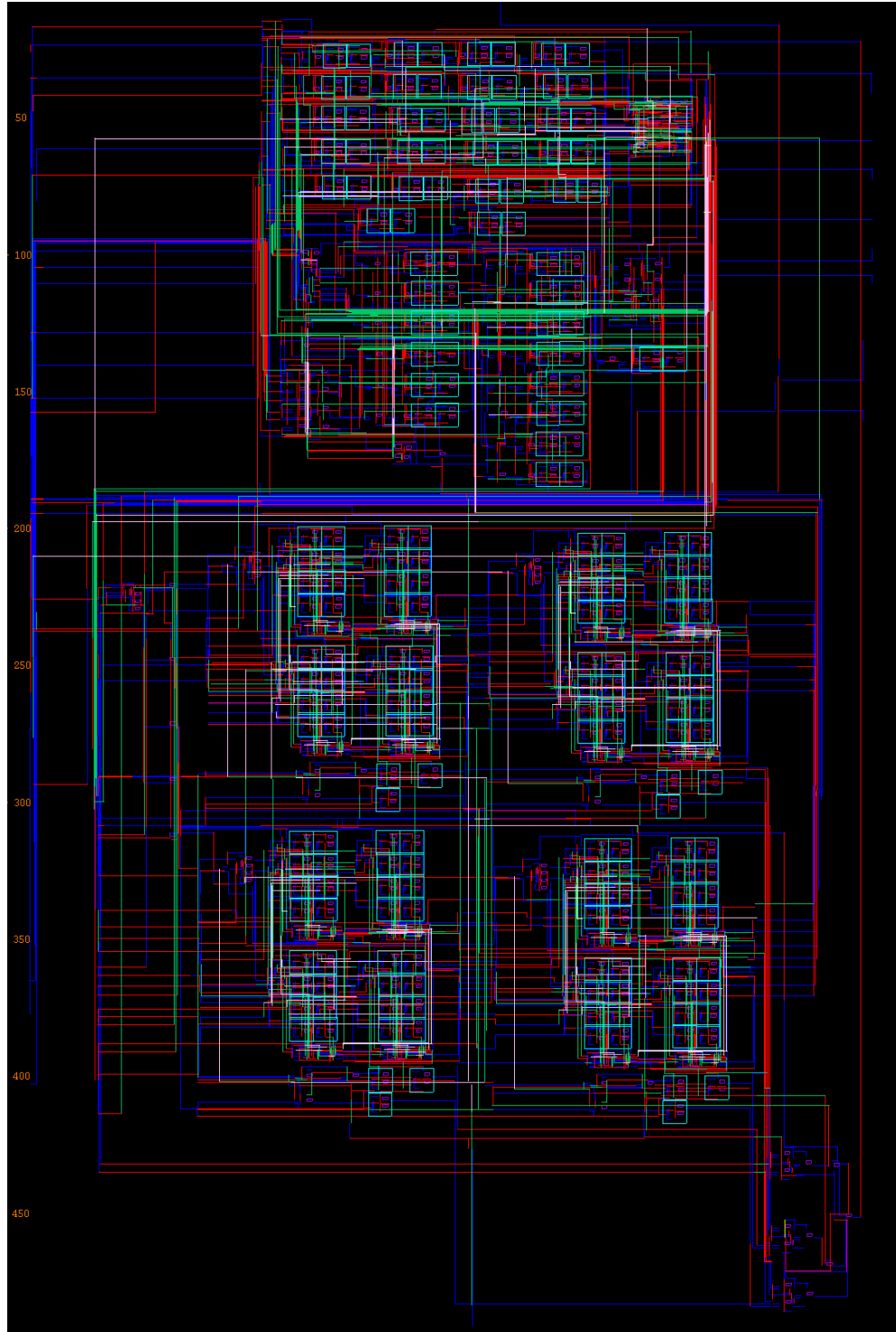


Figure 117: Cache Layout

Figure 117 shows the layout for the cache. Cache width 314 micro meters and height is 500 micro meters. The total area is 157 mm.

```

0: 1 / 1 121 16 3
INFO: Status: Finishing as of 2025-11-24 10:40:19 Elapsed: 00:00:04 CPU time[s]: 4
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 3 4 1 69.97 121 16
Worker resource usage summary: Total CPU[s]: 3 Run duration[s]: 4 Max peak memory[MB]: 121
Worker CPU utilization summary: 3 of 4 available CPU seconds (75.00%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache.sum
INFO: All selected rules completed
      Total CPU Time : 3(s)
      Total Real Time : 5(s)
      Total Original Geometry : 8771(350434)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 10:40:19

```

Figure 118: Cache DRC Test

Figure 118 shows the DRC Test for the cache layout and it was able to run with 0 errors.

```

0: 1 / 1 121 16 3
INFO: Status: Finishing as of 2025-11-24 10:40:19 Elapsed: 00:00:04 CPU time[s]: 4
INFO: Resource usage by worker at exit:
      CPU Elapsed CPUs CPU Peak Peak engine
Worker: time[s] time[s] Allocated usage[%] Memory[MB] Memory[MB]
0: 3 4 1 69.97 121 16
Worker resource usage summary: Total CPU[s]: 3 Run duration[s]: 4 Max peak memory[MB]: 121
Worker CPU utilization summary: 3 of 4 available CPU seconds (75.00%) with 1 average available CPU(s)
INFO: Generating summary in /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache.sum
INFO: All selected rules completed
      Total CPU Time : 3(s)
      Total Real Time : 5(s)
      Total Original Geometry : 8771(350434)
      Total DRC RuleChecks : 562
      Total DRC Results : 0 (0)
      ASCII report database is /afs/umbc.edu/users/r/r/r197/home/cadence/cadence_setup/drc_test/cache.drc_errors.ascii
INFO: Checking in all SoftShare licenses.

Pegasus finished normally. 2025-11-24 10:40:19

```

Figure 119: Cache LVS Test

Figure 119 shows the LVS test for the cache layout and it was able to pass.

5 Work Breakdown

Task	Person
Cache cell VHDL	Richard
Cache block VHDL	Richard
Cache array VHDL	Yomi
Cache FSM behavioral	Mostly Yomi and some from Richard
Cache FSM structural	Mostly Richard and some from Yomi
Cache Top VHDL	Yomi
Testbenches and Helper Modules	Richard and Yomi
Cache Cell Layout	Yomi
Cache Block Layout	Yomi
Cache Array Layout	Yomi
Cache FSM Layout	Richard
Lower Lever Module Layouts	Richard and Yomi

Note: The instructions for running the files is on the readme from the zipfile.